Model Approximation in Transport Dynamics Motivated by Biological Phenomena

C.M.T. (Mai) Nguyen

June 30, 2026

Bachelor Thesis

Supervisor: Dr. S.C. Hille



Leiden University
Mathematical Institute

"For a moment, nothing happened.
Then, after a second or so, nothing continued to happen."

Douglas Adams, The Hitchhiker's Guide to the Galaxy

Contents

1 Introduction **7**

2 Mathematical background for the Velocity-Jump Model **9**
2.1 Integral operators on Hilbert spaces . . . 9
2.2 Constructing density functions with the Law of Large Numbers . . . 13
2.3 Master Transport Equation . . . 14

3 Derivation of the general Velocity-Jump Model **15**
3.1 Constructing density functions . . . 15
3.2 Contribution of the initial distributions . . . 16
3.3 Deriving the general Velocity-Jump formulation . . . 16
3.3.1 Constructing a reorientation kernel . . . 18
3.3.2 Constructing a reorientation operator . . . 20
3.3.3 Forcing from the initial distribution . . . 21
3.4 The case of ageless initial conditions . . . 21
3.5 The case of Markovian dynamics . . . 22

4 The Velocity-Jump Model with Markovian dynamics **24**
4.1 Properties of the turning kernel . . . 24
4.2 Properties of the turning operator . . . 26
4.3 The diffusive limit of the Markovian Velocity-Jump Model . . . 28
4.3.1 Time and space scalings . . . 28
4.3.2 The formal diffusion limit . . . 29

5 Limitations of the classical Velocity-Jump Model **31**
5.1 Markovian turning, resting states and memory effects . . . 31
5.2 Symmetric velocity space and the lack of directional confinement . . . 31
5.3 Exclusion of memory-based behavior and agent-agent interactions . . . 32
5.4 Homogeneous population structure . . . 32
5.5 Interacting populations . . . 32

6 Derivation of the General Hitchhikers' model **34**
6.1 Constructing density functions . . . 34
6.2 Contribution of the initial distributions . . . 37
6.3 Deriving the general Hitchhikers' model . . . 38
6.3.1 Eliminating auxiliary variables . . . 40
6.3.2 Final model formulation . . . 42
6.3.3 The case of ageless initial conditions . . . 43
6.3.4 The case of Markovian Dynamics . . . 44

7 The Markovian Hitchhikers' Model **46**
7.1 Model formulation and assumptions . . . 46
7.2 Scaling the Hitchhikers' model . . . 47
7.2.1 Scaling in the case of fast transitions . . . 48
7.2.2 Scaling in the case of slow transitions . . . 49
7.3 Diffusive limit of the Hitchhikers' Model . . . 49
7.4 Numerical verification of the parabolic limit . . . 52
7.4.1 Numerical estimation of the diffusion tensor . . . 53
7.4.2 Numerical discretization of the PDE system . . . 53
7.4.3 The particle-based stochastic simulation . . . 56
7.4.4 Coupling of the stochastic simulation and PDE solution . . . 56
7.4.5 Density comparison . . . 58
7.4.6 Parameters in the simulation study . . . 58

7.4.7 Results of the simulation study 58

8 Mathematical background for moment equations 61
8.1 Moment equations 61
8.2 Mean Squared Displacement (MSD) 61
8.2.1 The ensemble MSD for multiple particles 62
8.2.2 The time-averaged MSD for single-particle tracking 62
8.2.3 The MSD for diffusive processes 63

9 Moment equations for the Markovian Hitchhikers' Model 65
9.1 The case of a homogeneous carrier density 66
9.1.1 The long-term MSD 66
9.1.2 The MSD for two-dimensional Brownian motion 68
9.2 Numerical verification of the MSD from the moment system for the Markovian case . 68
9.3 Numerical verification of the time-averaged single-particle MSD 69

10 Discussion and conclusions 71
10.1 Practical application of the Hitchhikers' Model 71
10.2 Limitations of the study and recommendations for future work 71
10.3 Novelty 72

Appendices 74

A Deriving moment equations for the Markovian Hitchhikers' Model 74
A.1 Deriving ODEs for the second spatial moments 74
A.2 Deriving ODEs for mixed moments 76
A.3 Deriving ODEs for the second velocity moments 77

B Code availability and computational details 78

References 79

1 Introduction

Transport is omnipresent in the living world. Whether one observes microbes searching for nutrients or birds migrating across continents, these organisms rely on movement to survive, reproduce, and interact with their environment. Despite the enormous differences in scale, physiology, and ecological context, the underlying transport mechanisms often reduce to the same abstract process, in which entities exhibit movement patterns that can be described in terms of alternating phases of directed motion and reorientation.

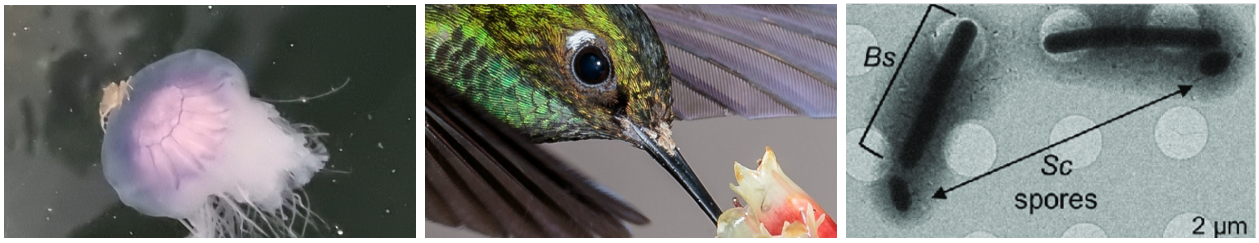
While movement on a single-entity level can seem unpredictable, observations on a larger scale often reveal its purpose. Mathematical models can form a bridge between individual-level behaviour on small time scales and slowly emerging population-level patterns. The study of transport dynamics therefore helps to better understand the rules of motion and transformation by tracking the evolution of a distribution of entities. These models are thus an essential tool for understanding dispersal, foraging, and collective phenomena [33].

A classical framework for modelling such behaviour is the velocity-jump process, in which individuals move with a constant velocity for a random duration before undergoing an instantaneous change in direction, speed or both. This individual-based framework yields a diffusive limit that describes the macroscopic spread of a population. The Velocity-Jump Model was introduced by Othmer et al. [33] and has been successful in explaining movement patterns in organisms that exhibit rapid changes in movement such as the bacterium *Escherichia coli*, where run-and-tumble dynamics are well approximated by Poisson turning events and exponential run times [34, 43].

Recent work by Taylor-King et al. extends the velocity-jump framework to include a resting state and generalised waiting-time distributions [47]. This leads to a coupled system of integro-differential equations with memory kernels that encode the history of the process. By relaxing the assumption of Poisson turning events, the model allows for memory effects in the reorientation process. This broadens the application of the Velocity-Jump Model from random movement to more complex movement patterns, as Taylor-King et al. demonstrated by successfully applying it to model migratory bird trajectories.

In many studies on velocity-jump models, only one species is taken into account, sometimes including a one-sided influence of its environment [14, 34, 43]. However, we propose that the velocity-jump framework has the potential to describe more complex interactions between multiple species. The motivation to formulate such a model is rooted in one peculiar ecological interaction that can be found in many biological systems. In the case of *phoresy*, a phoretic animal with limited ability to travel great distances, disperses by attaching itself to a moving interspecific host [49]. This method of transport, also referred to as *hitchhiking*, allows phoretic organisms to utilize taxis mechanisms of their hosts to reach their preferred environment. A prominent example of this phenomenon is microbial carrier-mediated transport, also referred to as *intermicrobial hitchhiking* [31]. Through recent work, it has become clear that intermicrobial hitchhiking is a widespread, understudied transport mechanism that occurs in all sorts of ecological spaces, such as in soil [24], plant tissues [19], abiotic surfaces [19, 37] and human tissues [41, 44].

Despite growing evidence of its ecological importance, phoresy remains difficult to study due to challenges in tracking interacting organisms over large spatial scales [3]. Most empirical studies have appeared only in the last 15 years, and a unified theoretical framework is still lacking. Nevertheless, carrier-mediated transport is increasingly recognised as an important mechanism in shaping microbial communities, facilitating pathogen spread, and potentially benefiting hosts through metabolic cooperation or access to new niches [3]. For example, the mite-humming bird interaction depicted in Figure 1(b) increases flower visitation and thus pollination, as the mite and bird use the same nectar source [16]. In the human oral microbiome, it has been shown that hitchhiking contributes to the spatial organization of polymicrobial communities [44]. In soil systems, hitchhiking enables bacterial spores to reach plant roots, as is depicted in Figure 1(c). Here, they may produce antibiotics and undergo symbioses that benefit both plant and soil ecosystem health [32].



(a) (b) (c)

Figure 1: Three instances of carrier-mediated transport. (a) A phoretic crab on a jelly fish [21]. (b) Phoretic mites on a humming bird's beak [30]. (c) Phoretic spores attached to motile bacteria, which is a form of intermicrobial hitchhiking [31].

Phoresy, and in particular intermicrobial hitchhiking, involves a wide range of transporter–host combinations, including bacteria, viruses, fungi, protozoa, and multicellular animals. The wide diversity, together with unresolved mechanistic details, makes it difficult to capture the underlying dynamics using classical observational approaches. Despite all the variations in the process, the general principle has been reduced to four general mechanisms, as described by Muok and Briegel [31]. We postulate that mathematical models can help distinguish between those general classes of mechanisms.

Another reason to believe that mathematical models offer a complementary route to advance this field of study is that direct observation is challenging and many underlying processes are not directly measurable. As such, we present an extension of the Velocity-Jump Model, which allows for an exploration of diverse phoretic interactions through inclusion of mechanisms of attachment, transport, and detachment from the host. With this model, phenomenological observations can provide hypotheses regarding the underlying, non-measurable mechanisms that allow this interaction to take place.

The aim of this thesis is to analyse the classical Velocity-Jump Model, explore its limitations and propose an extension of this model capable of capturing carrier-mediated dispersal. Although the motivation for this extension emerged from ecological interactions, the term hitchhiking has been coined to describe carrier-mediated transport across many biological contexts, from microbial intracellular transport on organelles and microtubules [10, 29, 40] to environmental systems such as pathogen dispersal on microplastics [6]. While the exact mechanisms differ greatly in this span of settings, we aim to formulate a model that can be easily adjusted to elucidate the dynamics of all these phenomena.

We start by providing some mathematical background for the study of the original Velocity-Jump Model in Section 2. Then, we will derive a general formulation of this model in Section 3 and we will study the Markovian version of this model in Section 4. Here, we obtain a deterministic macroscopic limit equation as an approximation of the stochastic process. The limitations of the Velocity-Jump Model will be discussed in Section 5. This motivates the construction of a general Hitchhiking model in Section 6. In Section 7, we will study the Hitchhiking model with Markovian dynamics and perform a simulation study to compare the deterministic limits found to the underlying stochastic system. Finally, we introduce a new approximation method based on moment equations in Section 8 and we will evaluate this approximation for the Markovian Hitchhiking model in Section 9.

2 Mathematical background for the Velocity-Jump Model

Throughout this thesis, we will study transport processes through population density functions, where we follow the analysis outlined by Hillen and Othmer [22]. In this section, we will introduce some general notions on the spaces in which our functions of interest reside, and derive some useful properties that will be applied in the construction of the general Velocity-Jump Model and in the analysis of the Markovian Velocity-Jump Model.

To analyse the density functions, it is convenient to work in the setting of a Hilbert space. We will start with defining such spaces. Additionally, we will introduce the Master Transport Equation as the general form of a transport equation. Lastly, we will study the Law of Large Numbers, to justify the construction of density functions from single-particle observations. The definitions, concepts and results from this section will be applied in the derivation and the analysis of the classical Velocity-Jump Model in Section 3 and 4.

2.1 Integral operators on Hilbert spaces

We will start by introducing Hilbert spaces, and some helpful properties for our analysis. To this end, we first define a broader class of spaces.

Definition 2.1 (Inner product space [46])**.** Let V be a vector space over a field $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. An *inner product* on V is a function

$$\langle \cdot, \cdot \rangle : V \times V \to \mathbb{F}$$

such that for all $x, y, z \in V$ and $\lambda, \mu \in \mathbb{F}$, the following hold:

1. $\langle \lambda x + \mu y, z \rangle = \lambda \langle x, z \rangle + \mu \langle y, z \rangle$,
2. $\langle x, y \rangle = \overline{\langle y, x \rangle}$,
3. $\langle x, x \rangle \geq 0$,
4. $\langle x, x \rangle = 0 \iff x = 0$,

where $\overline{\langle y, x \rangle}$ denotes the complex conjugate of $\langle y, x \rangle$.

A vector space V equipped with an inner product, is an *inner product space*.

Inner products give a natural notion of orthogonality and projection, which will be useful for our analysis. In this thesis, we will mainly work on real inner product spaces.

A Hilbert space is an inner product space with an extra property.

Definition 2.2 (Hilbert space)**.** A *Hilbert space* is an inner product space that is complete with respect to the norm induced by the inner product, $\|x\| := \sqrt{\langle x, x \rangle}$.

As we aim to study probability density functions, we want to find a Hilbert space based on a real vector space $V \subseteq \mathbb{R}^n$ for $n \in \{1, 2, 3\}$. The completeness of the Hilbert space ensures that the approximative limits stay in the function space. To model densities on spatial domains, we use standard function spaces. As we aim to formulate a deterministic equation as an approximation for a stochastic process, we specifically consider L^2-function spaces, which are included in the following definition.

Definition 2.3 (L^p function spaces [42])**.** Let $(X, \mathcal{A}, \mu)$ be a measure space. For $1 \leq p < \infty$ define

$$L^p(X, \mu) := \left\{ u : X \to \mathbb{R} \text{ measurable} : \int_X |u|^p \, d\mu < \infty \right\} / \sim,$$

where $u \sim v$ if $u = v$ μ-almost everywhere. The space $L^p(\mu)$ is equipped with the norm

$$\|u\|_{L^p} = \left(\int_X |u|^p \, d\mu \right)^{1/p}.$$

Throughout this thesis, we let the measure μ be the Lebesgue measure on domains $\Omega \subset \mathbb{R}^n$ unless stated otherwise. We then denote $L^2(\Omega)$ instead of $L^2(\Omega, \mu)$.

We will now confirm that the $L^2(\Omega)$ is indeed a Hilbert space.

Lemma 1. *For a bounded domain $\Omega \subset \mathbb{R}^n$ with Lebesgue measure, $L^2(\Omega, \mu)$ is a Hilbert space with inner product*

$$\langle u, v \rangle_{L^2(\Omega)} = \int_\Omega u(x)\, v(x)\, dx. \tag{1}$$

Proof. One can easily confirm that the inner product is well defined on $L^2(V)$ and that it adheres to the conditions from Definition 2.1. Completeness is proven as the Riesz-Fischer Theorem [42]. □

The inner product given in Equation (1) induces the norm given by

$$||f||_{L^2} = \sqrt{\langle f, f \rangle} = \left(\int_V |f|^2 \, d\mu \right)^{\frac{1}{2}}, \qquad f \in L^2(V). \tag{2}$$

In this thesis, $L^2(V)$ is the space we will use for population densities and for defining the linear operators that appear in the Velocity-Jump Model. We will use the notation $||f||_2$ rather than $||f||_{L^2}$ as the norm on $L^2(V)$.

Using the L^2-norm given in Equation (2), we define the class of integral operators.

Definition 2.4 (Integral operator)**.** Let $L^2(V)$ be a Hilbert space where V is a closed, bounded (subset of a) finite-dimensional vector space and let $\varphi : V \times V \to \mathbb{C}$ be continuous on $V \times V$. Then, for $x \in V$ and $f \in L^2(V)$, we define

$$(\psi_\varphi f)(x) = \int_V \varphi(x, y) f(y) \, dy,$$

where dy is the Lebesgue measure. The operator $\psi_\varphi : L^2(V) \to L^2(V)$ is an *integral operator* and the function φ is its *kernel*.

In order to better understand this integral operator, we introduce the operator norm, which describes to what extent it amplifies inputs.

Definition 2.5 (Bounded operator)**.** Let H be a real Hilbert space with norm $||\cdot||$. A linear operator $L : H \to H$ is *bounded* if there exists a $C \in \mathbb{R}_{>0}$ such that

$$||Lx|| \leq C||x||$$

holds for all $x \in H$. The smallest $C \in \mathbb{R}_{>0}$ for which this holds, is the *operator norm*, given by

$$||T|| := \sup_{x \in H \setminus \{0\}} \frac{||Tx||}{||x||},$$

or, equivalently

$$||T|| := \sup_{||x||=1} ||Tx||.$$

Boundedness ensures that an operator does not arbitrarily amplify an input. We will prove that the integral operator is indeed bounded.

Lemma 2. *The integral operator as defined in Definition 2.4 is a well defined, bounded linear operator on the function space $L^2(V)$.*

Proof. We note that as φ is continuous, so is $\varphi_x : V \to \mathbb{C}$ given by $y \mapsto \varphi(x, y)$ for all $x \in V$. So, we have $\varphi_x \in L^2(V)$. We then observe that $\int_V \varphi(x, y)f(y)\ dy = \langle \varphi_x, f\rangle$. In other words, $(\psi_\varphi f)(x)$ can be written as an inner product for all $f \in L^2(V), x \in V$. This means that $(\psi_\varphi f)$ is well defined on V.

Now we show that $(\psi_\varphi f)$ is a bounded operator on V. Applying the Cauchy-Schwarz inequality [42], gives

$$\begin{aligned}|(\psi_\varphi f)(x)|^2 &= \left|\int_V \varphi(x, y)f(y)\ dy\right|^2 \\ &= |\langle \varphi(x, y), f(y)\rangle|^2 \\ &\leq \langle \varphi(x, y), \varphi(x, y)\rangle\langle f(y), f(y)\rangle \\ &= \int_V |\varphi(x, y)|^2\ dy\ ||f||^2.\end{aligned}$$

As both φ and f are continuous, and V is bounded, we can conclude that $f \mapsto \psi_\varphi f$ is bounded on $L^2(V)$.

We will now prove that $f \mapsto \psi_\varphi f$ is continuous on $L^2(V)$.

We define $K := \int_V |f(x)|\ dx$, which is well defined, as

$$\begin{aligned}\int_V |f(x)|\ dx &= |\langle f, 1\rangle| \\ &\leq \sqrt{|\langle f, f\rangle|}\sqrt{|\langle 1, 1\rangle|} \\ &= |V|^{\frac{1}{2}} \left(\int_V |f(x)|^2\ dx\right)^{\frac{1}{2}},\end{aligned}$$

where we again applied the Cauchy-Schwarz inequality to obtain the inequality. Because $\int_V |f(x)|^2\ dx$ is bounded, so is $\int_V |f(x)|\ dx$.

So, we have $\psi_\varphi f \in L^2(V)$ and

$$||\psi_\varphi f||^2 \leq \int_V \int_V |\varphi(x, y)|^2\ dy\ dx\ ||f||^2.$$

As $\psi_\varphi f$ depends linearly on f, we can conclude that ψ_φ is a well-defined linear bounded operator on $L^2(V)$, and thus continuous [18]. □

One of the useful properties of bounded operators is the existence of an adjoint operator.

Definition 2.6 (Adjoint [46])**.** For inner product spaces V and W over the same field ($\mathbb{R}$ or $\mathbb{C}$) with the inner product $\langle \cdot, \cdot\rangle$, and a bounded linear map $f : V \to W$, the adjoint is the unique linear map $f^* : W \to V$ such that $\langle f(v), w\rangle = \langle v, f^*(w)\rangle$ for all $v \in V, w \in W$.

Adjoints can be used to obtain information on linear operators. In particular, we will use the property that real eigenvalues of the adjoint correspond with the real eigenvalues of the linear operator in $\mathbb{R}^n$, as we will show in the upcoming lemma.

Lemma 3. *Let $L : V \to V$ be a linear operator on a finite dimensional vector space V. Then, the eigenvalues of the adjoint operator L^* are given by the set*

$$\{\overline{\lambda} : \lambda \text{ is an eigenvalue of } L\}.$$

Proof. We will show that if λ is an eigenvalue of L, then $\overline{\lambda}$ is an eigenvalue of L^*. We note that λ is an eigenvalue of L if and only if $L - \lambda I$ is not invertible. Suppose $L - \lambda I$ is invertible. Then, there must exist some operator S such that letting S operate on $L - \lambda I$ equals the identity matrix. That is, we must have

$$S(L - \lambda I) = (L - \lambda I)S = I.$$

Taking the adjoint gives

$$S^*(L^* - \overline{\lambda}I) = (L^* - \overline{\lambda}I)S^* = I. \tag{3}$$

From this, we see that $L - \lambda I$ is invertible if and only if $L^* - \overline{\lambda}I$ is invertible. Hence, $L - \lambda I$ is not invertible if and only if $L^* - \overline{\lambda}I$ is not invertible. As shown in Equation (3), this is exactly the case if $\overline{\lambda}$ is an eigenvalue of L^*. Since λ is an eigenvalue of L, the claim follows. □

A specific type of operators that will turn out to be useful, are Hilbert-Schmidt operators.

Definition 2.7 (Hilbert-Schmidt operator)**.** Let H be a Hilbert space. An operator $L : H \to H$ is called a *Hilbert-Schmidt operator* if $\sum_{i\in\mathbb{N}} ||Le_i||^2 < \infty$ holds for any orthonormal basis $(e_i)_{i\in\mathbb{N}}$.

We will now see that the integral operator we defined earlier is a Hilbert-Schmidt operator.

Lemma 4. *The integral operator*

$$(\psi_\varphi f)(x) = \int_V \varphi(x,y)f(y)\ dy$$

as defined in Definition 2.4 is a Hilbert-Schmidt operator on $L^2(V)$ for $V \subset \mathbb{R}^n$.

Proof. Let $(f_i)_{i\in\mathbb{N}}$ be an orthonormal basis for $L^2(V)$. Then, for fixed $x \in V$, we have

$$\sum_{i\in\mathbb{N}} |\psi_\varphi f_i(x)|^2 = \sum_{i\in\mathbb{N}} \left| \int_V \varphi(x,y)f_i(y)\ dy \right|^2.$$

As $y \mapsto \varphi(x,y)$ is an element of $L^2(V)$, we can write

$$\psi_\varphi f_i(x) = \langle \varphi(x,\cdot), f_i \rangle,$$

which gives

$$\sum_{i\in\mathbb{N}} |\psi_\varphi f_i(x)|^2 = \sum_{i\in\mathbb{N}} |\langle \varphi(x,\cdot), f_i\rangle|^2.$$

As $(f_i)_{i\in\mathbb{N}}$ is an orthonormal basis, we find

$$\sum_{i\in\mathbb{N}} |\langle \varphi(x,\cdot), f_i\rangle|^2 = ||\varphi(x,\cdot)||^2 = \int_V |\varphi(x,y)|^2\ dy.$$

So, for almost every $x \in V$, we find

$$\sum_{i\in\mathbb{N}} |\psi_\varphi f_i(x)|^2 = \int_V |\varphi(x,y)|^2\ dy,$$

and integrating over x then gives us

$$\sum_{i\in\mathbb{N}} ||\psi_\varphi f_i||^2 = \int_V \int_V |\varphi(x,y)|^2\ dy\ dx.$$

As φ was taken to be an element of $L^2(V \times V)$, we know that the right hand side is finite. Thus, we find

$$\sum_{i\in\mathbb{N}} ||\psi_\varphi f_i||^2 < \infty,$$

meaning the integral operator meets the condition stated in Definition 2.7. □

We will use some properties of Hilbert-Schmidt operators to make the analysis more manageable.

Lemma 5. *Hilbert-Schmidt operators are bounded and compact. Additionally, a linear operator L and its adjoint L^* have a pure point spectrum and only nonzero eigenvalues with finite multiplicities.*

Proof. We refer to Theorem 4.5 and Lemma 14.11 in [20] for the proof. □

We use a special form of the Fredholm Alternative to solve equations involving linear operators on Hilbert spaces.

Theorem 2.1 (Fredholm Alternative for linear operators on Hilbert spaces)**.** *Let H be a Hilbert space and $L : \mathcal{D}(L) \subset H \to H$ a bounded linear operator. Consider the equation $Lu = f$. The following holds:*

1. *The equation $Lu = f$ has a solution $u \in H$ if and only if*
$$f \in \ker(L^*)^\perp$$
holds, where L^* is the adjoint of L.
2. *If the solvability condition given in Equation (1) holds, a solution exists and the set of solutions is given by*
$$\{u = u_0 + w, w \in \ker(L)\}$$
for some particular solution $u_0 \in H$.
3. *Exactly one of the following is true:*
   (a) $\ker(L) = \{0\}$. *Then, the equation $Lu = f$ has a unique solution for every $f \in H$.*
   (b) $\ker(L) \neq \{0\}$. *Then, the equation $Lu = f$ has a solution only when $f \perp \ker(L^*)$.*

Proof. We refer to Theorem 1.1 of [39] for the proof. □

2.2 Constructing density functions with the Law of Large Numbers

We now shift our attention from the function spaces to the density functions themselves. When modelling many identical particles, we often observe only the macroscopic density rather than individual trajectories. The Law of Large Numbers justifies studying the total density function, as according to this law, empirical averages of independent, identically distributed particles converge to the common expectation. We will give an intuition for this application of the Law of Large Numbers.

Let $X_1(t), \dots, X_N(t)$ be N independent and identically distributed random variables, describing positions at time t in a space Ω. We then define the empirical measure

$$\mu_N(t) := \frac{1}{N} \sum_{i=1}^{N} \delta_{X_i(t)},$$

where δ_X is the Dirac measure at position X. For a test function $\varphi : \Omega \to \mathbb{R}$, we define the empirical average density as follows:

$$\langle \mu_N, \varphi \rangle = \int \varphi \, d\mu_N = \frac{1}{N} \sum_{i=1}^{N} \varphi(X_i).$$

Applying the Law of Large Numbers on any bounded, measurable function φ gives

$$\langle \mu_N, \varphi \rangle \xrightarrow[\text{almost surely}]{N \to \infty} \mathbb{E}[\varphi(X_1)] = \int_\Omega \varphi \, d\nu = \langle \nu, \varphi \rangle$$

for some probability measure ν. If $\Omega \subseteq \mathbb{R}^n$ and ν is absolutely continuous with respect to the Lebesgue measure, we can apply the Radon-Nikodym Theorem [42], from which we conclude that there exists a density $p(\mathbf{x}, t)$ such that integrating φ against the measure ν is equivalent to integrating $\varphi(\mathbf{x})p(\mathbf{x}, t)$ against the Lebesgue measure for some density $p(\cdot, t)$ over Ω. In other words, we see

$$\int_\Omega \varphi \, d\nu = \int_\Omega \varphi(x) \, p(\mathbf{x}, t) \, d\mathbf{x}.$$

Hence the empirical measures μ_N converge weakly and almost surely to the macroscopic probability measure ν for $N \to \infty$. The function $p(\mathbf{x}, t)$ is the corresponding density that captures the almost-sure limit of the empirical particle distribution for $N \to \infty$.

2.3 Master Transport Equation

As we will study the constructed density functions over a large spatial and time scale, we introduce the framework of the Master Transport Equation. A Master Transport Equation describes the local conservation law for a scalar density. That is to say, the rate of change of mass or probability at a point equals the negative divergence of the flux of this density. Let $p(\mathbf{x}, t)$ denote the density of individuals or probability mass at position $\mathbf{x}$ and time t. The flux $J(p)$ encodes the direction and the rate at which the mass is transported through space. Its divergence $\operatorname{div}(J(p))$ measures the net outflow of the flux. For any region $\Omega \subseteq \mathbb{R}^n$, mass conservation requires

$$\frac{\partial}{\partial t}\int_\Omega p(\mathbf{x},t)\ d\mathbf{x} = -\int_{\partial\Omega} J(p(\mathbf{x},t))\cdot\mathbf{n}\ dS + \int_\Omega Q(\mathbf{x},t)\ d\mathbf{x},$$

where Q describes a gain or loss of particles on the full region Ω and the vector $\mathbf{n}$ denotes the outward normal. For a small surface element with normal $\mathbf{n}$, $J(p)\cdot\mathbf{n}$ is the rate (per unit area) at which mass crosses that surface in the outward direction. A no-flux boundary condition, given by $J(p)\cdot\mathbf{n} = 0$, enforces mass conservation in a closed domain.

Assuming sufficient smoothness of J and p, we can apply the Divergence Theorem [1] on the right hand side, which gives

$$\frac{\partial}{\partial t}\int_\Omega p(\mathbf{x},t)\ d\mathbf{x} = -\int_\Omega \operatorname{div}\left(J(p)\right)\ d\mathbf{x}.$$

Since this holds for all regions $\Omega \subseteq \mathbb{R}^n$, we obtain the general form

$$\frac{\partial}{\partial t}p(\mathbf{x},t) = -\operatorname{div}(J(p)) + Q(\mathbf{x},t). \tag{4}$$

This is a general framework that holds for all types of transport. All modelling choices are made through the expression for the flux $J(p)$. For example, in a pure diffusion process, the density flux is directly proportional to the gradient of the density, rendering $J(p) = -D\nabla_{\mathbf{x}}p$. For the process of advection, the flux is defined as the product of the vector $\mathbf{v}$ that describes the direction and the force of advection, and the density, which corresponds to the flux $J(p) = \mathbf{v}\cdot p$. For general processes, it is convenient to rewrite some expression for the transport to the form given in Equation (4) and relate terms from it to the standard forms, such as for diffusion and advection. This allows for an intuitive understanding of the flux of moving entities.

3 Derivation of the general Velocity-Jump Model

In a velocity-jump process, discontinuous changes in the directional movement or speed of a moving entity are generated by a random process. We will refer to these changes as jumps, turns or reorientations. The period between two jumps is referred to as a run. The population of entities at position $\mathbf{x} \in \Omega \subseteq \mathbb{R}^n$, moving with velocity $\mathbf{v} \in V \subseteq \mathbb{R}^n$ at time t is tracked by the density $p_t(\mathbf{x}, \mathbf{v})$. The justification of approximating the particle-based events by a density is given in Section 2.2. A turning kernel $T : V \times V \to [0, 1]$ assigns the probability that a particle moving with velocity $\mathbf{v}' \in V$ jumps to velocity $\mathbf{v}$.

In this section, we will derive a general stochastic, integro-partial differential equation that describes the evolution of $p_t(\mathbf{x}, \mathbf{v})$ over time. Then, we will study a situation with a specific initial condition, where the start of the runs correspond with the start of the observation window. We see that this assumption causes the general expression to collapse to a simpler version. We will then show that this is indeed a generalization of the Markovian model that is more broadly used, for example by Hillen and Othmer [22, 33]. This also reduces the model to a simpler form, making it more suitable for further analysis.

3.1 Constructing density functions

To construct a full expression for p_t, we introduce a random variable ρ for the duration of a run with the corresponding probability density function f_ρ, where $f_\rho(t)$ describes how likely a run is to end around time t, such that $f_\rho(t)\, dt$ gives the probability of the run ending in the arbitrarily small interval $[t, t + dt)$. Then,

$$F_\rho(t) := \text{Prob}(\rho > t) = \int_t^\infty f_\rho(s)\, ds \tag{5}$$

describes the probability of a run exceeding time t.

The density $p_t(\mathbf{x}, \mathbf{v})$ comprises of all particles that, at some time before t, have taken on velocity $\mathbf{v}$. We can relate the history of the system to time point t by introducing an auxiliary variable $\eta_t(\mathbf{x}, \mathbf{v})$ which describes the density of particles at position $\mathbf{x}$ at time t, having started a run with velocity $\mathbf{v}$. Then, particles that were not initialized in a run with velocity $\mathbf{v}$ can contribute to $p_t(\mathbf{x}, \mathbf{v})$ if they start a run with velocity $\mathbf{v}$ at some time point $t - s$ for some $s \in [0, t)$. In order to reach position $\mathbf{x}$ at time t, they must have taken on velocity $\mathbf{v}$ at position $\mathbf{x} - (t - s)\mathbf{v}$. They thus contributed to $\eta_s(\mathbf{x} - (t - s)\mathbf{v}', \mathbf{v}')$ for some time $s \in [0, t)$.

Additionally, we recognize that, depending on the type of reorientation behavior, the initial distribution of particles might affect the dynamics at time t. We therefore denote the contribution of the initial condition of p_t as the general function $b_t(\mathbf{x}, \mathbf{v})$. This consists of the particles that were initialized in a run with velocity $\mathbf{v}$, and have reached exactly position $\mathbf{x}$ at time t. We will introduce a precise formulation for b_t in Section 3.2. We then obtain the following general expression relating the density p_t to the forcing from the initial condition b_t and the auxiliary variable η:

$$p_t(\mathbf{x}, \mathbf{v}) = b_t(\mathbf{x}, \mathbf{v}) + \int_0^t F_\rho(t - s)\eta_s(\mathbf{x} - (t - s)\mathbf{v}, \mathbf{v})\, ds. \tag{6}$$

In order to study the system, we must find an expression for $\eta(\mathbf{x}, \mathbf{v})$. We denote the contribution of the initial condition of η_t as the general function $\beta_t(\mathbf{x}, \mathbf{v})$. This consists of the particles that were initialized in a run with velocity $\mathbf{v}$, and have reached exactly position $\mathbf{x}$ at time t. We will introduce a precise formulation for β_t in Section 3.2. Particles that are not initialized in a run with velocity $\mathbf{v}$ may make a turn at some time $s \in [0, t)$ from a different velocity $\mathbf{v}'$. In order for the particle to contribute to $\eta_t(\mathbf{x}, \mathbf{v})$, this turn than must have taken place at position $\mathbf{x} - (t - s)\mathbf{v}$. As such a turn might take place from any prior velocity $\mathbf{v}' \in V$, we integrate over the velocity space and we get the following expression:

$$\eta_t(\mathbf{x}, \mathbf{v}) = \beta_t(\mathbf{x}, \mathbf{v}) + \int_V T(\mathbf{v}, \mathbf{v}') \int_0^t f(t - s)\eta_s(\mathbf{x} - (t - s)\mathbf{v}', \mathbf{v}')\, ds\, d\mathbf{v}'. \tag{7}$$

3.2 Contribution of the initial distributions

We continue by finding expressions for the terms b_t and β_t in Equations (6) and (7), which describe the contribution of the initial distribution of particles to the densities p_t and η_t at time t, respectively.

The density $p_t(\mathbf{x}, \mathbf{v})$ describes a distribution of all particles in space and in velocity, with time $t = 0$ marking the start of the observation window. However, this description does not specify the system's initial state. To incorporate the initial conditions, one must account for the underlying distribution of running times of particles that are already in motion at $t = 0$. In other words, the model must track how long each mobile particle has been running prior to the start of the observation period. We call this running time of moving particles the age $a \in [0, \infty)$. We then consider p_0 to be an a-measure-valued distribution over the state space $\Omega \times V$, and we denote this distribution as $p_0(\mathbf{x}, \mathbf{v}, da)$. This density describes the running particles at time $t = 0$ at location $\mathbf{x} \in \Omega$ with velocity $\mathbf{v} \in V$, that have age $a \in [0, \infty)$ in this run.

From p_0, we derive an expression for b_t, which describes the contribution of the initial conditions to p_t for all times $t \in [0, \infty)$. This contribution consists of the particles that were initialized in a run with velocity $\mathbf{v}$, at position $\mathbf{x} - t\mathbf{v}$ and that have not jumped to a different velocity yet at time t. As this jumping process is dependent on the age of the run, the conditional probability of a particle with age a at time $t = 0$ contributing to the density p_t is given by

$$\text{Prob}(\rho > t + a | \rho > a) = \frac{\text{Prob}(\rho > t + a \wedge \rho > a)}{\text{Prob}(\rho > a)} = \frac{F_\rho(t + a)}{F_\rho(a)}.$$

We then find the following expression for b_t :

$$b_t(\mathbf{x}, \mathbf{v}) = \int_0^\infty \frac{F_\rho(t + a)}{F_\rho(a)} p_0(\mathbf{x} - t\mathbf{v}, \mathbf{v}, da). \tag{8}$$

In a similar manner, we use the initial distribution $p_0(\mathbf{x}, \mathbf{v}, da)$ to construct an expression for $\beta_t(\mathbf{x}, \mathbf{v})$, which describes the contribution of the initial distribution to η_t. We note that the particles that contribute to $\eta_t(\mathbf{x}, \mathbf{v})$ have turned to velocity $\mathbf{v}$ from a different velocity $\mathbf{v}' \in V$. The run of age a with velocity $\mathbf{v}'$ must thus have ended at exactly time $t + a$. This jump must have taken place at point $\mathbf{x} - t\mathbf{v}'$. As this could have taken place from any prior velocity $\mathbf{v}' \in V$, we integrate over the velocity space, and we obtain the following expression:

$$\beta_t(\mathbf{x}, \mathbf{v}) = \int_V T(\mathbf{v}, \mathbf{v}') \int_0^\infty \frac{f_\rho(t + a)}{F_\rho(a)} p_0(\mathbf{x} - t\mathbf{v}', \mathbf{v}', da) \, d\mathbf{v}'. \tag{9}$$

This represents the density of carries that were initially not running with velocity $\mathbf{v}$, but whose the first velocity-jump was to velocity $\mathbf{v}$ at time t at position $\mathbf{x}$.

Substituting the expressions for b_t and β_t in Equations (6) and (7) gives the full system

$$\begin{cases} p_t(\mathbf{x}, \mathbf{v}) = \displaystyle\int_0^\infty \frac{F_\rho(t + a)}{F_\rho(a)} p_0(\mathbf{x} - t\mathbf{v}, \mathbf{v}, da) \\ \qquad\qquad + \displaystyle\int_0^t F_\rho(t - s) \eta_s(\mathbf{x} - (t - s)\mathbf{v}, \mathbf{v}) \, ds, \\ \eta_t(\mathbf{x}, \mathbf{v}) = \displaystyle\int_V T(\mathbf{v}, \mathbf{v}') \int_0^\infty \frac{f_\rho(t + a)}{F_\rho(a)} p_0(\mathbf{x} - t\mathbf{v}', \mathbf{v}', da) \, d\mathbf{v}' \\ \qquad\qquad + \displaystyle\int_V T(\mathbf{v}, \mathbf{v}') \int_0^t f_\rho(t - s) \eta_s(\mathbf{x} - (t - s)\mathbf{v}', \mathbf{v}') \, ds \, d\mathbf{v}'. \end{cases} \tag{10}$$

3.3 Deriving the general Velocity-Jump formulation

We aim to eliminate the auxiliary variable η from the system given in Equation (10) in order to obtain one integro-partial differential equation that describes the density function p. To this end, we first

determine the derivate of p with respect to time. We will use the general formulation for p_t as given in Equation (6). We then have

$$\frac{\partial}{\partial t}p_t(\mathbf{x},\mathbf{v}) = \frac{\partial}{\partial t}b_t(\mathbf{x},\mathbf{v}) + \frac{\partial}{\partial t}\left(\int_0^t F_\rho(t-s)\eta_s(\mathbf{x}-(t-s)\mathbf{v},\mathbf{v})\ ds\right). \tag{11}$$

We start by calculating $\frac{\partial}{\partial t}b_t(\mathbf{x},\mathbf{v})$. Under the assumption that f_ρ is continuous, we can use Equation (5) and find

$$f_\rho(t) = -\frac{dF_\rho(t)}{dt}. \tag{12}$$

Furthermore, we find

$$\begin{aligned}\frac{\partial}{\partial t}p_0(\mathbf{x}-t\mathbf{v},\mathbf{v},da) &= \frac{\partial}{\partial t}(\mathbf{x}-t\mathbf{v})\cdot\frac{\partial}{\partial \mathbf{x}}p_0(\mathbf{x}-t\mathbf{v},\mathbf{v},da)\\ &= (-\mathbf{v}\cdot\nabla_{\mathbf{x}})p_0(\mathbf{x}-t\mathbf{v},\mathbf{v},da).\end{aligned} \tag{13}$$

Then, taking the derivative of b_t as defined in Equation (8) with respect to time gives

$$\begin{aligned}\frac{\partial}{\partial t}b_t(\mathbf{x},\mathbf{v}) &= \frac{\partial}{\partial t}\left(\int_0^\infty \frac{F_\rho(t+a)}{F_\rho(a)}p_0(\mathbf{x}-t\mathbf{v},\mathbf{v},da)\right)\\ &= \int_0^\infty \frac{\partial}{\partial t}\left(\frac{F_\rho(t+a)}{F_\rho(a)}p_0(\mathbf{x}-t\mathbf{v},\mathbf{v},da)\right)\\ &= -\int_0^\infty \frac{f_\rho(t+a)}{F_\rho(a)}p_0(\mathbf{x}-t\mathbf{v},\mathbf{v},da) + \int_0^\infty \frac{F_\rho(t+a)}{F_\rho(a)}(-\mathbf{v}\cdot\nabla_{\mathbf{x}})p_0(\mathbf{x}-t\mathbf{v},\mathbf{v},da),\end{aligned} \tag{14}$$

where in the last equation, the chain rule was applied, as well as the identities given in Equation (12) and (13).

For the second integral in Equation (11), we apply the Leibniz Integral Rule, given as follows.

Theorem 3.1 (Leibniz Integral Rule)**.** *Suppose $f(\mathbf{x},t)$ is a function such that $f(\mathbf{x},t)$ and its partial derivative $\frac{\partial}{\partial t}f(\mathbf{x},t)$ are continuous in $\mathbf{x}$ and t on a rectangle $\mathcal{R} = \{(\mathbf{x},t) : a(t) \le \mathbf{x} \le b, t_0 \le t \le t_1\}$. Also suppose that the functions $a, b : [t_0,t_1] \to \mathbb{R}$ are continuous, as are $\frac{d}{dt}a(t)$ and $\frac{d}{dt}b(t)$. Then, for $t \in [t_0,t_1]$, the following holds:*

$$\frac{d}{dt}\left(\int_{a(t)}^{b(t)} f(\mathbf{x},t)\ d\mathbf{x}\right) = \int_{a(t)}^{b(t)}\frac{\partial}{\partial t}f(\mathbf{x},t)\ d\mathbf{x} + f(b(t),t)\cdot\frac{d}{dt}b(t) - f(a(t),t)\cdot\frac{d}{dt}a(t).$$

Proof. We refer to Theorem 14 in [38] for the proof of the Leibniz Integral Rule. □

We assume that the conditions from Theorem 3.1 hold on $\Omega \times V \times [0,T]$ for some time $T > 0$ and we use Theorem 3.1 with

$$f(\mathbf{x},t) = F_\rho(t-s)\eta_s(\mathbf{x}-(t-s)\mathbf{v},\mathbf{v}), \qquad a(t) \equiv 0, \qquad b(t) = t.$$

Then, on the observation window $[0,T]$, we find

$$\begin{aligned}\frac{\partial}{\partial t}\left(\int_0^t F_\rho(t-s)\eta_s(\mathbf{x}-(t-s)\mathbf{v},\mathbf{v})\,ds\right) &= \int_0^t \frac{\partial}{\partial t}F_\rho(t-s)\eta_s(\mathbf{x}-(t-s)\mathbf{v},\mathbf{v})\,ds + \eta_t(\mathbf{x},\mathbf{v})\\ &= \eta_t(\mathbf{x},\mathbf{v}) - \int_0^t f_\rho(t-s)\eta_s(\mathbf{x}-(t-s)\mathbf{v},\mathbf{v})\,ds\\ &\quad + \int_0^t F_\rho(t-s)(-\mathbf{v}\cdot\nabla_{\mathbf{x}})\eta_s(\mathbf{x}-(t-s)\mathbf{v},\mathbf{v})\,ds,\end{aligned} \tag{15}$$

where in the first equality, we used that $F_\rho = \int_0^\infty f_\rho(s)\, ds = 1$, as f_ρ is a probability density function on $[0, \infty)$.

Then, substituting Equations (14) and (15) in Equation (11) gives

$$\begin{aligned}
\frac{\partial}{\partial} p_t(\mathbf{x}, \mathbf{v}) = \eta_t(\mathbf{x}, \mathbf{v}) &- \int_0^\infty \frac{f_\rho(t+a)}{F_\rho(a)} p_0(\mathbf{x} - t\mathbf{v}, \mathbf{v}, da) \\
&+ \int_0^\infty \frac{F_\rho(t+a)}{F_\rho(a)} (-\mathbf{v} \cdot \nabla_{\mathbf{x}}) p_0(\mathbf{x} - t\mathbf{v}, \mathbf{v}, da) \\
&- \int_0^t f_\rho(t-s) \eta_s(\mathbf{x} - (t-s)\mathbf{v}, \mathbf{v})\, ds \\
&+ \int_0^t F_\rho(t-s)(-\mathbf{v} \cdot \nabla_{\mathbf{x}}) \eta_s(\mathbf{x} - (t-s)\mathbf{v}, \mathbf{v})\, ds \\
= \eta_t(\mathbf{x}, \mathbf{v}) &- (\mathbf{v} \cdot \nabla_{\mathbf{x}}) \left(\int_0^\infty \frac{F_\rho(t+a)}{F_\rho(a)} p_0(\mathbf{x} - t\mathbf{v}, \mathbf{v}, da) + \int_0^t F_\rho(t-s) \eta_s(\mathbf{x} - (t-s)\mathbf{v}, \mathbf{v})\, ds \right) \\
&- \int_0^\infty \frac{f_\rho(t+a)}{F_\rho(a)} p_0(\mathbf{x} - t\mathbf{v}, \mathbf{v}, da) \\
&- \int_0^t f_\rho(t-s) \eta_s(\mathbf{x} - (t-s)\mathbf{v}, \mathbf{v})\, ds. \\
= \eta_t(\mathbf{x}, \mathbf{v}) &- (\mathbf{v} \cdot \nabla_{\mathbf{x}}) p_t(\mathbf{x}, \mathbf{v}) \\
&- \int_0^\infty \frac{f_\rho(t+a)}{F_\rho(a)} p_0(\mathbf{x} - t\mathbf{v}, \mathbf{v}, da) \\
&- \int_0^t f_\rho(t-s) \eta_s(\mathbf{x} - (t-s)\mathbf{v}, \mathbf{v})\, ds.
\end{aligned}$$

Thus, we get

$$\begin{aligned}
\frac{\partial}{\partial t} p_t(\mathbf{x}, \mathbf{v}) + (\mathbf{v} \cdot \nabla_{\mathbf{x}}) p_t(\mathbf{x}, \mathbf{v}) = &\eta_t(\mathbf{x}, \mathbf{v}) \\
&- \int_0^\infty \frac{f_\rho(t+a)}{F_\rho(a)} p_0(\mathbf{x} - t\mathbf{v}, \mathbf{v}, da) \\
&- \int_0^t f_\rho(t-s) \eta_s(\mathbf{x} - (t-s)\mathbf{v}, \mathbf{v})\, ds.
\end{aligned} \tag{16}$$

The left hand side of this equation describes the spatial and temporal changes of p over time. The right-hand side decomposes into contributions from newly initiated runs, from particles already in motion at time $t = 0$ and from the memory of past reorientation events. In the upcoming subsections, we will rewrite the right hand side in terms without the auxiliary variable η.

3.3.1 Constructing a reorientation kernel

In the last two terms of Equation (16), we recognize a structure similar to the expression for $\eta_t(\mathbf{x}, \mathbf{v})$ in Equation (7). In order to eliminate this auxiliary variable from our system we need to find a kernel $\Phi_\rho : \mathbb{R}_+ \to \mathbb{R}_+$ such that

$$\int_{s=\sigma}^t \Phi_\rho(t-s) F_\rho(s-\sigma)\, ds = f_\rho(t-\sigma), \qquad \sigma \in [0, t]. \tag{17}$$

We find such a kernel using the following lemma.

Lemma 6. *For a general density function $f : \mathbb{R} \to [0, 1]$ and $F(t) := \int_t^\infty f(s)\, ds$ with $\frac{d}{dt} f(t) = -F(t)$ and for all $\sigma \in [0, s]$, the kernel*

$$\overline{\Phi}(\lambda) = \frac{\lambda \overline{f}(\lambda)}{1 - \overline{f}(\lambda)}, \tag{18}$$

in Laplace space satisfies the condition

$$\int_{s=\sigma}^{t} \Phi(t-s)F(s-\sigma)\ ds = f(t-\sigma), \tag{19}$$

where we use the notation

$$\overline{g}(\lambda) := \mathscr{L}\{g(t)\}(\lambda) = \int_0^\infty \exp(-\lambda t)g(t)\ dt$$

for the Laplace transform of a general function $g(t)$.

The inverse of the expression given in Equation (18) *exists and provides a unique solution for Equation* (19)*.*

Proof. Write

$$F_\sigma := \begin{cases} F(s-\sigma) & s \geq \sigma, \\ 0 & \text{otherwise} \end{cases}, \qquad f_\sigma := \begin{cases} f(s-\sigma) & s \geq \sigma, \\ 0 & \text{otherwise.} \end{cases}$$

Then, Equation (19) can be written as

$$(\Phi * F_\sigma)(t) = f_\sigma.$$

We take the Laplace transform of this equation, using the property that the Laplace transform of a time convolution becomes a product. We then get

$$\overline{\Phi}(\lambda)\overline{F}_\sigma(\lambda) = \overline{f}_\sigma(\lambda). \tag{20}$$

With the definition of a Laplace transform, we find

$$\begin{aligned} \overline{f}_\sigma(\lambda) &= \int_0^\infty \exp(-\lambda t)f_\sigma(t)\ dt \\ &= \int_\sigma^\infty \exp(-\lambda t)f(t-\sigma)\ dt \\ &\overset{u:=t-\sigma}{=} \int_0^\infty \exp(-\lambda(u+\sigma))f(u)\ du \\ &= \exp(-\lambda\sigma)\int_0^\infty \exp(-\lambda u)f(u)\ du \\ &= \exp(-\lambda\sigma)\overline{f}(\lambda). \end{aligned}$$

With a similar calculation, we obtain

$$\overline{F}_\sigma(\lambda) = \exp(-\lambda\sigma)\overline{F}_\sigma.$$

We then use the assumption that $\frac{d}{dt}f(t) = -F(t)$ and we get

$$\begin{aligned} \overline{f}(\lambda) &= -\int_0^\infty \exp(-\lambda t)\frac{d}{dt}F(t)\ dt \\ &= [-\exp(-\lambda t)F(t)]_{t=0}^\infty + \int_0^\infty F(t)(-\lambda\exp(-\lambda t)\ dt \\ &= 1 - \lambda\overline{F}(\lambda), \end{aligned}$$

where the second equality is a result of partial integration and the third equality follows from the assumption that $F(0) = \int_0^\infty f(t)\ dt = 1$. From this equality, we get

$$\overline{F}(\lambda) = \frac{1-\overline{f}(\lambda)}{\lambda}.$$

Substituting this into Equation 20 gives:

$$\overline{\Phi}(\lambda) = \frac{\overline{f}(\lambda)}{\frac{1-\overline{f}(\lambda)}{\lambda}} = \frac{\lambda \overline{f}(\lambda)}{1-\overline{f}(\lambda)}.$$

So, indeed, $\overline{\Phi}(\lambda) = \frac{\lambda}{1-\overline{f}(\lambda)}$ satisfies the equality given in Equation (19).

For $\mathrm{Re}(\lambda) > 0$, we have $|\overline{f}(\lambda)| < |\overline{f}(0)| = 1$, ensuring $F(\lambda)$ exists for $\mathrm{Re}(\lambda) > 0$. Consequently, $\overline{\Phi}(\lambda)$ is uniquely determined. Since the Laplace transform is injective on the class of Laplace-transformable functions, this determines Φ uniquely. □

3.3.2 Constructing a reorientation operator

We continue eliminating the auxiliary variable η from Equation (16). Using Lemma 6, we define Φ_ρ such that

$$\overline{\Phi}_\rho(\lambda) := \frac{\lambda \overline{f}_\rho(\lambda)}{1-\overline{f}_\rho(\lambda)},$$

which we know exists and can be uniquely determined. We then let the kernel Φ_ρ operate on the density function p. Using Equation (6) we get:

$$\begin{aligned}\int_0^t \Phi_\rho(t-s)p_s(\mathbf{x}-(t-s)\mathbf{v},\mathbf{v})\,ds = &\int_0^t \Phi_\rho(t-s)b_s(\mathbf{x}-(t-s)\mathbf{v},\mathbf{v})\,ds \\ &+ \int_0^t \Phi_\rho(t-s)\int_0^s F_\rho(s-\sigma)\eta_\sigma(\mathbf{x}-(t-s)\mathbf{v}-(s-\sigma)\mathbf{v},\mathbf{v})\,d\sigma\,ds \\ = &\int_0^t \Phi_\rho(t-s)b_s(\mathbf{x}-(t-s)\mathbf{v},\mathbf{v})\,ds \\ &+ \int_0^t\int_\sigma^t \Phi_\rho(t-s)F_\rho(s-\sigma)\eta_\sigma(\mathbf{x}-(t-\sigma)\mathbf{v},\mathbf{v})\,ds\,d\sigma.\end{aligned}$$

The last equality follows from the Fubini's Theorem [42], as the integrand is assumed to be measurable and nonnegative on the triangular region $\{(s,\sigma) : 0 \le \sigma \le s \le t\}$. This means that the last term of the equation for p_t in the system given in Equation (10) can be substituted via

$$\int_0^t f_\rho(t-\sigma)\eta_\sigma(\mathbf{x}-(t-\sigma)\mathbf{v},\mathbf{v})\,d\sigma = \int_0^t \Phi_\rho(t-s)p_s(\mathbf{x}-(t-s)\mathbf{v},\mathbf{v})\,ds - \int_0^t \Phi_\rho(t-s)b_s(\mathbf{x}-(t-s)\mathbf{v},\mathbf{v})\,ds,$$

which no longer depends on the auxiliary variable η. Upon further inspection of Equation (16), one observes that the term $\eta_t(\mathbf{x},\mathbf{v})$ can be eliminated from the equation for p_t in Equation (10) by calculating

$$\begin{aligned}\int_0^t \Phi_\rho(t-s)\left(\int_V T(\mathbf{v},\mathbf{v}')p_s(\mathbf{x}-(t-s)\mathbf{v}',\mathbf{v}')\right)ds = &\int_0^t\int_V T(\mathbf{v},\mathbf{v}')\Phi_\rho(t-s)b_s(\mathbf{x}-(t-s)\mathbf{v}',\mathbf{v}')\,d\mathbf{v}'\,ds \\ &+ \int_0^t\int_V T(\mathbf{v},\mathbf{v}')f_\rho(t-\sigma)\eta_\sigma(\mathbf{x}-(t-\sigma)\mathbf{v}',\mathbf{v}')\,d\mathbf{v}',\end{aligned}$$

where we recognize the second term as a term for the expression for $\eta_t(\mathbf{x},\mathbf{v})$ in Equation (10), and we will indeed use

$$\begin{aligned}\int_0^t\int_V T(\mathbf{v},\mathbf{v}')f_\rho(t-\sigma)\eta_\sigma(\mathbf{x}-(t-\sigma)\mathbf{v}',\mathbf{v}')\,d\mathbf{v}' = &\int_0^t \Phi_\rho(t-s)\left(\int_V T(\mathbf{v},\mathbf{v}')p_s(\mathbf{x}-(t-s)\mathbf{v}',\mathbf{v}')\right)ds \\ &- \int_0^t\int_V T(\mathbf{v},\mathbf{v}')\Phi_\rho(t-s)b_s(\mathbf{x}-(t-s)\mathbf{v}',\mathbf{v}')\,d\mathbf{v}'\,ds.\end{aligned} \tag{21}$$

We can then write Equation (11) as follows, using Equation (7) and (21):

$$\begin{aligned}\frac{\partial}{\partial t}p_t(\mathbf{x},\mathbf{v}) + (\mathbf{v}\cdot\nabla_{\mathbf{x}})p_t(\mathbf{x},\mathbf{v}) = \beta_t(\mathbf{x},\mathbf{v}) \\ &+ \int_0^t \Phi_\rho(t-s)\left(\int_V T(\mathbf{v},\mathbf{v}')p_s(\mathbf{x}-(t-s)\mathbf{v}',\mathbf{v}')\ d\mathbf{v}'\right)\ ds \\ &- \int_0^t \int_V T(\mathbf{v},\mathbf{v}')\Phi_\rho(t-s)b_s(\mathbf{x}-(t-s)\mathbf{v}',\mathbf{v}')\ d\mathbf{v}'\ ds \\ &- \int_0^\infty \frac{f_\rho(t+a)}{F_\rho(a)}p_0(\mathbf{x}-t\mathbf{v},\mathbf{v},da) \\ &- \int_0^t \Phi_\rho(t-s)p_s(\mathbf{x}-(t-s)\mathbf{v},\mathbf{v})\ ds \\ &+ \int_0^t \Phi_\rho(t-s)b_s(\mathbf{x}-(t-s)\mathbf{v},\mathbf{v})\ ds,\end{aligned} \tag{22}$$

which does not contain the auxiliary variable η. We then define the Turning operator $\mathcal{L}^T_{\Psi_\rho}[p]$ as follows:

$$\begin{aligned}\mathcal{L}^T_{\Phi_\rho}[p_t(\cdot,\cdot)](\mathbf{x},\mathbf{v}) := &- \int_0^t \Phi_\rho(t-s)p_s(\mathbf{x}-(t-s)\mathbf{v},\mathbf{v})\ ds \\ &+ \int_0^t \Phi_\rho(t-s)\left(\int_V T(\mathbf{v},\mathbf{v}')p_s(\mathbf{x}-(t-s)\mathbf{v}',\mathbf{v}')\ d\mathbf{v}'\right)\ ds,\end{aligned} \tag{23}$$

which describes the process of particles taking on a different velocity after a run with velocity $\mathbf{v}$, or particles reorientating and taking on velocity $\mathbf{v}$ after a run with a different velocity $\mathbf{v}'$.

3.3.3 Forcing from the initial distribution

From Equations (22) and (23), we observe that in addition to a contribution in displacement from the reorientation process, there are terms with contributions from the initial distribution. We define a forcing term term by collecting all terms in Equation (22) that are dependent on the initial conditions. This gives

$$\begin{aligned}B_t(\mathbf{x},\mathbf{v}) := \beta_t(\mathbf{x},\mathbf{v}) \\ &- \int_V T(\mathbf{v},\mathbf{v}')\int_0^t \Phi_\rho(t-s)b_s(\mathbf{x}-(t-s)\mathbf{v}',\mathbf{v}')\ ds\ d\mathbf{v}' \\ &- \int_0^\infty \frac{f_\rho(t+a)}{F_\rho(a)}p_0(\mathbf{x}-t\mathbf{v},\mathbf{v},da) \\ &+ \int_0^t \Phi_\rho(t-s)b_s(\mathbf{x}-(t-s)\mathbf{v},\mathbf{v})\ ds,\end{aligned} \tag{24}$$

with β_t as defined in Equation (9). We strongly suspect that $\lim_{t\to\infty} B_t(\mathbf{x},\mathbf{v}) = 0$ must hold for all distributions f_ρ, which would mean that the contribution for the initial distribution would dissipate over time. However, this remains to be proven.

We can now formulate the general Velocity-Jump model:

$$\frac{\partial}{\partial t}p_t(\mathbf{x},\mathbf{v}) + (\mathbf{v}\cdot\nabla_{\mathbf{x}})p_t(\mathbf{x},\mathbf{v}) = \mathcal{L}^T_{\Phi_\rho}[p_t(\cdot,\cdot)](\mathbf{x},\mathbf{v}) + B_t(\mathbf{x},\mathbf{v}), \tag{25}$$

with $\mathcal{L}^T_{\Phi_\rho}[p_\cdot(\cdot,\cdot)](\mathbf{x},\mathbf{v};t)$ as in Equation (23) and $B_t(\mathbf{x},\mathbf{v})$ as in Equation (24).

3.4 The case of ageless initial conditions

We now consider a special case of the system given in Equation (25). In particular, we assume that all particles are initiated with running age 0. That is to say, at $t = 0$, the observations start simultaneously with the start of each initial run. This is different to the more general situation, where

it is assumed that the observations start when the particles have possibly already been in motion. Under this assumption, we can model the initial conditions using the Dirac measure, where the age is no longer relevant. This then allows us to define

$$p_0(\mathbf{x},\mathbf{v}) := \int_0^\infty p_0(\mathbf{x},\mathbf{v},da) = p_0(\mathbf{x},\mathbf{v},[0,\infty)]).$$

Then, we get

$$b_t(\mathbf{x},\mathbf{v}) = F_\rho(t)p_0(\mathbf{x}-t\mathbf{v},\mathbf{v}) \tag{26}$$

and

$$\beta_t(\mathbf{x},\mathbf{v}) = f(t)\int_V T(\mathbf{v},\mathbf{v}')p_0(\mathbf{x}-t\mathbf{v}',\mathbf{v}')\ d\mathbf{v}'. \tag{27}$$

We also get

$$\begin{aligned}\int_0^t \Phi_\rho(t-s)b_s(\mathbf{x}-(t-s)\mathbf{v},\mathbf{v})\ ds &= \int_0^t \Phi_\rho(t-s)F(s)p_0(\mathbf{x}-(t-s)\mathbf{v}-s\mathbf{v},\mathbf{v})\ ds\\ &= \int_0^t \Phi_\rho(t-s)F(s)\ ds\ p_0(\mathbf{x}-t\mathbf{v},\mathbf{v})\\ &= f_\rho(t)p_0(\mathbf{x}-t\mathbf{v},\mathbf{v}),\end{aligned}$$

where we used the definition of Φ_ρ in the last equality. Furthermore, we find that

$$\int_0^\infty \frac{f_\rho(t+a)}{F_\rho(a)}p_0(\mathbf{x}-t\mathbf{v},\mathbf{v},da) = \frac{f_\rho(t)}{F_\rho(0)}p_0(\mathbf{x}-t\mathbf{v},\mathbf{v}) = f_\rho(t)p_0(\mathbf{x}-t\mathbf{v},\mathbf{v}).$$

So, in this case, we can calculate Equation (24) as follows:

$$\begin{aligned}B_t(\mathbf{x},\mathbf{v}) =& f_\rho(t)\int_V T(\mathbf{v},\mathbf{v}')p_0(\mathbf{x}-t\mathbf{v}',\mathbf{v}')\ d\mathbf{v}'\\ &- f_\rho(t)\int_V T(\mathbf{v},\mathbf{v}')p_0(\mathbf{x}-t\mathbf{v}',\mathbf{v}')\ d\mathbf{v}'\\ &- f_\rho(t)p_0(\mathbf{x}-t\mathbf{v},\mathbf{v})\\ &+ f_\rho(t)p_0(\mathbf{x}-t\mathbf{v},\mathbf{v})\\ =& 0.\end{aligned}$$

Hence, under the assumption of ageless initial conditions, the forcing of the initial distribution disappears, and we find

$$\frac{\partial}{\partial t}p_t(\mathbf{x},\mathbf{v}) + (\mathbf{v}\cdot\nabla_{\mathbf{x}})p_t(\mathbf{x},\mathbf{v}) = \mathcal{L}^T_{\Phi_\rho}[p_\cdot(\cdot,\cdot)](\mathbf{x},\mathbf{v},t), \tag{28}$$

with $\mathcal{L}^T_\Phi[p_\cdot(\cdot,\cdot)](\mathbf{x},\mathbf{v},t)$ as in Equation (23).

3.5 The case of Markovian dynamics

We will now study Equation (25) in the case of exponentially distributed running times. That is, the random variable ρ follows an exponential distribution with some parameter $\lambda \in \mathbb{R}_{>0}$. We then get,

$$f_\rho(t) = \lambda\exp(-\lambda t), \qquad F_\rho(t) = \exp(-\lambda t),$$

and Laplace transforming f, with transformed variable z, gives

$$\begin{aligned}\overline{f}_\rho(z) &:= \mathscr{L}\{f_\rho(t)\}(z)\\ &= \int_0^\infty \exp(-zt)f_\rho(t)\ dt\\ &= \lambda\int_0^\infty \exp(-(z+\lambda)t)\ dt\\ &= \lambda\left[-\frac{1}{z+\lambda}\exp(-(z+\lambda)t)\right]_{t=0}^\infty\\ &= \frac{\lambda}{z+\lambda}\end{aligned}$$

for $\mathrm{Re}(z) > -\lambda$. With Lemma 6, we find

$$\begin{aligned}\overline{\Phi}_\rho(z) &= \frac{z\overline{f}_\rho(z)}{1-\overline{f}_\rho(z)} \\ &= \frac{\frac{z\lambda}{z+\lambda}}{1-\frac{\lambda}{z+\lambda}} \\ &= \lambda.\end{aligned}$$

So, we get a kernel in Laplace space that is independent of the complex time variable. Laplace-inverting the constant λ gives

$$\Phi(t) = \lambda\delta_0(t) = \begin{cases}\lambda & t = 0, \\ 0 & \text{otherwise.}\end{cases}$$

In other words, Φ is the product of the parameter λ and the Diract measure at 0. With Equation (23), we then find that the reorientation kernel reduces to

$$\mathcal{L}_\lambda^T[p](\mathbf{x},\mathbf{v};t) = -\lambda p_t(\mathbf{x},\mathbf{v}) + \lambda\int_V T(\mathbf{v},\mathbf{v}')p_t(\mathbf{x},\mathbf{v}'). \tag{29}$$

In order to study the forcing term B_t as given in Equation (24), we first find that

$$\frac{f_\rho(t+a)}{F_\rho(a)} = \frac{\lambda\exp(-\lambda(t+a))}{\exp(-\lambda a)} = \lambda\exp(-\lambda t).$$

We then see

$$\beta_t(\mathbf{x},\mathbf{v}) = \lambda\exp(-\lambda t)\int_0^\infty\int_V T(\mathbf{v},\mathbf{v}')p_0(\mathbf{x}-t\mathbf{v}',\mathbf{v}',da)\ d\mathbf{v}'.$$

With

$$\frac{F_\rho(t+a)}{F_\rho(a)} = \frac{\exp(-\lambda(t+a))}{\exp(-\lambda a)} = \exp(-\lambda t),$$

we see that

$$b_t(\mathbf{x},\mathbf{v}) = \exp(-\lambda t)\int_0^\infty p_0(\mathbf{x}-t\mathbf{v},\mathbf{v},da).$$

We also find

$$\begin{aligned}\int_0^t \Phi_\rho(t-s)b_s(\mathbf{x}-(t-s)\mathbf{v},\mathbf{v})\ ds &= \int_0^t \lambda\delta_0(t-s)b_s(\mathbf{x}-(t-s)\mathbf{v},\mathbf{v})\ ds \\ &= \lambda b_t(\mathbf{x},\mathbf{v}).\end{aligned}$$

So, Equation (24) in this case becomes

$$\begin{aligned}B_t(\mathbf{x},\mathbf{v}) = {}& \lambda\exp(-\lambda t)\int_{[0,\infty]}\int_V T(\mathbf{v},\mathbf{v}')p_0(\mathbf{x}-t\mathbf{v}',\mathbf{v}',da)\ d\mathbf{v}' \\ & - \lambda\exp(-\lambda t)\int_V T(\mathbf{v},\mathbf{v}')\int_0^\infty p_0(\mathbf{x}-t\mathbf{v},\mathbf{v},da)\ d\mathbf{v}' \\ & - \lambda\exp(-\lambda t)\int_0^\infty p_0(\mathbf{x}-t\mathbf{v},\mathbf{v},da) \\ & + \lambda\exp(-\lambda t)\int_0^\infty p_0(\mathbf{x}-t\mathbf{v},\mathbf{v},da). \\ = {}& 0.\end{aligned}$$

So, the Markovian Velocity-jump process is described by

$$\frac{\partial}{\partial t}p_t(\mathbf{x},\mathbf{v}) + (\mathbf{v}\cdot\nabla_{\mathbf{x}})p_t(\mathbf{x},\mathbf{v}) = -\lambda p_t(\mathbf{x},\mathbf{v}) + \lambda\int_V T(\mathbf{v},\mathbf{v}')p_t(\mathbf{x},\mathbf{v}'). \tag{30}$$

Many studies applying or extending the velocity-jump process assume Markovian dynamics, as the model was first introduced using this assumption [22, 33]. This simpler integro partial differential equation allows for more analytical insights than the general form given in Equation (25). Therefore, we will continue studying the Markovian case in the next section.

4 The Velocity-Jump Model with Markovian dynamics

In Section 3.5, we derived a formulation for the Velocity-Jump Model with an exponentially distributed running time. This model describes a velocity-jump process, where the discontinuous changes in directional movement or speed are generated by a Poisson process. We concluded that under this assumption, the initial distribution of particles does not contribute to the dynamics, making the model simpler to study compared to the general formulation given in Equation (25). This is also the model originally reported as the Velocity-Jump model [33]. We will study the Markovian model given in Equation (30) in closer detail, using the approach outlined by Hillen and Othmer [22] and by Hillen and Swan [23]. No external stimuli or other behavior-changing influences are incorporated in this model. We assume that the particles move within a region $\Omega \subseteq \mathbb{R}^n$. Additionally, we assume that the velocity space $V \subseteq \mathbb{R}^n$ is symmetric in the origin. The consequences of this assumption are discussed in Section 5.2.

For the general model formulation in Section 3, we denoted the population density as $p_t(\mathbf{x}, \mathbf{v})$, as this was convenient in the operator formulation. However, as we have seen in Section 3.5, no operator kernels remain in the model formulation. Therefore, we will proceed to use the standard notation $p(\mathbf{x}, \mathbf{v}, t)$ in this section. The general operator kernel Φ_ρ can be replaced by the rate parameter $\lambda \in \mathbb{R}_{>0}$, as Lemma 6 in Section 3.5 reveals that this is the unique kernel for the case of Markovian dynamics. Then, $\frac{1}{\lambda}$ describes the mean run length between velocity jumps. As before, a turning kernel $T : V \times V \to [0, 1]$ assigns the probability that a cell moving with velocity $\mathbf{v}' \in V$ jumps to velocity $\mathbf{v}$. We will study the following model:

$$\frac{\partial}{\partial t} p(\mathbf{x}, \mathbf{v}, t) + \mathbf{v} \cdot \nabla_{\mathbf{x}} p(\mathbf{x}, \mathbf{v}, t) = -\lambda p(\mathbf{x}, \mathbf{v}, t) + \lambda \int_V T(\mathbf{v}, \mathbf{v}') p(\mathbf{x}, \mathbf{v}', t) \, d\mathbf{v}'. \tag{31}$$

To better understand the microscopic reorientation process, we study the turning kernel T in Section 4.1, which we use to construct a turning operator in Section 4.2. We then zoom out and study the collective behaviour of the population by deriving a macroscopic limit equation for the velocity-jump process in Section 4.3.

4.1 Properties of the turning kernel

We consider the function space $L^2(V)$, as introduced in Definition 2.3 in Section 2.1. For fixed $\mathbf{x} \in \Omega$ and fixed time t, we consider the population's velocity profile $p(\mathbf{x}, \cdot, t)$ as an element of $L^2(V)$. To study the reorientation process of the velocity-jump process in Equation 31, we introduce the integral operator $\mathcal{T} : L^2(V) \to L^2(V)$ as follows:

$$\mathcal{T}[p(\mathbf{x}, \cdot, t)](\mathbf{v}) = \int_V T(\mathbf{v}, \mathbf{v}') p(\mathbf{x}, \mathbf{v}', t) \, d\mathbf{v}'. \tag{32}$$

We can find the adjoint of the integral operator $\mathcal{T}$, as defined in Definition 2.6, by fixing $\mathbf{x}$ and t and taking $p(\mathbf{x}, \cdot, t), q(\mathbf{x}, \cdot, t) \in L^2(V)$ and calculating:

$$\begin{aligned}
\big\langle \mathcal{T}[p](\mathbf{v}), q(\mathbf{v}) \big\rangle &= \left\langle \int_V T(\mathbf{v}, \mathbf{v}') p(\mathbf{x}, \mathbf{v}', t) \, d\mathbf{v}', q(\mathbf{v}) \right\rangle \\
&= \int_V \left(\int_V T(\mathbf{v}, \mathbf{v}') p(\mathbf{x}, \mathbf{v}', t) \, d\mathbf{v}' \right) q(\mathbf{v}) \, d\mathbf{v} \\
&= \int_V \left(\int_V T(\mathbf{v}, \mathbf{v}') p(\mathbf{x}, \mathbf{v}', t) q(\mathbf{v}) \, d\mathbf{v} \right) \, d\mathbf{v}' \\
&= \int_V p(\mathbf{x}, \mathbf{v}', t) \left(\int_V T(\mathbf{v}, \mathbf{v}') q(\mathbf{v}) \, d\mathbf{v} \right) \, d\mathbf{v}' \\
&= \left\langle p(\mathbf{x}, \mathbf{v}', t), \int_V T(\mathbf{v}, \mathbf{v}') q(\mathbf{v}) \, d\mathbf{v} \right\rangle \\
&= \left\langle p(\mathbf{x}, \mathbf{v}, t), \int_V T(\mathbf{v}', \mathbf{v}) q(\mathbf{v}') \, d\mathbf{v}' \right\rangle,
\end{aligned}$$

where we apply Fubini's Theorem [42] in the third equality. As, per Definition 2.6 in Section 2.1, this expression must be equal to $\langle p, \mathcal{T}^* q \rangle$. Then, we conclude that the adjoint $\mathcal{T}^*$ is given by

$$\mathcal{T}^* p(\mathbf{x}, \mathbf{v}, t) = \int_V T(\mathbf{v}', \mathbf{v}) p(\mathbf{x}, \mathbf{v}', t) \, d\mathbf{v}'. \tag{33}$$

We will use the adjoint $\mathcal{T}^*$ to obtain some insights in the operator $\mathcal{T}$. To this end, we introduce some assumptions on the kernel T. For $\mathbf{v}, \mathbf{v}' \in V$, the following must hold:

(T1) $T(\mathbf{v}, \mathbf{v}') \geq 0$

(T2) $\int_V T(\mathbf{v}, \mathbf{v}') \, d\mathbf{v} = 1$,

(T3) $\int_V \int_V T^2(\mathbf{v}, \mathbf{v}') \, d\mathbf{v}' \, d\mathbf{v} < \infty$,

(T4) There is a function $u_0 \in L^2(V) \backslash \{0\}, \rho > 0$ and an $N > 0$ such that for all $(\mathbf{v}, \mathbf{v}') \in V \times V$, one of the following holds:

(a) $u_0(\mathbf{v}) \leq T^N(\mathbf{v}, \mathbf{v}') \leq \rho u_0(\mathbf{v})$,

(b) $u_0(\mathbf{v}) \leq T^N(\mathbf{v}', \mathbf{v}) \leq \rho u_0(\mathbf{v})$,

where T^N represents the N-th iterate of T:

$$T^N(\mathbf{v}, \mathbf{v}') = \int_V \cdots \int_V T(\mathbf{v}', \mathbf{w}) \dots T(\mathbf{w}_{N-1}, \mathbf{v}) \, d\mathbf{w}_1 \dots \, d\mathbf{w}_{N-1}.$$

(T5) $||\mathcal{T}||_{\langle 1 \rangle^\perp} < 1$, with $\langle 1 \rangle^\perp$ the orthogonal complement in $L^2(V)$ of the space spanned by 1.

(T6) $\int_V T(\mathbf{v}, \mathbf{v}') \, d\mathbf{v}' = 1$.

Assumptions (T1) and (T2) ensure that the turning kernel T describes a probability distribution on V. Assumption (T3) assures that the integral operators $\mathcal{T}$ and $\mathcal{T}^*$ are Hilbert-Schmidt operators, as defined in Definition 2.7 and the subsequent lemma in Section 2.1.

Assumption (T4) ensures that the integral operator $\mathcal{T}$ is u_0-positive and has the Krein-Rutman property. For the details of this assumption, we refer to the analysis by Hillen and Swan [23]. In our analysis, we will use the result that the operator $\mathcal{T}$ has a simple, leading eigenvalue. When we let $\mathcal{T}$ operate on the constant density $p(\mathbf{x}, \mathbf{v}, t) \equiv 1$, we find

$$\mathcal{T}^* 1 = \int_V T(\mathbf{v}, \mathbf{v}') \, d\mathbf{v}' = 1, \tag{34}$$

due to Assumption (T2). Therefore we find that $p(\mathbf{x}, \mathbf{v}, t) \equiv 1$ is the leading eigenfunction of $\mathcal{T}^*$ with corresponding eigenvalue equal to one. Assumption (T6) forces that the same holds for operator $\mathcal{T}$. While the technical details of the u_0-positivity property lie beyond the scope of this thesis, it is worth noting that Assumption (T4) has an interpretation that prevents various contexts from being studied with the Velocity-Jump model. The assumption states that some iterate of T must cover the full symmetric velocity space V. This excludes situations in which the movement is restricted with respect to direction, for example. This limitation will be further explored in Section 5.2. An example of an operator adhering to Assumption (T4) is given in Figure 2.

Assumption (T5) is necessary to prove that the limiting equation is parabolic by ensuring a spectral gap between the leading eigenvector 1 and the remainder of the spectrum of $\mathcal{T}$. For details of this property, we refer to Lemma 3.3 by Hillen and Othmer [22].

Assumption (T6) describes the conservation of mass for the adjoint integral operator $\mathcal{T}^*$. This describes condition under which isotropic diffusion limits arise, as we will see in the upcoming analysis. When this assumption is relaxed, other analysis methods are required to study the system.



Figure 2: Four iterates of a turning operator T for $V = \mathbb{S}^1$, where T can map a velocity $\mathbf{v}' \in V$ to any velocity $\mathbf{v} \in V$ that has an angle up to 45 degrees with velocity $\mathbf{v}'$. In the left circle, it is indicated after how many iterations each region of V is covered. It takes four regions to cover whole velocity space V. This figure is inspired by Figure 6 by Hillen and Swan [23].

4.2 Properties of the turning operator

We will use the results from the previous section to analyse the right hand side of Equation (31). We refer to this expression as the linear turning operator $\mathcal{L} : L^2(V) \to L^2(V)$. For fixed $\mathbf{x} \in \Omega$ and fixed time t, we view the population's velocity profile $p(\mathbf{x}, \cdot, t)$ as an element of $L^2(V)$. The operator $\mathcal{L}$ is then given by

$$\mathcal{L}[p(\mathbf{x}, \cdot, t)](\mathbf{v}) = -\lambda p(\mathbf{x}, \mathbf{v}, t) + \lambda \mathcal{T} p(\mathbf{x}, \mathbf{v}, t), \tag{35}$$

with the integral operator $\mathcal{T}$ given in Equation (32). The turning operator has adjoint

$$\mathcal{L}^*[p(\mathbf{x}, \cdot, t](\mathbf{v}) = -\lambda p(\mathbf{x}, \mathbf{v}, t) + \lambda \mathcal{T}^* p(\mathbf{x}, \mathbf{v}, t).$$

As we had concluded in Equation (34), we have $\mathcal{T}^* 1 = 1$ due to Assumption (T6), so for all velocities $\mathbf{v}$, we find

$$\mathcal{L}^*[1](\mathbf{v}) = -\lambda + \lambda = 0. \tag{36}$$

Therefore, zero is a simple eigenvalue of $\mathcal{L}^*$ with the constant eigenfunction $p(v) \equiv 1$. From Lemma 3, we know that $\mathcal{L}$ and $\mathcal{L}^*$ have the same real eigenvalues, and we can thus conclude that $\mathcal{L}$ has eigenvalue zero.

We can obtain information on the remaining eigenvalues of operator $\mathcal{L}$ by studying its adjoint. Let $\nu \in \mathbb{R}$ be an eigenvalue of the adjoint turning operator $\mathcal{T}^*$. That is, for a corresponding eigenfunction ψ, we have

$$\mathcal{T}^*[\psi] = \nu\psi.$$

Then, using Equation (35), we find

$$\begin{aligned} \mathcal{L}^*[\psi] &= -\lambda\psi + \lambda\mathcal{T}^*\psi \\ &= -\lambda\psi + \lambda\nu\psi \\ &= \psi(-\lambda + \lambda\nu). \end{aligned}$$

Therefore, the eigenvalues of $\mathcal{L}^*$, and by Lemma 5 in Section 2.1, the eigenvalues of $\mathcal{L}$, are the elements of the set

$$\{\mu = \lambda(-1 + \nu) : \nu \text{ eigenvalue of } \mathcal{T}^*\}.$$

As $|\mathrm{Re}(\nu)| \leq 1$ holds for all eigenvalues ν of $\mathcal{T}^*$ due to Assumption (T4), we can conclude that

$$-2\lambda < \mathrm{Re}(\mu) < 0$$

holds for all eigenvalues μ of $\mathcal{L}$.

Let $q(\mathbf{v})$ now be an eigenfunction of $\mathcal{L}$ with eigenvalue $\mu \neq 0$. Then we get

$$\begin{aligned}
-\lambda q(\mathbf{v}) + \lambda \int_V T(\mathbf{v},\mathbf{v}')q(\mathbf{v}')\ d\mathbf{v}' &= \mu q(\mathbf{v}) \\
\int_V T(\mathbf{v},\mathbf{v}')q(\mathbf{v}')\ d\mathbf{v}' &= \left(1+\frac{\mu}{\lambda}\right) q(\mathbf{v}) \\
\int_V \int_V T(\mathbf{v},\mathbf{v}')q(\mathbf{v}')\ d\mathbf{v}'\ d\mathbf{v} &= \int_V \left(1+\frac{\mu}{\lambda}\right) q(\mathbf{v})\ d\mathbf{v} \\
\int_V \left(q(\mathbf{v}')\left(\int_V T(\mathbf{v},\mathbf{v}')\ d\mathbf{v}\right)\right) d\mathbf{v}' &= \left(1+\frac{\mu}{\lambda}\right)\int_V q(\mathbf{v})\ d\mathbf{v} \qquad (37)
\end{aligned}$$

Here, we applied Fubini's theorem in the last step, which is justifiable, as $V \subset \mathbb{R}^n$ is assumed to be compact and any compact subset of $\mathbb{R}^n$ with the Lebsgue measure is finite, and therefore σ-finite. Hence, $V \times V$ is σ-finite and we may apply Fubini's theorem. From Assumption (T2), we substitute $\int_V T(\mathbf{v},\mathbf{v}')\ d\mathbf{v} = 1$. Equation (37) can then be written as

$$\int_V q(\mathbf{v}')\ d\mathbf{v}' = \left(1+\frac{\mu}{\lambda}\right)\int_V q(\mathbf{v})\ d\mathbf{v}.$$

Therefore, we must have

$$\frac{\mu}{\lambda}\int_V q(\mathbf{v})\ d\mathbf{v} = 0.$$

As μ was taken to be unequal to zero, we must have

$$\int_V q(\mathbf{v})\ d\mathbf{v} = 0.$$

Therefore, we can conclude that any eigenfunction of $\mathcal{L}$ corresponding to a nonzero eigenvalue has mean zero over V. As

$$\langle 1\rangle^{\perp} = \left\{ f \in L^2(V) : \langle f, 1\rangle = \int_V f(v)\ dv = 0 \right\}$$

contains all eigenfunctions of $\mathcal{L}$ corresponding to a non-zero eigenvalue, and the space $\langle 1\rangle$ contains all eigenfunctions of $\mathcal{L}$ corresponding to eigenvalue 0, we find the decomposition $L^2(V) = \langle 1\rangle \oplus \langle 1\rangle^{\perp}$.

We recall from Definition 2.5 that

$$||\mathcal{T}||_{\langle 1\rangle^{\perp}} = \sup_{\psi \in \langle 1\rangle^{\perp}\setminus\{0\}} \frac{||\mathcal{T}[\psi]||_{L^2(V)}}{||\psi||_{L^2(V)}} < 1,$$

with the inequality holding due to Assumption (T5). Let $\psi \in \langle 1\rangle^{\perp}$. Then we get:

$$\begin{aligned}
\int_V \psi(\mathbf{v})\mathcal{L}[\psi](\mathbf{v})\ d\mathbf{v} &= \langle \mathcal{L}[\psi], \psi\rangle \\
&= -\lambda||\psi||^2 + \lambda\langle \mathcal{T}[\psi], \psi\rangle \\
&= -\lambda||\psi||^2 + \lambda\int_V \psi\mathcal{T}[\psi]\ d\mathbf{v}. \qquad (38)
\end{aligned}$$

This expression describes the action of the turning operator $\mathcal{L}$ on functions orthogonal to the constant function. The negative term represents uniform decay. The positive term represents redistribution due to the turning-actions. We can use the decomposition of $L^2(V)$ to introduce another constant, κ, which provides information on the dissipativity of the turning operator. In other words, this new constant κ describes to what extent perturbations are contracted. With the Cauchy-Schwarz inequality, we get

$$|\langle \mathcal{T}[\psi], \psi\rangle| \leq ||\mathcal{T}[\psi]|| \cdot ||\psi|| \leq ||\mathcal{T}||_{\langle 1\rangle^{\perp}}||\psi||^2_{L^2(V)},$$

where Cauchy-Schwarz is applied a second time on $\langle 1\rangle^{\perp}$. Due the the non-negativity constraints, we get

$$\langle \mathcal{T}[\psi], \psi\rangle \leq ||\mathcal{T}||_{\langle 1\rangle^{\perp}}||\psi||_{L^2(V)}.$$

Substituting this inequality in Equation (38) gives:

$$\int_V \psi(\mathbf{v})\mathcal{L}[\psi](\mathbf{v})\ d\mathbf{v} \leq \lambda(-1 + ||\mathcal{T}||_{\langle 1\rangle^\perp})||\psi||^2_{L^2(V)}.$$

We define

$$\kappa := 1 - ||\mathcal{T}||_{\langle 1\rangle^\perp},$$

and recall that $||\mathcal{T}||_{\langle 1\rangle^\perp} < 1$. Then, we get

$$\int \psi\mathcal{L}[\psi]\ d\mathbf{v} \leq -\kappa||\psi||^2_{L^2(V)}$$

for all $\psi \in \langle 1\rangle^\perp$.

We conclude that

$$\int_V \psi\mathcal{L}[\psi]\ d\mathbf{v}$$

is dissipative. This means that the repeated turning smooths out directional variations, ensuring that the system displays isotropic diffusion under the assumptions that we imposed on the turning kernel, as listed in (T1)-(T6). The rate at which this happens, is described by κ.

4.3 The diffusive limit of the Markovian Velocity-Jump Model

We now shift our focus to the macroscopic scale and briefly recall how the individual-level dynamics of the velocity-jump process determine the evolution of the macroscopic density. To this end, we assume N independent and identical particles. For each particle $i \in \{1, \dots, N\}$, we denote its position and velocity at time t by $(\mathbf{x}_i(t), \mathbf{v}_i(t)) \in \mathbb{R}^n \times V$. We assume that $(\mathbf{x}_i(t), \mathbf{v}_i(t))$ admits a density $q^{(i)}(\mathbf{x}_i, \mathbf{v}_i, t)$ with respect to $d\mathbf{x}\ d\mathbf{v}$. Then, we can find the spatial marginal of particle i by $\int_V q^{(i)}(\mathbf{x}, \mathbf{v}, t)\ d\mathbf{v}$. The expected number of particles at location $\mathbf{x}$ moving at time t, is then given by the following expression:

$$p^{(N)}(\mathbf{x}, t) = \sum_{i=1}^{N} \int_V q^{(i)}(\mathbf{x}, \mathbf{v}, t)\ d\mathbf{v}.$$

As discussed in Section 2.2, we can interpret the limiting function

$$\tilde{p}(\mathbf{x}, t) := \lim_{N\to\infty} \frac{1}{N} \sum_{i=1}^{N} \int_V q^{(i)}(\mathbf{x}, \mathbf{v}, t)\ d\mathbf{v}$$

as the probability density for the position of a particle at position $\mathbf{x}$ at time t. We scale this with some constant p_0, describing the total number of particles in a system, and we obtain an expression for the density $\overline{p}(\mathbf{x}, t)$:

$$\overline{p}(\mathbf{x}, t) := p_0\tilde{p}(\mathbf{x}, t). \tag{39}$$

In the upcoming analysis, we will study this density in scaled coordinates and derive a macroscopic limit equation for this density over time in the Markovian Velocity-Jump Model.

4.3.1 Time and space scalings

In order to relate stochastic small-scale processes to large-scale observations, we will introduce space and time scales. In this scaling, we assume that many velocity jumps occur in one time unit, but the net displacement in one spatial unit is small. If observed over a large amount of reorientations, the movement should appear directed. As we have concluded in Section 4.2, for even larger amounts of time, the movement should resemble an isotropic diffusive process. We thus choose an appropriate $\varepsilon > 0$ such that the timescale of the turning process is $\tau_{\text{turn}} \sim \mathcal{O}(1)$, the timescale of the drift is $\tau_{\text{drift}} \sim \mathcal{O}\left(\frac{1}{\varepsilon}\right)$ and the timescale of diffusion is $\tau_{\text{diff}} \sim \mathcal{O}\left(\frac{1}{\varepsilon^2}\right)$. We may find a range of the appropriate value of ε through observations of the studied system.

We incorporate the scales by nondimensionalising the position, velocity and time variables. We first identify a length scale $L \in \mathbb{R}_{>0}$ that is characteristic of the macroscopic evolution, a speed $s \in \mathbb{R}_{>0}$ and a time scale $\sigma \in \mathbb{R}_{>0}$, which is the macroscopic time scale of observation. We introduce the dimensionless variables

$$\tilde{\mathbf{v}} := \frac{\mathbf{v}}{s}, \qquad \tilde{\mathbf{x}} := \frac{\mathbf{x}}{L}, \qquad \tilde{t} := \frac{t}{\sigma}.$$

Then, we introduce a notation for the density function in the scaled system:

$$p(\tilde{\mathbf{v}}, \tilde{\mathbf{x}}, \tilde{t}) := \overline{p}(\mathbf{x}, \mathbf{v}, t),$$

where $\overline{p}$ is defined in Equation (39). A dimensionless formulation of Equation (31) is then given by

$$\frac{1}{\sigma}\frac{\partial p}{\partial \tilde{t}} + \left(\frac{s}{L}\right)\tilde{\mathbf{v}} \cdot \nabla_{\tilde{\mathbf{x}}} p = -\lambda p + \lambda \int_V T(\mathbf{v}, \mathbf{v}') p(\tilde{\mathbf{x}}, \tilde{\mathbf{v}}', \tilde{t})\, d\mathbf{v}'. \tag{40}$$

In order to match the time scales τ_{turn}, τ_{drift} and τ_{diff}, we take $\sigma\lambda \sim \mathcal{O}(\varepsilon^{-2})$ and $\frac{\sigma}{L} \sim \mathcal{O}(s/\varepsilon)$. Then, Equation (40) can be rewritten as follows:

$$\varepsilon^2 \frac{\partial p}{\partial \tilde{t}} + \varepsilon \tilde{\mathbf{v}} \cdot \nabla_{\tilde{\mathbf{x}}} p = \mathcal{L} p, \tag{41}$$

with

$$\tilde{t} = \varepsilon^2 t, \qquad \tilde{\mathbf{x}} = \frac{\varepsilon \mathbf{x}}{s}.$$

This is the system that we will be using when obtaining a macroscopic limit equation for the velocity-jump process;

4.3.2 The formal diffusion limit

We will now show that the scaling introduced in Section 4.3.1 yields a diffusion equation when performing a regular perturbation expansion of the density p with respect to the scaling parameter ε. We write the solution $p(\tilde{\mathbf{x}}, \tilde{\mathbf{v}}, \tilde{t})$ as the sum of functions with decreasing contributions, until the contribution becomes negligible:

$$p(\tilde{\mathbf{x}}, \tilde{\mathbf{v}}, \tilde{t}) = \sum_{i=0}^{k} \varepsilon^i p_i(\tilde{\mathbf{x}}, \tilde{\mathbf{v}}, \tilde{t}) + \varepsilon^{k+1} p_{k+1}(\tilde{\mathbf{x}}, \tilde{\mathbf{v}}, \tilde{t}). \tag{42}$$

We can use the linearity of the turning operator $\mathcal{L}$ and substitute this expansion in Equation (41) to obtain a system of equalities:

$$\begin{aligned}
\varepsilon^0 : &\quad \mathcal{L}[p_0] = 0, \\
\varepsilon^1 : &\quad \mathcal{L}[p_1] = \tilde{\mathbf{v}} \cdot \nabla p_0, \\
\varepsilon^2 : &\quad \mathcal{L}[p_2] = \frac{\partial p_0}{\partial \tilde{t}} + \tilde{\mathbf{v}} \cdot \nabla p_1, \\
&\qquad \vdots \\
\varepsilon^i : &\quad \mathcal{L}[p_i] = \frac{\partial p_{i-2}}{\partial \tilde{t}} + \tilde{\mathbf{v}} \cdot \nabla p_{i-1}
\end{aligned} \tag{43}$$

From the ε^0 term in the system in Equation (43), we find that $p_0 \in \ker(\mathcal{L})$ must hold. We know from Equation (36) that $\ker(\mathcal{L}) = \langle 1 \rangle$, as the eigenvalue corresponding to constant functions is zero. In particular, we can conclude that p_0 describes isotropic movement: there is no directional bias causing a flux, so p_0 does not depend on velocity $\tilde{\mathbf{v}}$.

From the ε^1 term in Equation (43), we find a solvability equation, which is obtained from the Fredholm Alternative, given in Theorem 2.1. As $\ker(\mathcal{L}) = \ker(\mathcal{L}^*) = \langle 1 \rangle \neq \{0\}$, we must have

$$\langle \tilde{\mathbf{v}} \cdot \nabla p_0, \phi \rangle = 0$$

for all $\phi \in \ker(\mathcal{L}^*) = \langle 1 \rangle$ in order for the equation for the ε^1 term in Equation (43) to be solvable.

Taking $\phi \equiv 1$ in this conditions gives the necessary solvability condition:

$$\int_V \tilde{\mathbf{v}} \cdot \nabla p_0 \; d\tilde{\mathbf{v}} = 0. \tag{44}$$

When applying the previous result that p_0 is independent of $\tilde{\mathbf{v}}$ and the assumption that V is symmetric, we can see that indeed, we have

$$\begin{aligned} \int_V \tilde{\mathbf{v}} \cdot \nabla p_0 \; d\tilde{\mathbf{v}} &= \nabla p_0 \int_V \tilde{\mathbf{v}} \cdot \; d\tilde{\mathbf{v}} \\ &= \nabla p_0 \cdot 0 \\ &= 0. \end{aligned}$$

So, under these assumptions, the solvability condition in Equation (44) holds and if we let $\mathcal{F}$ be the pseudo-inverse of $\mathcal{L}$ on $\langle 1 \rangle^\perp$. That is, we define

$$\mathcal{F} = \left(\mathcal{L}_{\langle 1 \rangle^\perp} \right)^{-1},$$

and with this, we find

$$p_1 = \mathcal{F}(\tilde{\mathbf{v}} \cdot \nabla p_0). \tag{45}$$

Similarly, for the $\mathcal{O}(\varepsilon^2)$ equality in Equation (43), we get

$$\int_V \frac{\partial p_0}{\partial \tilde{t}} + \tilde{\mathbf{v}} \cdot \nabla p_1 \; d\tilde{\mathbf{v}} = 0.$$

When substituting $p_1 = \mathcal{F}(\tilde{\mathbf{v}} \cdot \nabla p_0)$ in this equality, we get

$$\int_V \frac{\partial p_0}{\partial \tilde{t}} + \tilde{\mathbf{v}} \cdot \nabla(\mathcal{F}(\tilde{\mathbf{v}} \cdot \nabla p_0)) \; d\tilde{\mathbf{v}} = 0. \tag{46}$$

We recall that p_0 is independent of $\tilde{\mathbf{v}}$ and that $\nabla p_0 = \nabla_{\tilde{\mathbf{x}}} p_0$ is constant in the velocity-field. Therefore, we find

$$\int_V \frac{\partial p_0}{\partial \tilde{t}} + \tilde{\mathbf{v}} \cdot \nabla(\mathcal{F}(\tilde{\mathbf{v}}) \cdot \nabla p_0) \; d\tilde{\mathbf{v}} = 0,$$

which we can write as

$$|V| \frac{\partial p_0}{\partial \tilde{t}} + \nabla \int_V \tilde{\mathbf{v}} \mathcal{F} \tilde{\mathbf{v}} \; d\tilde{\mathbf{v}} \cdot \nabla p_0 = 0.$$

We let

$$D := -\frac{1}{|V|} \int_V \tilde{\mathbf{v}} \mathcal{F} \tilde{\mathbf{v}}. \tag{47}$$

Then, we find that we have obtained the equation

$$\frac{\partial p_0}{\partial \tilde{t}} = \nabla(D \nabla p_0), \tag{48}$$

which is the diffusion equation as mentioned in the framework of the Master Transport in Section 2.3. Therefore, we conclude that the Markovian Velocity-Jump model, with restrictions (T1)-(T6) has the isotropic diffusion equation as its the macroscopic limit equation. Equation (47) reveals a link from the microscopic reorientation rate to the observable diffusion rate.

5 Limitations of the classical Velocity-Jump Model

The classical velocity-jump model considered in Section 4 is a single-population, Markovian transport model with a symmetric velocity space. These assumptions make the model analytically tractable and lead to a diffusion limit, but they also restrict the class of motions it can represent. The classical model has been thoroughly studied and extended, widening the range of applications. In this section, we discuss several limitations imposed by the assumptions used in the construction and analysis of the Markovian Velocity-Jump Model and survey known extensions and remaining gaps.

5.1 Markovian turning, resting states and memory effects

The classical Velocity-Jump model is formulated with the assumption that reorientations occur according to a Poisson process with a constant rate. As a result, the successive run length and orientation are assumed to be independently sampled once the current velocity is known. While the Markovian framework is mathematically convenient, it excludes several movement patterns observed in applications. In particular, it cannot represent non-exponential run-time distributions, explicit resting phases, or settings in which moving entities adapt their movement based on earlier events. The assumption of Markovian dynamics is relaxed in the general formulation given in Section 3. In this model formulation, general distributions for the running time can be used, allowing for memory-dependent dynamics. This general form of the model has, to our knowledge, not yet been reported in other studies.

The classical model also excludes motility patterns of the form *move–rest–reorient–move*, unless resting is introduced as an additional state variable or an explicit waiting-time distribution is built into the model. Taylor-King et al. introduced a generalised velocity-jump framework with resting states, allowing both running and waiting times to follow general distributions, together with arbitrary angular and speed distributions [47]. Their analysis showed that the long-time diffusion coefficient depends on the moments of the running-time distribution and on the mean waiting time.

Another form of memory involves internal state dynamics. In chemotaxis models, for instance, each cell has an intracellular signalling state that adapts to stimuli. Erban and Othmer coupled the velocity-jump process to an intracellular signalling variable whose spread is described by an ordinary differential equation, so that the turning rate depends on both internal and external variables [13]. In that setting, the cell effectively retains a memory of past stimuli through the evolution of its internal state, and this memory influences the macroscopic chemotactic drift. They derive macroscopic chemotaxis PDEs, showing that cell memory of past signals affects motion. In general, any internal hidden variable introduces correlations in turning that break with the Markovian assumption.

Although these extensions capture important forms of memory, resting, and dependence on the particles' internal state, a fully general Velocity-Jump Model that simultaneously incorporates such effects is, as far as we know, still lacking.

5.2 Symmetric velocity space and the lack of directional confinement

The classical Velocity-Jump Model assumes a symmetric velocity space V. That is to say, for any $\mathbf{v} \in V$, we must have $-\mathbf{v} \in V$. Additionally, it assumes that any velocity can be reached from any other under the turning kernel. This geometrical symmetry excludes truly one-sided or unidirectional motion. This limitation is exemplified in the simplest one-dimensional case, where a particle cannot be constrained to move only to the right. If we take some velocity $\mathbf{v}^+ \in V\backslash\{0\}$, then $\{\mathbf{v}^-, 0, \mathbf{v}^+\} \subseteq V$ must hold for $\mathbf{v}^- := -\mathbf{v}^+$. As V is symmetric around zero, Assumption 2 given in Section 4.1, stating that $\int_V T(\mathbf{v}, \mathbf{v}'), d\mathbf{v}' = 1$ must hold for all $\mathbf{v}$, forces the existence of leftward velocities even if $\mathbf{v}^+$ is the only "forward" speed. That is, there must be a subset $V' \subseteq V$ such that $\int_{\mathbf{v}' \in V'} T(\mathbf{v}^-, \mathbf{v}')\, d\mathbf{v}' = 1$. As this holds for every $\mathbf{v} \in V$, the whole left side of the velocity space is accessible, meaning that there is no possibility of particles strictly moving to the right. The same principle holds in higher dimensions.

Despite the model prohibiting exclusion of a velocity subspace, it is possible to make it very unlikely for certain velocities to be sampled. This is, for example done, in the previously described models

including chemotaxis [14, 34]. In such models, the turning kernel is dependent on a chemical gradient so that particles prefer certain directions. In the diffusion limit this induces an advective drift or anisotropic diffusion tensor. The state space for the particle densities does not consist only of positions and velocities, but also of concentration of the stimulus. The bias on the velocity space is induced as the particles prefer jumps towards higher concentrations of a stimulus. However, even such biased models still technically permit motion in all directions, just with low probabilities. A strict directional confinement would require a different formalism. Extending the classical Velocity-Jump Model to allow strictly anisotropic or one-sided velocity supports remains unresolved. Such an extension would alter the transport operator and the resulting diffusion limit.

5.3 Exclusion of memory-based behavior and agent-agent interactions

The classical Velocity-Jump Model treats each particle as an independent entity, whose turning events are memoryless and unaffected by other particles. This cannot represent the steric blocking and density-dependent motility that arise in crowded systems. This is a significant limitation in biological settings where motion is strongly influenced by local occupancy, such as dense cell monolayers, wound-healing fronts, tumor invasion, and other experiments in which aborted moves or local blocking are observed [17].

Gavagnin and Yates adressed this limitation by constructing an on-lattice, volume-excluding velocity-jump model based on individual-level interactions between particles [17]. They succeeded in deriving a continuum description for this model. By making volume-exclusion explicit at the microscopic level, the derivation produces nonlinear, density-dependent diffusion terms and advective corrections in the macroscopic PDEs. This showed that phenomena driven by local crowding are no longer washed out by the linear diffusion that follows from noninteracting assumptions.

5.4 Homogeneous population structure

The construction of the particle density function relies on the assumption that all particles are independent and have identically distributed running times. This excludes heterogeneity in run-time distributions, turning behaviour, and motility parameters. It also excludes direct coupling between distinct subpopulations. As a result, the classical model cannot represent leader-follower effects, cargo-carrier transport, competing cell types, or other mechanisms in which the movement of one subpopulation depends explicitly on the state of another.

Some of these restrictions have been relaxed in the literature, but only in specific settings. Painter and Sherratt presented a one-dimensional velocity-jump model including interacting populations, in which the density influences the turning rate [35]. This study, however, does not develop a full velocity-jump model, but shows that, under a parabolic scaling, the continuum equation reduces to the same diffusion-type equations obtained from their space-jump approach. Although closely related to the velocity-jump framework, this derivation is based on movement modelled as periods of smooth motion punctuated by reorientations, rather than a sequence of discrete velocity jumps in space. More recently, Erban and Haskovec formulated coupled velocity-jump processes for locusts and derived a limit equation for the collective behaviour, in which alignment between individuals is taken into account [12].

These modelling studies shows that population coupling and interaction have be incorporated into the velocity-jump framework. However, the proposed extensions are specific for certain biological mechanisms and do not provide a general framework for interacting populations. A fully general velocity-jump theory for multiple interacting populations has, to our knowledge, not been published.

5.5 Interacting populations

The previously described models are limited to a single-population velocity-jump process. To our knowledge, fully coupled multi-dimensional models formulations for interacting species moving according to the velocity-jump process have not been rigorously derived. In the next section, we will

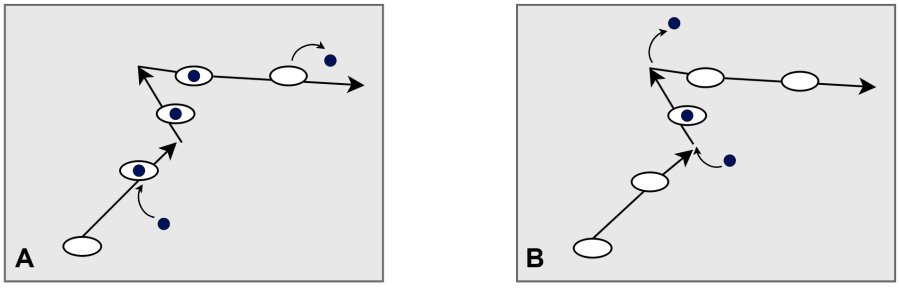


Figure 3: A schematic overview of two hitchhiking mechanisms between one transporter (white oval) and one hitchhiker (blue circle), as described in this section. The transporter moves according to the large straight arrows. The hitchhiking mechansims for the cargo are described by the small curved arrows. **A**: particles can jump on carriers either during reorientations or during runs. Their behavior does not affect the movement of the transporter. **B**: particles can only jump on or off carriers when a carrier reorientates. During the runs of the transporter, the hitchhiker cannot attach to the transporters.

present a general model, based on the Velocity-Jump Model, that represents two interacting populations. In particular, one population functions as transporter and the other as cargo. In other words, this is a two-species carrier-mediated transport model. We describe three general forms of attachment mechanisms:

[A] Particles jump on carriers without influencing their movement.

[B] Particles can only jump on carriers when the carriers are performing a velocity-jump.

[C] Particles jump on carriers, causing a velocity-jump. This resembles a startle reaction, meaning the turning kernel is different from the one that describes the regular velocity-jump process.

A schematic overview of the distinction between situation [A] and [B] is given in Figure 3. We will build a model that describes the populations of mobile and immobile hitchhikers in type [A]. This type of behavior is observed in dynamics such as intermicrobial hitchhiking between two motile microbes. For example, the immotile spores in Figure 1(c) in Section 1 must attach to a motile bacterium to enter the mobile state. This attachment can place at any point in the velocity-jump process of the carrier. Transport of type [B] applies, for example, in situations where the hitchhikers' positions are restricted to unconnected subspaces of Ω, and the hitchhikers are dependent on the transporters to move between the separate areas. An example of such dependency is given in Figure 1(b), where the mites cannot move to different flowers without jumping on the humming birds, and the birds' visit of a flower corresponds to a velocity-jump. Situation [C] will not be considered, due to the difficulty in constructing a turning kernel that describes the startled reaction of the carrier caused by the attachment of a hitchhiker. We also suspect that this type of interaction is considered to be an exception in nature rather than an evolved mechanism.

6 Derivation of the General Hitchhikers' model

We will now use the theory discussed in the previous sections to derive a general formulation of a model describing carrier-mediated transport. We will refer to this process as hitchhiking. In the model, a population of transporters, also called carriers, is moving according to the velocity-jump process. A population of hitchhikers is either attached to a transporter and following its same velocity-jump process, or detached and immobile. We assume that the positions of the carriers and hitchhikers are restricted to the same area $\Omega \subseteq \mathbb{R}^n$, and they can both occupy the whole area. In contrast to the derivation of the generalized Velocity-jump model in Section 3, we now assume that the carrier reorientate according to a Poisson process with rate $\lambda > 0$. The resulting exponential run-time distribution is memoryless, so the probability of a carrier reorienting is independent of how long it has already been travelling. As a consequence, no age variable associated with carrier runs is required, which makes the derivation more notationally manageable and eliminates the need for age-structured initial conditions with respect to carrier run times. Relaxing this assumption would require the same techniques that will be applied in the upcoming analysis, as we do not make any assumptions yet on the distributions of the attachment and detachment rates.

For the density of carriers at position $\mathbf{x} \in \Omega$ moving with velocity $\mathbf{v} \in V$ at time t, we use a general transport equation, based on the formulation discussed in Section 2.3. Additionally, we have to explicitly model the velocity-jump process as described in Section 4. We get

$$\frac{\partial}{\partial t}u(\mathbf{x}, \mathbf{v}, t) = -\mathrm{div}_{\mathbf{x}}(\mathbf{v} \cdot u(\mathbf{x}, \mathbf{v}, t)) + \mathcal{L}[u](\mathbf{x}, \mathbf{v}, t),$$

where, for some $\lambda \in \mathbb{R}_{>0}$ and a turning kernel T as described in Section 4, we have

$$\mathcal{L}[u](\mathbf{x}, \mathbf{v}, t) = -\lambda u(\mathbf{x}, \mathbf{v}, t) + \lambda \int_V T(\mathbf{v}, \mathbf{v}')u(\mathbf{x}, \mathbf{v}', t)\ d\mathbf{v}'.$$

Attached hitchhikers follow the carriers' movement, described by the exponential distribution with parameter λ and the turning kernel T.

6.1 Constructing density functions

We define $s_m(\mathbf{x}, \mathbf{v}, t)$ as the density of hitchhikers at position $\mathbf{x}$, running with velocity $\mathbf{v}$ at time t. Similarly, we define $s_i(\mathbf{x}, t)$ as the density of immobile particles at position $\mathbf{x}$ at time t. Note that we eliminate the velocity-dependence in the immobile state. This allows for a cleaner model formulation and analysis, though the dependence could be included, should that be necessary for the model application. We start formulating a model for s_m and s_i by constructing renewal equations for these variables. To this end, we introduce some auxiliary variables:

- $\eta(\mathbf{x}, \mathbf{v}, t)$ is the density of particles at position $\mathbf{x}$ at time t, moving with velocity $\mathbf{v}$ after having just left a resting state.
- $\zeta(\mathbf{x}, \mathbf{v}, t)$ is the density of particles at position $\mathbf{x}$ at time t, moving with velocity $\mathbf{v}$, having just started a new run after a velocity-jump on a carrier.
- $\kappa(\mathbf{x}, t)$ is the density of particles at position $\mathbf{x}$ at time t and having just entered the resting state. We do not keep track of the previous velocity in this model.

When formulating an expression for $s_m(\mathbf{x}, \mathbf{v}, t)$, we have to consider the two types of paths from which hitchhiking particles can enter a run with velocity $\mathbf{v}$:

- Particles start a run from some resting state. That is to say, they enter s_m from $\eta(\mathbf{x}-(t-s)\mathbf{v}, \mathbf{v}, s)$ for some $s \in [0, t]$. They then travel with velocity $\mathbf{v}$ to position $\mathbf{x}$. The duration of this part of the run is $(t - s)$.
- Particles switch from moving with velocity $\mathbf{v}' \in V$ to moving with velocity $\mathbf{v}$, i.e., they enter from $\zeta(\mathbf{x} - (t - s)\mathbf{v}', \mathbf{v}', s)$ for some $s \in [0, t]$. During the part of the run with velocity $\mathbf{v}$ of duration s, the particle reaches position $\mathbf{x}$.

Furthermore, in order to contribute to $s_m(\mathbf{x}, \mathbf{v}, t)$, the moving particles in a running state must not start a rest before time t. The mobile period is defined as the total time of attachment. This could involve partial, a single or multiple runs of the velocity-jump process before (re)-entering the immobile state. The total time spent continuously attached to the carrier, before entering rest, is denoted as the random variable τ with probability density function f_τ. We introduce

$$F_\tau(t) := \text{Prob}(\tau \geq t) = \int_t^\infty f_\tau(s)\ ds$$

as the probability that a mobile period exceeds time t. We assume that this random variable does not depend on factors such as the location, velocity or the movement history of the particle in question.

During a running period, the hitchhiker can reorientate, following the velocity-jump process of its carrier. The time until the next velocity jump of the carrier, either with or without an attached hitchhiker, is described by the random variable ρ. As we assumed that the carriers follow a Markovian velocity-jump process, we have

$$f_\rho(t) = \lambda \exp(-\lambda t), \qquad F_\rho(t) = \exp(-\lambda t).$$

Note that the mobile period, whose duration is described by the random variable τ, is defined for the hitchhikers, and describes the duration of the co-transport. The running period, with corresponding random variable ρ, is used to describe the velocity-jump process for the carriers. We assume that these processes do not intervene with each other. That is to say, the hitchhikers' attachment does not affect the velocity-jump process of the carriers, and the duration of the carriers' movement does not affect the hitchhikers' ability to attach. In order for particles to stay mobile with velocity $\mathbf{v}$, it must not have entered a resting state or changed velocities before reaching position $\mathbf{x}$. We denote

$$G(t) = \text{Prob}(\tau > t, \rho > t) =: F_\tau(t)F_\rho(t) = \exp(-\lambda t)F_\tau(t) \tag{49}$$

as the probability that neither event has occured at time t. The multiplication of probabilities follows from the assumption that τ and ρ are independent random variables, and the last equality is a result of the assumption that carriers follow a Markovian velocity-jump process.

We define $s_i(\mathbf{x}, t)$ as the density of immobile particles at position $\mathbf{x}$ at time t. We use the random variable ω to describe the duration of the immobile period between two mobile periods. With the corresponding probability density function f_ω, we define

$$F_\omega(t) := \text{Prob}(\omega \geq t) = \int_t^\infty f_\omega(s)\ ds$$

as the probability that a resting period exceeds time t. As in Section 3, we include a general forcing term as a result of the initial conditions. We denotes these as $b_t^m(\mathbf{x}, \mathbf{v})$ and $b_t^i(\mathbf{x}, \mathbf{v})$ for the mobile and immobile hitchhikers, respectively. We will introduce a formulation for these forcings in Section 6.2.

We can then define the following system of renewal equations:

$$\begin{cases} s_m(\mathbf{x}, \mathbf{v}, t) = b_t^m(\mathbf{x}, \mathbf{v}) + \int_0^t G(t-s)\left[\eta(\mathbf{x} - (t-s)\mathbf{v}, \mathbf{v}, s) + \zeta(\mathbf{x} - (t-s)\mathbf{v}, \mathbf{v}, s)\right]\ ds \\ s_i(\mathbf{x}, t) = b_t^i(\mathbf{x}) + \int_0^t F_\omega(t-s)\kappa(\mathbf{x}, s)\ ds. \end{cases} \tag{50}$$

In order to study this system of two equations, we must find expressions for $\eta(\mathbf{x}, \mathbf{v}, t)$, $\zeta(\mathbf{x}, \mathbf{v}, t)$ and $\kappa(\mathbf{x}, t)$, which contribute to the actual renewal equations.

We find an expression for $\eta(\mathbf{x}, \mathbf{v}, t)$ by collecting all particles that were in a resting state $\kappa(\mathbf{x}, s)$ which ended at time $s < t$ and that then took on velocity $\mathbf{v}$ at time $t - s$. The number of particles entering the mobile state at position $\mathbf{x}$ at time t is given by

$$\int_0^t f_\omega(t-s)\kappa(\mathbf{x}, s)\ ds.$$

In order to actually take on velocity $\mathbf{v}$, the particle has to attach to a transporter that moves with velocity $\mathbf{v}$. We thus define a carrier-dependent attachment kernel, denoted as $P[u_t(\mathbf{x},\cdot)]$. This describes the probability of a particle going in direction $\mathbf{v}$, given that it initiates an attachment period. We must have

$$\int_V P[u_t(\mathbf{x},\cdot)](\mathbf{v})\ d\mathbf{v} = 1$$

for all $\mathbf{x} \in \Omega$ and all times t. An example of such a kernel is given by

$$P[u_t(\mathbf{x},\cdot)](\mathbf{v}) := \frac{u(\mathbf{x},\mathbf{v},t)}{\int_V u(\mathbf{x},\mathbf{v}',t)\ d\mathbf{v}'.},$$

which describes the proportion of transporters that move with velocity $\mathbf{v}$ at position $\mathbf{x}$ and at time t.

We denote the contribution of the initially mobile hitchhikers as $\beta_t^\eta(\mathbf{x},\mathbf{v})$, and we will derive a more specific expression for this in Section 6.2. We thus get the following expression for η:

$$\eta(\mathbf{x},\mathbf{v},t) = \beta_t^\eta(\mathbf{x},\mathbf{v}) + P[u_t(\mathbf{x},\cdot)](\mathbf{v}) \int_0^t f_\omega(t-s)\kappa(\mathbf{x},s)\ ds.$$

We find an expression for $\zeta(\mathbf{x},\mathbf{v},t)$ by collecting all particles that were in a mobile state and have changed their velocity from some other velocity $\mathbf{v}'$ to velocity $\mathbf{v}$ at a time before t. For the velocity-jump to take place, the duration of attachment must be sufficiently long, and the previous run must have ended at some point before time t. We thus define a velocity-jump kernel

$$K_\zeta(t) := F_\tau(t) f_\rho(t) = \lambda \exp(-\lambda t) F_\tau(t) \tag{51}$$

as the probability density function of a velocity-jump event. The last equality only holds under the assumption of a Markovian velocity-jump process for the carriers. Then, $K_\zeta(t)\left(\eta(\mathbf{x},\mathbf{v},t) + \zeta(\mathbf{x},\mathbf{v},t)\right)$ describes the particle mass at time t that is undergoing a velocity-jump from velocity $\mathbf{v}$ to a different velocity at position $\mathbf{x}$. We denote the contribution of the initial distributions to ζ as $\beta_t^\zeta(\mathbf{x},\mathbf{v})$. We get the following expression for ζ:

$$\zeta(\mathbf{x},\mathbf{v},t) = \beta_t^\zeta(\mathbf{x},\mathbf{v}) + \int_V \int_0^t T(\mathbf{v},\mathbf{v}') K_\zeta(t-s) \left[\eta(\mathbf{x}-(t-s)\mathbf{v}',\mathbf{v}',s) + \zeta(\mathbf{x}-(t-s)\mathbf{v}',\mathbf{v}',s)\right]\ ds\ d\mathbf{v}'.$$

Particles arrive in $\kappa(\mathbf{x},t)$ after a running period has ended. This can happen after either the first run since a rest or after later runs. The running period in total has to end at exactly time $t-s$ for some $s \in [0,t]$, and the particles have to move exactly to point $\mathbf{x}$ with velocity $\mathbf{v}$. To describe the probability of particles entering a resting state, we define the resting kernel:

$$K_\kappa(t) := F_\rho(t) f_\tau(t) = \exp(-\lambda t) f_\tau(t). \tag{52}$$

For the resting state, we do not keep track of the previous velocities. That is to say, the probability of continuing movement with velocity $\mathbf{v}$ after reorienting in a resting state is the same for all velocities $\mathbf{v}'$. There is no forcing contribution in the equation for $\kappa(\mathbf{x},t)$ from the initially immobile population. The quantity $\kappa(\mathbf{x},t)$ describes the flux of particles entering the immobile state at time t, and particles that are already immobile at $t=0$ do not enter this state at later times. Their contribution is accounted for by the term b_i^t in the equation for $s_i(\mathbf{x},t)$. We thus obtain the expression

$$\kappa(\mathbf{x},t) = \int_V \int_0^t K_\kappa(t-s) \left[\eta(\mathbf{x}-(t-s)\mathbf{v},\mathbf{v},s) + \zeta(\mathbf{x}-(t-s)\mathbf{v},\mathbf{v},s)\right]\ ds\ d\mathbf{v}.$$

Combining all the equations gives the following general system for the auxiliary variables:

$$\begin{cases} \eta(\mathbf{x},\mathbf{v},t) = \beta_t^\eta(\mathbf{x},\mathbf{v}) + P[u_t(\mathbf{x},\cdot)](\mathbf{v}) \int_0^t f_\omega(t-s)\kappa(\mathbf{x},s)\ ds, \\ \zeta(\mathbf{x},\mathbf{v},t) = \beta_t^\zeta(\mathbf{x},\mathbf{v}) + \int_V \int_0^t T(\mathbf{v},\mathbf{v}') K_\zeta(t-s) \left[\eta(\mathbf{x}-(t-s)\mathbf{v}',\mathbf{v}',s) + \zeta(\mathbf{x}-(t-s)\mathbf{v}',\mathbf{v}',s)\right]\ ds\ d\mathbf{v}', \\ \kappa(\mathbf{x}) = \int_V \int_0^t K_\kappa(t-s) \left[\eta(\mathbf{x}-(t-s)\mathbf{v},\mathbf{v},s) + \zeta(\mathbf{x}-(t-s)\mathbf{v},\mathbf{v},s)\right]\ ds\ d\mathbf{v}. \end{cases} \tag{53}$$

6.2 Contribution of the initial distributions

We will now find expressions for the forcings caused by the initial distribution. In Equations (50) and (53), these forcings are denoted as general terms, for which we will now find explicit expressions.

As in Section 3.2, we prescribe an initial condition in which each particle carries an age variable describing how long its current run has already lasted at $t = 0$, when the observation begins. Because the Hitchhikers' model also includes a resting state, we must likewise specify an initial age distribution for the immobile hitchhikers. For the upcoming analysis, it is convenient to define the initial densities as measures. We thus have

$$s_m^0(\mathbf{x}, \mathbf{v}, da), \qquad s_i^0(\mathbf{x}, d\alpha),$$

which represent, respectively, the initial age measures over the mobile and immobile states. That is, for fixed $(\mathbf{x}, \mathbf{v})$, the measure $s_m^0(\mathbf{x}, \mathbf{v}, \cdot)$ assigns mass to measurable subsets of the running-period domain $[0, \infty)$ and encodes the initial distribution of run ages. In other words, this captures for how long the particles have been mobile at $t = 0$. Similarly, for fixed $\mathbf{x}$, the measure $s_i^0(\mathbf{x}, \cdot)$ describes the initial distribution of resting ages.

For the forcing $b_t^m(\mathbf{x}, \mathbf{v}, t)$, we note that all participating particles have to be initiated in a resting state before starting an attachment period at exactly time t. The probability of this event occurring is described by $F_\rho(t)$. Then, these particles have to move in direction $\mathbf{v}$ at location $\mathbf{x}$. For this probability, the same logic applies as for the forcing b_t in Section 3.2. Assuming independence of these events, we find

$$\begin{aligned} b_t^m(\mathbf{x}, \mathbf{v}) &= F_\rho(t) \int_0^\infty \frac{F_\tau(t+a)}{F_\tau(a)} s_m^0(\mathbf{x} - t\mathbf{v}, \mathbf{v}, da) \\ &= \exp(-\lambda t) \int_0^\infty \frac{F_\tau(t+a)}{F_\tau(a)} s_m^0(\mathbf{x} - t\mathbf{v}, \mathbf{v}, da), \end{aligned}$$

where the second equality is a result from the assumed Markovian dynamics for the carriers.

We can find the forcing $b_t^i(\mathbf{x}, t)$ by collecting all the particles that were initiated in the resting state at position $\mathbf{x}$ and have not left this state yet at time t. The probability of this is event is given by

$$\mathrm{Prob}(\omega > t + \alpha \,|\, \omega > \alpha) = \frac{\mathrm{Prob}(\omega > t + \alpha \wedge \omega > \alpha)}{\mathrm{Prob}(\omega > \alpha)} = \frac{F_\omega(t+\alpha)}{F_\omega(\alpha)}.$$

Then, we find

$$b_t^i(\mathbf{x}) = \int_0^\infty \frac{F_\omega(t+\alpha)}{F_\omega(\alpha)} s_i^0(\mathbf{x}, d\alpha).$$

We will now derive expressions for the initial forcings for the auxiliary variables.

The forcing $\beta_t^\eta(\mathbf{x}, \mathbf{v})$ is caused by the particles that were initialized in a resting state and have left this state at time t to start a run with velocity $\mathbf{v}$. The total run duration is then given by $t + \alpha$ with α denoting the resting age at time $t = 0$. We thus get

$$\beta_t^\eta(\mathbf{x}, \mathbf{v}) = P[u_t(\mathbf{x}, \cdot)](\mathbf{v}) \int_0^\infty \frac{f_\omega(t+\alpha)}{F_\omega(\alpha)} s_i^0(\mathbf{x}, d\alpha).$$

For the expression for $\beta_t^\zeta(\mathbf{x}, \mathbf{v})$, we can follow the logic used to construct β_t in Section 3.2. This forcing is caused by particles that have turned to velocity $\mathbf{v}$ from a different velocity $\mathbf{v}' \in V$. The jump must have taken place at time t and therefore, the initial run of age a with velocity $\mathbf{v}'$ must have ended at exactly time $t + a$. In order to have reached position $\mathbf{x}$, the particle must have been initiated at point $\mathbf{x} - t\mathbf{v}'$ with velocity $\mathbf{v}'$. As this could have taken place from any prior velocity $\mathbf{v}' \in V$, we find

$$\begin{aligned} \beta_t^\zeta(\mathbf{x}, \mathbf{v}) &= \int_V \int_0^\infty T(\mathbf{v}, \mathbf{v}') \frac{f_\rho(t+a)}{F_\rho(a)} s_m^0(\mathbf{x} - t\mathbf{v}', \mathbf{v}', da) \; d\mathbf{v}' \\ &= \lambda \exp(-\lambda t) \int_V T(\mathbf{v}, \mathbf{v}') s_m^0(\mathbf{x} - t\mathbf{v}', \mathbf{v}', [0, \infty)) \; d\mathbf{v}', \end{aligned}$$

where the second equality only holds in our specific case of carriers with a Poisson-distributed velocity-jump transport mechanism.

We now observe that in all the expressions, the auxiliary variables η and ζ only appear summed. We can thus introduce the density

$$\xi(\mathbf{x}, \mathbf{v}, t) := \eta(\mathbf{x}, \mathbf{v}, t) + \zeta(\mathbf{x}, \mathbf{v}, t),$$

and under the assumption of mass conservation, we get

$$\beta_t^{\xi}(\mathbf{x}, \mathbf{v}) = \beta_t^{\eta}(\mathbf{x}, \mathbf{v}) + \beta_t^{\zeta}(\mathbf{x}, \mathbf{v}).$$

The full system is then given by

$$\begin{cases} s_m(\mathbf{x}, \mathbf{v}, t) = \exp(-\lambda t) \displaystyle\int_0^\infty \frac{F_\tau(t+a)}{F_\tau(a)} s_m^0(\mathbf{x} - t\mathbf{v}, \mathbf{v}, da) \\ \qquad\qquad + \displaystyle\int_0^t G(t-s)\xi(\mathbf{x} - (t-s)\mathbf{v}, \mathbf{v}, s)\ ds \\ s_i(\mathbf{x}, t) = \displaystyle\int_0^\infty \frac{F_\omega(t+\alpha)}{F_\omega(\alpha)} s_i^0(\mathbf{x}, d\alpha) \\ \qquad\qquad + \displaystyle\int_0^t F_\omega(t-s)\kappa(\mathbf{x}, s)\ ds. \end{cases} \tag{54}$$

with auxiliary variables

$$\begin{cases} \xi(\mathbf{x}, \mathbf{v}, t) = P[u_t(\mathbf{x}, \cdot)](\mathbf{v}) \displaystyle\int_0^\infty \frac{f_\omega(t+\alpha)}{F_\omega(\alpha)} s_i^0(\mathbf{x}, d\alpha) \\ \qquad\qquad + \lambda \exp(-\lambda t) \displaystyle\int_V T(\mathbf{v}, \mathbf{v}') s_m^0(\mathbf{x} - t\mathbf{v}', \mathbf{v}', [0, \infty))\ d\mathbf{v}' \\ \qquad\qquad + P[u_t(\mathbf{x}, \cdot)](\mathbf{v}) \displaystyle\int_0^t f_\omega(t-s)\kappa(\mathbf{x}, s)\ ds \\ \qquad\qquad + \displaystyle\int_V \int_0^t T(\mathbf{v}, \mathbf{v}') K_\zeta(t-s)\xi(\mathbf{x} - (t-s)\mathbf{v}', \mathbf{v}', s)\ ds\ d\mathbf{v}', \\ \kappa(\mathbf{x}, t) = \displaystyle\int_V \int_0^t K_\kappa(t-s)\xi(\mathbf{x} - (t-s)\mathbf{v}, \mathbf{v}, s)\ ds\ d\mathbf{v}. \end{cases} \tag{55}$$

6.3 Deriving the general Hitchhikers' model

We will now eliminate the auxiliary variables ξ and κ from the system given in Equation (54). This process takes several steps. We start by taking the derivative of the equation for $s_m(\mathbf{x}, \mathbf{v}, t)$ in Equation (54). We get

$$\begin{aligned} \frac{\partial}{\partial t} b_t^m(\mathbf{x}, \mathbf{v}) &= \frac{\partial}{\partial t} \left(\exp(-\lambda t) \int_0^\infty \frac{F_\tau(t+a)}{F_\tau(a)} s_m^0(\mathbf{x} - t\mathbf{v}, \mathbf{v}, da) \right) \\ &\quad - \lambda \exp(-\lambda t) \int_0^\infty \frac{F_\tau(t+a)}{F_\tau(a)} s_m^0(\mathbf{x} - t\mathbf{v}, \mathbf{v}, da) \\ &= -\exp(-\lambda t) \int_0^\infty \frac{f_\tau(t+a)}{F_\tau(a)} s_m^0(\mathbf{x} - t\mathbf{v}, \mathbf{v}, da) \\ &\quad - \exp(-\lambda t)(\mathbf{v} \cdot \nabla_{\mathbf{x}}) \int_0^\infty \frac{F_\tau(t+a)}{F_\tau(a)} s_m^0(\mathbf{x} - t\mathbf{v}, \mathbf{v}, da). \\ &= -\lambda b_t^m(\mathbf{x}, \mathbf{v}) - (\mathbf{v} \cdot \nabla_{\mathbf{x}}) b_t^m(\mathbf{x}, \mathbf{v}) - \exp(-\lambda t) \int_0^\infty \frac{f_\tau(t+a)}{F_\tau(a)} s_m^0(\mathbf{x} - t\mathbf{v}, \mathbf{v}, da). \end{aligned}$$

For the second derivative of the equation for $s_m(\mathbf{x}, \mathbf{v}, t)$, we use

$$\begin{aligned}\frac{d}{dt}G(t) &= \frac{d}{dt}F_\tau(t)F_\rho(t)\\ &= \frac{d}{dt}\left(\int_t^\infty f_\tau(s)\ ds \int_t^\infty f_\rho(s)\ ds\right)\\ &= -f_\tau(t)F_\rho(t) - F_\tau(t)f_\rho(t)\\ &= -K_\kappa(t) - K_\zeta(t).\end{aligned} \tag{56}$$

Using Theorem 3.1, we find

$$\begin{aligned}\frac{\partial}{\partial t}\left(\int_0^t G(t-s)\xi(\mathbf{x}-(t-s)\mathbf{v}, \mathbf{v}, s)\ ds\right) &= \int_0^t \frac{\partial}{\partial t}\left(G(t-s)\xi(\mathbf{x}-(t-s)\mathbf{v}, \mathbf{v}, s)\ ds\right) + \xi(\mathbf{x}, \mathbf{v}, t)\\ &= \xi(\mathbf{x}, \mathbf{v}, t) + \int_0^t \left(\frac{\partial}{\partial t}\big(G(t-s)\big)\right)\xi(\mathbf{x}-(t-s)\mathbf{v}, \mathbf{v}, s)\ ds\\ &\qquad + \int_0^t G(t-s)\frac{\partial}{\partial t}\left(\xi(\mathbf{x}-(t-s)\mathbf{v}, \mathbf{v}, s)\right)\ ds\Big)\\ &= \xi(\mathbf{x}, \mathbf{v}, t) - \int_0^t K_\kappa(t-s)\xi(\mathbf{x}-(t-s)\mathbf{v}, \mathbf{v}, s)ds\\ &\qquad - \int_0^t K_\zeta(t-s)\xi(\mathbf{x}-(t-s)\mathbf{v}, \mathbf{v}, s)ds\\ &\qquad + \int_0^t G(t-s)(-\mathbf{v}\cdot\nabla_{\mathbf{x}})\xi(\mathbf{x}-(t-s)\mathbf{v}, \mathbf{v}, s)\ ds,\end{aligned}$$

where Equation (56) is applied in the last equality. We then take the derivative of the equation for $s_i(\mathbf{x}, t)$ in Equation (54). With $\frac{\partial}{\partial t}F_\omega(t) = -f_\omega(t)$, we find

$$\begin{aligned}\frac{\partial}{\partial t}b_t^i(\mathbf{x}) &= \frac{\partial}{\partial t}\left(\int_0^\infty \frac{F_\omega(t+\alpha)}{F_\omega(\alpha)}s_i^0(\mathbf{x}, d\alpha)\right)\\ &= -\int_0^\infty \frac{f_\omega(t+\alpha)}{F_\omega(\alpha)}s_i^0(\mathbf{x}, d\alpha)\end{aligned}$$

and for the second term, we apply Theorem 3.1 and we get

$$\frac{\partial}{\partial t}\left(\int_0^t F_\omega(t-s)\kappa(\mathbf{x}, s)\ ds\right) = -\int_0^t f_\omega(t-s)\kappa(\mathbf{x}, s)\ ds + \kappa(\mathbf{x}, t),$$

and as we recognize the equation for $s_m(\mathbf{x}, \mathbf{v}, t)$ in the equation of $\frac{\partial}{\partial t}s_m(\mathbf{x}, \mathbf{v}, t)$, we can then write

$$\begin{cases}\frac{\partial}{\partial t}s_m(\mathbf{x}, \mathbf{v}, t) + (\mathbf{v}\cdot\nabla_{\mathbf{x}})s_m(\mathbf{x}, \mathbf{v}, t) = \xi(\mathbf{x}, \mathbf{v}, t)\\ \qquad\qquad - \lambda\exp(-\lambda t)\displaystyle\int_0^\infty \frac{F_\tau(t+a)}{F_\tau(a)}s_m^0(\mathbf{x}-t\mathbf{v}, \mathbf{v}, da)\\ \qquad\qquad - \exp(-\lambda t)\displaystyle\int_0^\infty \frac{f_\tau(t+a)}{F_\tau(a)}s_m^0(\mathbf{x}-t\mathbf{v}, \mathbf{v}, da)\\ \qquad\qquad - \displaystyle\int_0^t K_\kappa(t-s)\xi(\mathbf{x}-(t-s)\mathbf{v}, \mathbf{v}, s)ds\\ \qquad\qquad - \displaystyle\int_0^t K_\zeta(t-s)\xi(\mathbf{x}-(t-s)\mathbf{v}, \mathbf{v}, s)ds\\ \\ \frac{\partial}{\partial t}s_i(\mathbf{x}, t) = -\displaystyle\int_0^\infty \frac{f_\omega(t+\alpha)}{F_\omega(\alpha)}s_i^0(\mathbf{x}, d\alpha)\\ \qquad\qquad - \displaystyle\int_0^t f_\omega(t-s)\kappa(\mathbf{x}, s)\ ds\\ \qquad\qquad + \kappa(\mathbf{x}, t).\end{cases} \tag{57}$$

6.3.1 Eliminating auxiliary variables

We will eliminate the auxiliary variables from the system given in Equation (57) by defining the kernels Φ_ρ, Φ_τ and Φ_ω such that we have

$$\int_\sigma^t \Phi_\rho(t-s)F_\rho(s-\sigma)\,ds = f_\rho(t-\sigma), \quad \int_\sigma^t \Phi_\tau(t-s)F_\tau(s-\sigma)\,ds = f_\tau(t-\sigma), \quad \int_\sigma^t \Phi_\omega(t-s)F_\omega(s-\sigma)\,ds.$$

According to Lemma 6 in Section 3.3.1, these kernels exist and can be uniquely determined.

Through $\overline{\Phi}(z) = \lambda$ as derived in Section 3.3.1, we find $\Phi_\rho = \lambda\delta_0$. We then have

$$\begin{aligned}
\int_0^t \Phi_\rho(t-s)\, s_m(\mathbf{x}-(t-s)\mathbf{v},\mathbf{v},s)\,ds &= \lambda s_m(\mathbf{x},\mathbf{v},t) \\
&= \lambda b_t^m(\mathbf{x},\mathbf{v}) + \lambda\int_0^t G(t-s)\,(\xi(\mathbf{x}-(t-s)\mathbf{v},\mathbf{v},s))\;ds \\
&= \lambda b_t^m(\mathbf{x},\mathbf{v}) + \lambda\int_0^t F_\rho(t-s)F_\tau(t-s)\,(\xi(\mathbf{x}-(t-s)\mathbf{v},\mathbf{v},s))\;ds \\
&= \lambda b_t^m(\mathbf{x},\mathbf{v}) + \int_0^t \lambda\exp(-\lambda(t-s))F_\tau(t-s)\,(\xi(\mathbf{x}-(t-s)\mathbf{v},\mathbf{v},s))\;ds \\
&= \lambda b_t^m(\mathbf{x},\mathbf{v}) + \int_0^t K_\zeta(t-s)\,(\xi(\mathbf{x}-(t-s)\mathbf{v},\mathbf{v},s))\;ds,
\end{aligned}$$

where we used the definitions introduced in Equation (49) in the third equality and Equation (51) in the last equality. Based on this, we can write

$$\int_0^t K_\zeta(t-s)\,(\xi(\mathbf{x}-(t-s)\mathbf{v},\mathbf{v},s))\;ds = \lambda s_m(\mathbf{x},\mathbf{v},t) - \lambda b_t^m(\mathbf{x},\mathbf{v}).$$

In a similar manner, we determine

$$\begin{aligned}
\int_0^t K_\kappa(t-s)\,(\xi(\mathbf{x}-(t-s)\mathbf{v},\mathbf{v},s))\;ds &= \int_0^t \Phi_\tau(t-s)s_m(\mathbf{x}-(t-s)\mathbf{v},\mathbf{v},s)\,ds \\
&\quad - \int_0^t \Phi_\tau(t-s)b_s^m(\mathbf{x}-(t-s)\mathbf{v},\mathbf{v})\,ds.
\end{aligned}$$

Substituting these terms and the expression for ξ in the equation for $s_m(\mathbf{x},\mathbf{v},t)$ in Equation (57) gives

$$\begin{aligned}
\frac{\partial}{\partial t}s_m(\mathbf{x},\mathbf{v},t) + (\mathbf{v}\cdot\nabla_{\mathbf{x}})s_m(\mathbf{x},\mathbf{v},t) &= P[u_t(\mathbf{x},\cdot)](\mathbf{v})\int_0^\infty \frac{f_\omega(t+\alpha)}{F_\omega(\alpha)}s_i^0(\mathbf{x},d\alpha) \\
&\quad + \lambda\exp(-\lambda t)\int_V T(\mathbf{v},\mathbf{v}')s_m^0(\mathbf{x}-t\mathbf{v}',\mathbf{v}',[0,\infty))\;d\mathbf{v}' \\
&\quad + P[u_t(\mathbf{x},\cdot)](\mathbf{v})\int_0^t f_\omega(t-s)\kappa(\mathbf{x},s)\,ds \\
&\quad + \int_V\int_0^t T(\mathbf{v},\mathbf{v}')K_\zeta(t-s)\,\xi(\mathbf{x}-(t-s)\mathbf{v}',\mathbf{v}',s)\,ds\,d\mathbf{v}' \\
&\quad - \lambda\exp(-\lambda t)\int_0^\infty \frac{F_\tau(t+a)}{F_\tau(a)}s_m^0(\mathbf{x}-t\mathbf{v},\mathbf{v},da) \\
&\quad - \exp(-\lambda t)\int_0^\infty \frac{f_\tau(t+a)}{F_\tau(a)}s_m^0(\mathbf{x}-t\mathbf{v},\mathbf{v},da) \\
&\quad - \int_0^t \Phi_\tau(t-s)\,s_m(\mathbf{x}-(t-s)\mathbf{v},\mathbf{v},s)\,ds \\
&\quad + \int_0^t \Phi_\tau(t-s)\,b_s^m(\mathbf{x}-(t-s)\mathbf{v},\mathbf{v})\,ds \\
&\quad - \lambda s_m(\mathbf{x},\mathbf{v},t) \\
&\quad + \lambda b_t^m(\mathbf{x},\mathbf{v}).
\end{aligned} \tag{58}$$

We then note that there are only two terms that involve the auxiliary variables. We can eliminate the term containing κ by calculating

$$\int_0^t \Phi_\omega(t-s)s_i(\mathbf{x},s)\ ds = \int_0^t \Phi_\omega(t-s)b_s^i(\mathbf{x})\ ds + \int_0^t f_\omega(t-s)\kappa(\mathbf{x},s)\ ds.$$

so we see that

$$\begin{aligned} P[u_t(\mathbf{x},\cdot)](\mathbf{v})\int_0^t f_\omega(t-s)\kappa(\mathbf{x},s)\ ds =& P[u_t(\mathbf{x},\cdot)](\mathbf{v})\bigg(\int_0^t \Phi_\omega(t-s)s_i(\mathbf{x},s)\bigg) \\ &- P[u_t(\mathbf{x},\cdot)](\mathbf{v})\bigg(\int_0^t \Phi_\omega(t-s)b_s^i(\mathbf{x})\ ds\bigg) \end{aligned}$$

We can eliminate the term involving ξ from Equation (58) via

$$\begin{aligned} \int_0^t\int_V T(\mathbf{v},\mathbf{v}')K_\zeta(t-s)\left(\xi(\mathbf{x}-(t-s)\mathbf{v}',\mathbf{v}',s)\right)\ ds &= \int_0^t\int_V T(\mathbf{v},\mathbf{v}')\Phi_\rho(t-s)s_m(\mathbf{x}-(t-s)\mathbf{v}',\mathbf{v}',t)\ d\mathbf{v}'\ ds \\ &\quad - \int_0^t\int_V T(\mathbf{v},\mathbf{v}')\Phi_\rho(t-s)b_s^m(\mathbf{x}-(t-s)\mathbf{v}',\mathbf{v}')\ d\mathbf{v}'\ ds \\ &= \lambda\int_V T(\mathbf{v},\mathbf{v}')s_m(\mathbf{x},\mathbf{v}',t)\ d\mathbf{v}' \\ &\quad - \lambda\int_V T(\mathbf{v},\mathbf{v}')b_t^m(\mathbf{x},\mathbf{v}')\ d\mathbf{v}'. \end{aligned}$$

We also recall that $b_t^m(\mathbf{x},\mathbf{v}) = \exp(-\lambda t)\int_0^\infty \frac{F_\tau(t+a)}{F_\tau(a)}s_m^0(\mathbf{x}-t\mathbf{v},\mathbf{v},da)$. Then, we can rewrite Equation (58) as

$$\begin{aligned} \frac{\partial}{\partial t}s_m(\mathbf{x},\mathbf{v},t) + (\mathbf{v}\cdot\nabla_{\mathbf{x}})s_m(\mathbf{x},\mathbf{v},t) =& P[u_t(\mathbf{x},\cdot)](\mathbf{v})\int_0^\infty \frac{f_\omega(t+\alpha)}{F_\omega(\alpha)}s_i^0(\mathbf{x},d\alpha) \\ &+ P[u_t(\mathbf{x},\cdot)](\mathbf{v})\bigg(\int_0^t \Phi_\omega(t-s)s_i(\mathbf{x},s)\bigg) \\ &- P[u_t(\mathbf{x},\cdot)](\mathbf{v})\bigg(\int_0^t \Phi_\omega(t-s)b_s^i(\mathbf{x})\ ds\bigg) \\ &+ \lambda\int_V T(\mathbf{v},\mathbf{v}')s_m(\mathbf{x},\mathbf{v}',t)\ d\mathbf{v}' \\ &- \int_0^t \Phi_\tau(t-s)s_m(\mathbf{x}-(t-s)\mathbf{v},\mathbf{v},s)\ ds \\ &+ \int_0^t \Phi_\tau(t-s)b_s^m(\mathbf{x}-(t-s)\mathbf{v},\mathbf{v})\ ds \\ &- \lambda s_m(\mathbf{x},\mathbf{v},t) \\ &- \exp(-\lambda t)\int_0^\infty \frac{f_\tau(t+a)}{F_\tau(a)}s_m^0(\mathbf{x}-t\mathbf{v},\mathbf{v},da) \end{aligned} \tag{59}$$

and we note that this expression depends only on our model variables $s_m(\mathbf{x},\mathbf{v},t)$ and $s_i(\mathbf{x},t)$ and on the initial distributions. We will categorize the terms on the right hand side in Section 6.3.2 to get a better understanding of the main processes in the system.

For now, we proceed by eliminating all the auxiliary variables in the equation for $\frac{\partial}{\partial t}s_i(\mathbf{x},t)$ in Equation (57). We can see that

$$\int_0^t f_\omega(t-s)\kappa(\mathbf{x},s)\ ds = \int_0^t \Phi_\omega(t-s)s_i(\mathbf{x},s)\ ds - \int_0^t \Phi_\omega(t-s)b_s^i(\mathbf{x})\ ds,$$

which is a form without auxiliary variables. We can also write

$$\kappa(\mathbf{x},t) = \int_V \int_0^t K_\kappa(t-s)\left(\xi(\mathbf{x}-(t-s)\mathbf{v},\mathbf{v}',s)\right)\, ds\, d\mathbf{v} = \int_V \int_0^t \Phi_\tau(t-s)s_m(\mathbf{x}-t\mathbf{v},\mathbf{v},t)\, ds\, d\mathbf{v} - \int_V \int_0^t \Phi_\tau(t-s)b_s^m(\mathbf{x}-(t-s)\mathbf{v},\mathbf{v})\, ds\, d\mathbf{v}$$

Which gives us

$$\begin{aligned}\frac{\partial}{\partial t}s_i(\mathbf{x},t) = &-\int_0^\infty \frac{f_\omega(t+\alpha)}{F_\omega(\alpha)}s_i^0(\mathbf{x},d\alpha)\\ &-\int_0^t \Phi_\omega(t-s)s_i(\mathbf{x},s)\, ds\\ &+\int_0^t \Phi_\omega(t-s)b_s^i(\mathbf{x})\, ds,\\ &+\int_V\int_0^t \Phi_\tau(t-s)s_m(\mathbf{x}-(t-s)\mathbf{v},\mathbf{v},s)\, ds\, d\mathbf{v}\\ &-\int_V\int_0^t \Phi_\tau(t-s)b_s^m(\mathbf{x}-(t-s)\mathbf{v},\mathbf{v})\, ds\, d\mathbf{v}.\end{aligned} \tag{60}$$

6.3.2 Final model formulation

In order to distinguish between terms that result from the initial conditions and all the other terms, we rewrite Equations (59) and (60) as follows:

$$\begin{cases}\frac{\partial}{\partial t}s_m(\mathbf{x},\mathbf{v},t) + (\mathbf{v}\cdot\nabla_{\mathbf{x}})s_m(\mathbf{x},\mathbf{v},t) = \mathcal{L}_\lambda[s_m(\mathbf{x},\cdot)](\mathbf{v},t)\\ \qquad\qquad + P[u_t(\mathbf{x},\cdot)](\mathbf{v})\left(\int_0^t \Phi_\omega(t-s)s_i(\mathbf{x},s)\right)\\ \qquad\qquad - \int_0^t \Phi_\tau(t-s)s_m(\mathbf{x}-(t-s)\mathbf{v},\mathbf{v},s)\, ds\\ \qquad\qquad + B_m(\mathbf{x},\mathbf{v},t)\\ \frac{\partial}{\partial t}s_i(\mathbf{x},t) = -\int_0^t \Phi_\omega(t-s)s_i(\mathbf{x},s)\, ds\\ \qquad + \int_V\int_0^t \Phi_\tau(t-s)s_m(\mathbf{x}-(t-s)\mathbf{v},\mathbf{v},s)\, ds\, d\mathbf{v}\\ \qquad + B_i(\mathbf{x},t).\end{cases} \tag{61}$$

for the reorientation kernel, as defined in Equation (23),

$$\mathcal{L}_\lambda^T[p](\mathbf{x},\mathbf{v},t) := -\lambda p_t(\mathbf{x},\mathbf{v}) + \lambda\int_V T(\mathbf{v},\mathbf{v}')p_t(\mathbf{x},\mathbf{v}')\, d\mathbf{v}'\ ,$$

which is the kernel derived for the general Velocity-Jump Model in Section 3.

We then have the forcings as a result of the initial distributions, given by

$$\begin{aligned}B_m(\mathbf{x},\mathbf{v},t) := &P[u_t(\mathbf{x},\cdot)](\mathbf{v})\int_0^\infty \frac{f_\omega(t+\alpha)}{F_\omega(\alpha)}s_i^0(\mathbf{x},d\alpha)\\ &- P[u_t(\mathbf{x},\cdot)](\mathbf{v})\left(\int_0^t \Phi_\omega(t-s)b_s^i(\mathbf{x})\, ds\right)\\ &+\int_0^t \Phi_\tau(t-s)b_s^m(\mathbf{x}-(t-s)\mathbf{v},\mathbf{v})\, ds\\ &-\exp(-\lambda t)\int_0^\infty \frac{f_\tau(t+a)}{F_\tau(a)}s_m^0(\mathbf{x}-t\mathbf{v},\mathbf{v},da)\end{aligned} \tag{62}$$

and

$$B_i(\mathbf{x},t) := -\int_0^\infty \frac{f_\omega(t+\alpha)}{F_\omega(\alpha)} s_i^0(\mathbf{x},d\alpha) \tag{63}$$
$$+\int_0^t \Phi_\omega(t-s) b_s^i(\mathbf{x})\ ds$$
$$-\int_V \int_0^t \Phi_\tau(t-s) b_s^m(\mathbf{x}-(t-s)\mathbf{v}',\mathbf{v}')\ ds\ d\mathbf{v}'.$$

6.3.3 The case of ageless initial conditions

As in Section 3.4, we consider a case in which each initial run and each initial rest starts at the beginning of the observations, at $t = 0$. We thus eliminate the age variable from out system. Since

$$s_m^0(\mathbf{x},\mathbf{v},a) = s_m^0(\mathbf{x},\mathbf{v})\delta(a), \qquad s_i^0(\mathbf{x},\alpha) = s_i^0(\mathbf{x})\delta(\alpha),$$

we can denote the initial distributions as

$$s_m^0(\mathbf{x},\mathbf{v}) := s_m^0(\mathbf{x},\mathbf{v},[0,\infty)), \qquad s_i^0(\mathbf{x}) := s_i^0(\mathbf{x},[0,\infty)).$$

To find an expression for $B_m(\mathbf{x},\mathbf{v},t)$ under the assumption of ageless initial conditions, we note that

$$\frac{f_\omega(t+\alpha)}{F_\omega(\alpha)} s_i^0(\mathbf{x},d\alpha) = f_\omega(t) s_i^0(\mathbf{x})\delta(\alpha),$$

$$\int_0^t \Phi_\omega(t-s) b_s^i(\mathbf{x},\mathbf{v})\ ds = f_\omega(t) s_i^0(\mathbf{x},).$$

So, with Equation (62), we find

$$B_m(\mathbf{x},\mathbf{v},t) = \int_0^t \Phi_\tau(t-s) b_s^m(\mathbf{x}-(t-s)\mathbf{v},\mathbf{v})\ ds$$
$$-\exp(-\lambda t) f_\tau(t) s_m^0(\mathbf{x}-t\mathbf{v},\mathbf{v}).$$

Similarly, Equation (63) reduces to

$$B_i(\mathbf{x},t) = -\int_V \int_0^t \Phi_\tau(t-s) b_s^m(\mathbf{x}-(t-s)\mathbf{v},\mathbf{v})\ ds\ d\mathbf{v}.$$

Then, when studying the total population density $s := s_i + s_m$ of the hitchhikers, the contribution of the initial conditions over time becomes

$$B(\mathbf{x},t) := \int_V B_m(\mathbf{x},\mathbf{v},t)\ d\mathbf{v} + B_i(\mathbf{x},t) = -\exp(-\lambda t) f_\tau(t) \int_V s_m^0(\mathbf{x}-t\mathbf{v},\mathbf{v})\ d\mathbf{v}.$$

Since $s_m^0(\mathbf{x}-t\mathbf{v},\mathbf{v},[0,\infty))$ is bounded for all $\mathbf{x}\in\Omega, \mathbf{v}\in V$, we have

$$\lim_{t\to\infty}\left(-\exp(-\lambda t) f_\tau(t) \int_V s_m^0(\mathbf{x}-t\mathbf{v},\mathbf{v})\ d\mathbf{v}\right) = 0.$$

In other words, the contribution of the initial distribution for the total hitchhiker population disappears over time.

6.3.4 The case of Markovian Dynamics

Assuming that all processes occur according to Poisson processes, we let

$$f_\tau(t) = \nu \exp(-\nu t), \qquad f_\omega(t) = \mu \exp(-\mu t)$$

for some parameters $\mu, \nu \in \mathbb{R}_{>0}$. This gives

$$F_\tau(t) = \exp(-\nu t), \qquad F_\omega(t) = \exp(-\mu t).$$

In particular, we assume that the duration of the hitchhikers' resting times are independent of the local carrier density. That is to say, f_ω does not depend on the density function u. This is representative of situations where the carrier density is high enough that a hitchhiker's attempt to attach to a carrier is always successful. This also means that the carriers can carry arbitrarily many hitchhikers at a given moment. The hitchhikers distribute over the carriers according to the carriers' distribution of $\mathbf{v}$ at time t, given by $P[u_t(\mathbf{x}, \cdot)](\mathbf{v})$.

With Lemma 6 in Section 3.3.1, the assumed distributions give

$$\Phi_\tau = \nu\delta_0, \quad \Phi_\omega = \mu\delta_0.$$

With this, the convolutions in Equation (61) reduce to

$$\int_0^t \Phi_\omega(t-s) s_i(\mathbf{x}, s)\ ds = \mu s_i(\mathbf{x}, t)$$

and

$$\int_0^t \Phi_\tau(t-s) s_m(\mathbf{x} - (t-s)\mathbf{v}, \mathbf{v}, s)\ ds = \nu s_m(\mathbf{x}, \mathbf{v}, t).$$

To calculate $B_m(\mathbf{x}, \mathbf{v}, t)$ for the case of Markovian dynamics, we calculate

$$\begin{aligned}
&\int_0^\infty \frac{f_\omega(t+\alpha)}{F_\omega(\alpha)} s_i^0(\mathbf{x}, d\alpha) = \mu \exp(-\mu t) s_i^0(\mathbf{x}, [0, \infty)),\\
&\int_0^t \Phi_\omega(t-s) b_s^i(\mathbf{x})\ ds = \mu b_t^i(\mathbf{x}),\\
&\int_0^t \Phi_\tau(t-s) b_s^m(\mathbf{x} - (t-s)\mathbf{v}, \mathbf{v})\ ds = \nu b_t^m(\mathbf{x}, \mathbf{v}).
\end{aligned}$$

For Equation (62), we then get

$$\begin{aligned}
B_m(\mathbf{x}, \mathbf{v}, t) = \mu \exp(-\mu t) P[u_t(\mathbf{x}, \cdot)](\mathbf{v}) s_i^0(\mathbf{x}, [0, \infty))\\
- \mu P[u_t(\mathbf{x}, \cdot)](\mathbf{v}) b_t^i(\mathbf{x})\\
+ \nu b_t^m(\mathbf{x}, \mathbf{v})\\
- \nu b_t^m(\mathbf{x}, \mathbf{v})\\
= 0,
\end{aligned}$$

where we used $b_t^i(\mathbf{x}) = \exp(-\mu t) s_i^0(\mathbf{x}, [0, \infty))$ under the assumption that ω is exponentially distributed with parameter μ. From this, we conclude that the initial distribution of the mobile hitchhikers does not affect the system's dynamics.

Similarly, for Equation (63), we find

$$\begin{aligned}
B_i(\mathbf{x}, t) = &- \mu \exp(-\mu t) s_i^0(\mathbf{x}, [0, \infty))\\
&+ \mu b_t^i(\mathbf{x})\\
&- \nu \int_V b_t^m(\mathbf{x}, \mathbf{v})\ d\mathbf{v}\\
= &-\nu \int_V b_t^m(\mathbf{x}, \mathbf{v})\ d\mathbf{v}.
\end{aligned}$$

We also see that the total contribution of the initial conditions for all time t and position $\mathbf{x}$ is determined entirely by the distribution of the immobile hitchhikers, as we have

$$\begin{aligned} B(\mathbf{x},t) &:= \int_V B_m(\mathbf{x},\mathbf{v},t)\ d\mathbf{v} + B_i(\mathbf{x},t) \\ &= -\nu \int_V b_t^m(\mathbf{x},\mathbf{v})\ d\mathbf{v}. \end{aligned}$$

As in the case of the ageless conditions, we note that

$$\lim_{t\to\infty} \nu b_t^m(\mathbf{x},\mathbf{v}) = \lim_{t\to\infty} \nu \exp(-\lambda t) \int_0^\infty \frac{F_\tau(t+a)}{F_\tau(a)} s_m^0(\mathbf{x}-t\mathbf{v},\mathbf{v},da) = 0,$$

as $s_m^0(\mathbf{x}-t\mathbf{v},\mathbf{v},[0,\infty))$ is bounded for all $\mathbf{x}\in\Omega, \mathbf{v}\in V$. So, in the case of Markovian dynamics, the contribution of the initial distribution for the total hitchhiker population vanishes over time. This is to be expected. Since we are mainly interested in studying the system's behavior over time, we neglect the forcing caused by the initial distributions, which we can justify by using the calculations performed in this section. In the next section, we assume that the forcings B_m and B_i are negligible and we will analyse the Hitchhikers' Model for Markovian dynamics given by

$$\begin{cases} \frac{\partial}{\partial t} s_m(\mathbf{x},\mathbf{v},t) + (\mathbf{v}\cdot\nabla_{\mathbf{x}}) s_m(\mathbf{x},\mathbf{v},t) = \mathcal{L}_\lambda[s_m](\mathbf{x},\mathbf{v},t) \\ \qquad\qquad\qquad\qquad\qquad\qquad + \mu P[u_t(\mathbf{x},\cdot)](\mathbf{v}) s_i(\mathbf{x},t) \\ \qquad\qquad\qquad\qquad\qquad\qquad - \nu s_m(\mathbf{x},\mathbf{v},t) \\ \\ \frac{\partial}{\partial t} s_i(\mathbf{x},t) = -\mu s_i(\mathbf{x},t) + \nu \int_V s_m(\mathbf{x},\mathbf{v},t). \end{cases} \tag{64}$$

7 The Markovian Hitchhikers' Model

As shown in Section 6.3.4, the general Hitchhikers' Model reduces to a much simpler form when Markovian dynamics are assumed. In this section, we will further study the model given in Equation (64) and derive a diffusive limit equation for the Markovian Hitchhiker's Model. We begin by giving a structured overview of the model by collecting and restating all assumptions used in its construction. Then, we derive a diffusive limit equation using similar techniques as those applied for obtaining the diffusive limit of the Markovian Velocity-Jump Model in Section 4.3. We will then present a simulation study, in which we the density solution of the macroscopic limit equation is compared to the empirical density obtained from an agent-based simulation over time.

7.1 Model formulation and assumptions

We will formulate a two-species hitchhiking model in which the population density of carriers is denoted as $u(\mathbf{x}, t)$. The PDE $\frac{\partial u}{\partial t}$ is assumed to have a closed expression, independent of the hitchhikers. Additionally, we further that u is a given sufficiently smooth and strictly positive function, bounded away from zero on the time interval of interest $[0, T]$. In particular, we let

$$u \in C^{2,1}(\Omega \times [0,T]), \qquad 0 < u_{\min} \leq u(\mathbf{x}, t) \leq u_{\max} < \infty.$$

As the velocity of the moving particles is not relevant for the density of the diffusing mass, we will work with

$$u(\mathbf{x}, t) = \int_V u(\mathbf{x}, \mathbf{v}, t)\ d\mathbf{v},$$

in which $u(\mathbf{x}, \mathbf{v}, t)$ represents the phase-space density of the carriers. In this section, we assume that the carriers follow a Markovian velocity-jump process and, with the result obtained in Section 4, we approximate $u(\mathbf{x}, t)$ by the diffusion equation. The diffusion coefficient can be derived as in Section 4.3.2 and is denoted by D. In other words, we can describe the transport of the population of carriers by

$$\frac{\partial}{\partial t} u(\mathbf{x}, t) = \nabla_{\mathbf{x}} \cdot (D \nabla_{\mathbf{x}} u(\mathbf{x}, t)). \tag{65}$$

For the populations of hitchhikers, we distinguish between an immobile population, with density s_i and a mobile population with density s_m. We will no longer model velocity-dependence, as all transport takes place with the velocity given by the diffusive limit of the population of hitchhikers. To this end, we define the velocity-independent densities

$$s_i(\mathbf{x}, t) := \int_V s_i(\mathbf{x}, \mathbf{v}, t)\ d\mathbf{v}, \qquad s_m(\mathbf{x}, t) := \int_V s_m(\mathbf{x}, \mathbf{v}, t)\ d\mathbf{v}. \tag{66}$$

We assume that the transition rate to the resting state is constant over time and space, as mobile hitchhikers can detach from the carrier at any time. In Equation (64), we assumed that the distribution of the duration of the hitchhikers' immobile state is independent of the carriers' density and can be modelled by the parameter μ in the Markovian case. However, we postulate that the transition rate should depend not only on the density of immobile hitchhikers but also on the local density of available carriers. Therefore, we introduce the u-dependent rate parameter $\mu(u)$. We write the transition rate as $\mu(u)u$ rather than a general function of u, since this separates carrier availability from the attachment rate. This formulation is convenient both for interpretation and for the subsequent mathematical analysis, in particular for the variable scaling we will introduce later when analysing the model. As a result of the velocity-independent densities given in Equation (66), we can integrate the probability distribution $P[u_t(\mathbf{x}, \cdot)](\mathbf{v})$ from Equation (64) over the velocity space V and find the attachment rate $\mu(u)u = \mu \int_V P_t[u_t(\mathbf{x}, \cdot)](\mathbf{v})$.

We do not take into account external stimuli or internal state variables. A schematic overview of the Markovian Hitchhikers' model with the introduced notation is given in Figure 4.

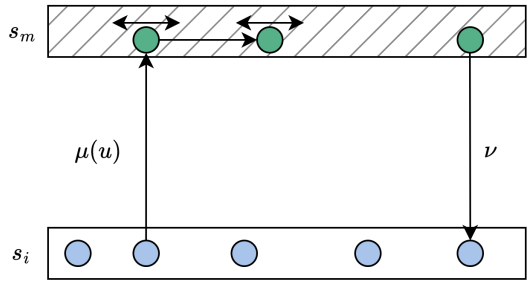


Figure 4: A schematic of the Markovian Hitchhikers' Model, where particles move in one dimension in a rectangular area. The blue circles represent immobile hitchhikers. The green circles represent individuals represent mobile hitchhikers. The arrows represent their motility. The mobile hitchhikers are moving on the mobile population of carriers, which are represented by the grey lines. Their density is assumed to evolve according to the diffusion equation given in Equation (65). The immobile hitchhikers jump on the carrier population with the carrier-dependent rate $\mu(u)$ and return to the immobile state by detaching with constant rate ν.

We neglect the contribution of the initial conditions, which we can justify with the results obtained in Section 6.3.4. The Markovian Hitchhikers' Model with a carrier-dependent attachment rate is then given by

$$\begin{cases} \frac{\partial}{\partial t} s_i(\mathbf{x},t) = \nu s_m(\mathbf{x},t) - \mu(u) u\, s_i(\mathbf{x},t), \\ \frac{\partial}{\partial t} s_m(\mathbf{x},t) + \mathbf{v}\cdot\nabla_{\mathbf{x}} s_m(\mathbf{x},t) - \nu s_m(\mathbf{x},t) + \mu(u) u\, s_i(\mathbf{x},t) + \mathcal{L}[s_m(\mathbf{x},t)]. \end{cases} \tag{67}$$

The mobile hitchhikers diffuse with the same rate as the carriers. Using the linearity of the transport operator $\mathcal{L}$, and mirroring the transport term in Equation (65), we obtain the following model:

$$\begin{cases} \frac{\partial}{\partial t} s_i(\mathbf{x},t) = \nu s_m(\mathbf{x},t) - \mu(u) u(\mathbf{x},t)\, s_i(\mathbf{x},t), \\ \frac{\partial}{\partial t} s_m(\mathbf{x},t) = -\nu s_m(\mathbf{x},t) + \mu(u) u\, s_i(\mathbf{x},t) + \nabla_x \cdot \big(D \nabla_x s_m(\mathbf{x},t)\big). \end{cases} \tag{68}$$

In this section, we will derive a macroscopic limit equation for the total hitchhiker density $s := s_i + s_m$, resulting from the system given in Equation (68) and perform simulations to quantify the agreement between the derived limit and the underlying stochastic process.

7.2 Scaling the Hitchhikers' model

We nondimensionalize the system by scaling the temporal and spatial variables by some characteristic factor. These scaling factors are chosen such that they allow for the derivation of a macroscopic, long-term limit equation. For the spatial vector, we denote this as the length L. Its value depends on the context in which the model is applied. In certain settings, for example in lab experiments, may be taken as the maximal length of the finite domain in which the hitchhikers and carriers reside.or larger spatial domains, such as those arising in studies of bird migration , the value of L can be chosen as the maximal observed length from a starting point. More generally, any suitable characteristic length may be selected, which then defines the spatial unit of the scaled system. The spatial scaling is therefore not unique. However, the chosen length unit does determine the corresponding diffusive time scale [4]. The scaled time unit represents the diffusive time for length L, which is given by $\frac{L^2}{D}$. We thus introduce the following scaled variables:

$$\hat{\mathbf{x}} = \frac{\mathbf{x}}{L}, \qquad \hat{t} := \frac{t}{\frac{L^2}{D}} = \frac{D}{L^2} t.$$

We find

$$\frac{\partial}{\partial \mathbf{x}} = \frac{1}{L}\frac{\partial}{\partial \hat{\mathbf{x}}}, \qquad \frac{\partial}{\partial t} = \frac{D}{L^2}\frac{\partial}{\partial \hat{t}}.$$

We also scale the densities by some reference density. This ensures that all dependent variables are dimensionless and of comparable magnitude. We define

$$\hat{s_i}(\hat{\mathbf{x}},\hat{t}) = \frac{s_i(\mathbf{x},t)}{s_0}, \qquad \hat{s_m}(\hat{\mathbf{x}},\hat{t}) = \frac{s_m(\mathbf{x},t)}{s_0}, \qquad \hat{u}(\hat{\mathbf{x}},\hat{t}) = \frac{u(\mathbf{x},t)}{u_0}.$$

If we substitute this in the system in Equation (68), we get:

$$\begin{cases} \frac{\partial}{\partial \hat{t}}\hat{s_i} = \frac{\nu L^2}{D}\hat{s_m} - \frac{\mu(\hat{u})\hat{u}L^2}{D}u_0\,\hat{s_i}, \\ \frac{\partial}{\partial \hat{t}}\hat{s_m} = -\frac{\nu L^2}{D}\hat{s_m} + \frac{\mu(\hat{u})\hat{u}L^2}{D}u_0\hat{s_i} + \nabla_{\hat{\mathbf{x}}}\cdot\left(\nabla_{\hat{\mathbf{x}}}\hat{s_m}\right). \end{cases} \tag{69}$$

We identify the two parameters that describe this dimensionless system:

$$\hat{\nu} := \frac{\nu L^2}{D}, \qquad \hat{\mu}(\hat{u}) := \frac{\mu(\hat{u})L^2}{D}u_0.$$

From this, we can see that the new system is defined to be relative to the diffusivity D. Since time has been scaled with the characteristic diffusive time $\frac{L^2}{D}$, the diffusion coefficient is absorbed into the nondimensionalization and no longer appears separately. The resulting system is therefore expressed relative to the diffusion time scale. This gives our system:

$$\begin{cases} \frac{\partial}{\partial \hat{t}}\hat{s_i} = \hat{\nu}\hat{s_m} - \hat{\mu}(\hat{u})\hat{u}\hat{s_i}, \\ \frac{\partial}{\partial \hat{t}}\hat{s_m} = -\hat{\nu}\hat{s_m} + \hat{\mu}(\hat{u})\hat{u}\hat{s_i} + \nabla_{\hat{\mathbf{x}}}\cdot\left(\nabla_{\hat{\mathbf{x}}}\hat{s_m}\right). \end{cases} \tag{70}$$

The interpretation of this scaled system now depends on the orders of magnitude of the parameter values. The parameter $\hat{\nu}$ describes the rate of the transition from the mobile state to the immobile state relative to the diffusion rate, and the parameter $\hat{\mu}(\hat{u})$ describes the rate of the reverse transition. Via the scaling in time, we defined a time scale in which the diffusion takes place in order $\mathcal{O}(1)$. If both transition parameters are also of order $\mathcal{O}(1)$, transitions between the mobile and immobile state occur on the same macroscopic time scale as diffusion. If, on the other hand, the scaling causes the transition parameters to be of order $O(\varepsilon^{-2})$ for some $0 < \varepsilon << 1$ instead, this means there are many transitions per unit time, and the duration of the mobile and immobile periods are short relative to the diffusion process. We will study both cases, and conclude that although the microscopic processes differ, they can be represented by the same mathematical model.

7.2.1 Scaling in the case of fast transitions

Under the assumption that $\hat{\nu}$ and $\hat{\mu}$ are of order $\mathcal{O}(\varepsilon^{-2})$, there are frequent jumps and the mobile and immobile periods are of short duration compared to the macroscopic diffusion. By setting

$$\hat{\nu} = \frac{\nu L^2}{D} \stackrel{!}{=} \varepsilon^{-2}, \qquad \hat{\mu}(\hat{u}) = \frac{\mu(u)L^2}{D}u_0 \stackrel{!}{=} \varepsilon^{-2} \tag{71}$$

we can solve for u_0, as μ_0, L and D are assumed to be known. This elucidates the scaling of the transporter density via $\hat{u}(\hat{\mathbf{x}},\hat{t}) = \frac{u(\mathbf{x},t)}{u_0}$. Substituting these new parameters in the system given in Equation (70) and then multiplying by ε^2 gives

$$\begin{cases} \varepsilon^2\frac{\partial}{\partial \hat{t}}\hat{s_i} = \hat{s_m} - \hat{u}\hat{s_i}, \\ \varepsilon^2\frac{\partial}{\partial \hat{t}}\hat{s_m} = -\hat{s_m} + \hat{u}\hat{s_i} + \varepsilon^2\nabla_{\hat{\mathbf{x}}}\cdot\left(\nabla_{\hat{\mathbf{x}}}\hat{s_m}\right). \end{cases} \tag{72}$$

7.2.2 Scaling in the case of slow transitions

Under the assumption that the jumping process takes place on the same spatial and temporal scales as the diffusive process, the durations of the runs and immobile periods are of the same order of magnitude as the carriers' diffusion rate. That is, we have $\hat{\nu}, \hat{\mu}(\hat{u}) \sim O(1)$.

To capture the system's long-time behavior on a large spatial scale, we introduce a parabolic scaling to Equation (68). As in Section 4, we choose an $\varepsilon > 0$ such that the diffusion is observed in order $\mathcal{O}(1)$.

We introduce the parabolic scaling

$$\chi := \varepsilon x, \qquad \tau := \varepsilon^2 t. \tag{73}$$

Instead of nondimensionalising the system as we did in the case of fast transitions, one may write the system given in Equation (69) directly as

$$\begin{cases} \varepsilon^2 \frac{\partial}{\partial \tau} \overline{s_i} = \nu \overline{s_m} - \mu(\bar{u}) \bar{u} \bar{s_i}, \\ \varepsilon^2 \frac{\partial}{\partial \tau} \overline{s_m} = -\nu \overline{s_m} + \mu(\bar{u}) \bar{u} \bar{s_i} + \varepsilon^2 \nabla_\chi \cdot \left(\nabla_\chi \overline{s_m} \right), \end{cases} \tag{74}$$

where

$$\overline{s_i}(\chi, \tau) := \hat{s_i}(\hat{\mathbf{x}}, \hat{t}), \quad \overline{s_m}(\chi, \tau) := \hat{s_m}(\hat{\mathbf{x}}, \hat{t}), \quad \overline{u}(\chi, \tau) := \hat{u}(\hat{\mathbf{x}}, \hat{t}). \tag{75}$$

We can see that despite the two scalings arising from different interpretations, the systems in Equation (72) and Equation (74) are mathematically equivalent via the identifications $\hat{s_i}(\chi, \tau) = \overline{s_i}(\hat{\mathbf{x}}, \hat{t})$ and $\hat{s_m}(\chi, \tau) = \overline{s_m}(\hat{\mathbf{x}}, \hat{t})$ and via the definitions the scaled parameters as given in Equation (71).

7.3 Diffusive limit of the Hitchhikers' Model

We continue studying the system given in Equation (74). We take example from the method described by Witelski and Bowen [50]. We rewrite Equation (74) by introducing

$$\mathscr{R}(u) := \begin{pmatrix} -\mu(\overline{u})\overline{u} & \nu \\ \mu(\overline{u})\overline{u} & -\nu \end{pmatrix}, \qquad \mathscr{D}\overline{s} := \begin{pmatrix} 0 \\ \nabla_\chi \cdot \left(D \nabla_\chi \overline{s_m} \right) \end{pmatrix},$$

which gives the system

$$\partial_\tau \overline{s} = \varepsilon^{-2} \mathscr{R}(u)\, \overline{s} + \mathscr{D}\overline{s}, \qquad \overline{s} := \begin{pmatrix} \overline{s_i} \\ \overline{s_m} \end{pmatrix}. \tag{76}$$

In this formulation, the factor ε^{-2} reveals that the reaction operator $\mathscr{R}$ acts on a faster time scale than the diffusion operator $\mathscr{D}$. When examining the reactor dynamics, we note that

$$(1, 1)\, \mathscr{R}(u) = 0,$$

which indicates that the total density

$$s := \overline{s_i} + \overline{s_m}$$

is conserved by the reaction term. Furthermore, we have

$$\mathscr{R}(u) \begin{pmatrix} \nu \\ \mu(\overline{u})\overline{u} \end{pmatrix} = 0. \tag{77}$$

so the one-dimensional kernel of $\mathscr{R}(u)$ is spanned by the equilibrium vector

$$r(u) := \begin{pmatrix} \nu \\ \mu(\bar{u})\bar{u} \end{pmatrix}.$$

Vectors in the kernel correspond to compositions of the mobile and immobile states caused by the transition process. In other words, if the proportions of mobile and immobile individuals are given by $r(u)$, then the reaction dynamics no longer change these proportions. Since

$$\operatorname{tr} \mathscr{R}(u) = -(\nu + \mu(\bar{u})\bar{u}), \qquad \det \mathscr{R}(u) = 0,$$

the nonzero eigenvalue of $\mathscr{R}(u)$ is $-(\nu+\mu(\bar{u})\bar{u})$, which is strictly negative. The zero eigenvalue indicates mass conservation of the total population, while the negative eigenvalue implies that deviations from the equilibrium composition decay exponentially fast. For fixed values of $u(\mathbf{x}, t)$, the internal state composition relaxes rapidly toward the equilibrium manifold

$$\mathcal{E}(u) := \{\rho\, r(u) : \rho \in \mathbb{R}\},$$

which consists of all equilibrium state distributions.

Thus, on the fast time scale, the proportions of mobile and immobile individuals equilibrate, whereas the total density changes on the slower diffusive time scale. In the upcoming analysis, we will capture this feature in the macroscopic limit equation for the hitchhiking process.

As in Section 4, we expand $\overline{s_i}(\chi, \tau)$ and $\overline{s_m}(\chi, \tau)$ in terms with decreasing contributions:

$$\overline{s_i} = \sum_{j=0}^{k} \varepsilon^j \overline{s_i}^{(j)}, \qquad \overline{s_m} = \sum_{j=0}^{k} \varepsilon^j \overline{s_m}^{(j)}.$$

Substituting the expansion for $\overline{s_i}$ in the system in Equation (74) gives:

$$\begin{aligned}
\varepsilon^0 :&\quad 0 = \nu \overline{s_m}^{(0)} - \mu(\bar{u})\bar{u} \cdot \overline{s_i}^{(0)} \\
\varepsilon^1 :&\quad 0 = \nu \overline{s_m}^{(1)} - \mu(\bar{u})\bar{u} \cdot \overline{s_i}^{(1)} \\
\varepsilon^2 :&\quad \frac{\partial}{\partial \tau} \overline{s_i}^{(0)} = \nu \overline{s_m}^{(2)} - \mu(\bar{u})\bar{u} \cdot \overline{s_i}^{(2)} \\
&\ \ \vdots
\end{aligned}$$

Similarly, the expansion for s_m gives:

$$\begin{aligned}
\varepsilon^0 :&\quad 0 = -\nu \overline{s_m}^{(0)} + \mu(\bar{u})\bar{u} \cdot \overline{s_i}^{(0)} \\
\varepsilon^1 :&\quad 0 = -\nu \overline{s_m}^{(1)} + \mu(\bar{u})\bar{u} \cdot \overline{s_i}^{(1)} \\
\varepsilon^2 :&\quad \frac{\partial}{\partial \tau} \overline{s_m}^{(0)} = -\nu \overline{s_m}^{(2)} + \mu(\bar{u})\bar{u} \cdot \overline{s_i}^{(2)} + \nabla_\chi(\nabla_x \overline{s_m}^{(0)}) \\
&\ \ \vdots
\end{aligned}$$

At leading order, the $\mathcal{O}(\varepsilon^0)$ terms in Equation (76) yield

$$\mathscr{R}(u)\overline{s}^{(0)} = 0.$$

Therefore, we find

$$\overline{s}^{(0)}(\mathbf{x}, t) \in \ker \mathscr{R}(u).$$

As we had concluded that the kernel of $\mathscr{R}$ consists of vectors corresponding to equilibrium proportions of hitchhikers in the mobile and immobile state, we conclude that there exists a scalar function $s(\mathbf{x}, t)$ such that

$$\begin{pmatrix} \overline{s_i}^{(0)} \\ \overline{s_m}^{(0)} \end{pmatrix} (\chi, \tau) = \begin{pmatrix} \nu \\ \mu(\bar{u})\bar{u}(\chi, \tau) \end{pmatrix} \frac{s(\chi, \tau)}{\mu(\bar{u})\bar{u}(\chi, \tau) + \nu}.$$

Here $s(\mathbf{x}, t)$ denotes the total density of the hitchhiker population at leading order, and we define

$$s := \overline{s}_i^{(0)} + \overline{s}_m^{(0)}$$

the leading-order total density of the hitchhiker population. This essentially means that the behavior of the complete hitchhiker population will be primarily determined by s. Let

$$\alpha(\chi,\tau) := \frac{\mu(\bar{u})\bar{u}(\chi,\tau)}{\mu(\bar{u})\bar{u}(\chi,\tau)+\nu}, \tag{78}$$

which is the overall magntide of the equilibrium vector. Then, we find

$$\overline{s_m}^{(0)}(\chi,\tau) = \alpha(\chi,\tau)\, s(\chi,\tau). \tag{79}$$

Adding the equalities for the $\mathcal{O}(\varepsilon^2)$ term gives:

$$\frac{\partial}{\partial\tau}\left(\overline{s}_i^{(0)} + \overline{s}_m^{(0)}\right) = \nabla_\chi \cdot \left(\nabla_\chi(\overline{s}_m^{(0)}\right).$$

Rewriting the left hand side as $\frac{\partial}{\partial\tau}s$ and substituting Equation (79) in this equality gives:

$$\partial_\tau s = \nabla_\chi \cdot \left(\nabla_\chi \overline{s_m}^{(0)}\right) = \nabla_\chi \cdot \left(\nabla_\chi(\alpha s)\right).$$

Since $\alpha = \alpha(\chi,\tau)$ may vary in space and time, the right-hand side is not a pure diffusion operator with constant coefficient. In order to interpret this expression, we rewrite it to fit the framework of the Master Transport Equation as described in Section 2.3, and we work out the chain rule:

$$\partial_\tau s = -\nabla_\chi \cdot (-\alpha\nabla_\chi s + (-\nabla_\chi\alpha)s) \tag{80}$$

The first term on the right hand side describes the apparent Fickian diffusive flux with a diffusivity that varies in space and time through α. The second term represents the transport caused by the hitchhiking with velocity $-\nabla_\chi(\alpha(\chi,\tau))$.

This reveals that the leading-order macroscopic equation contains a local diffusion coefficient

$$\tilde{D} := D\frac{\mu(\bar{u})\bar{u}(\chi,\tau)}{\nu+\mu(\bar{u})\bar{u}(\chi,\tau)}, \tag{81}$$

and an advection-like term with local coefficient

$$A(\mathbf{x},t) := -D\nabla_\chi\alpha.$$

From Equation (77), we find that the point-wise ratio

$$\frac{\overline{s_i}^{(0)}(\chi,\tau)}{\overline{s_m}^{(0)}(\chi,\tau)} = \frac{\nu}{\mu(\bar{u})\bar{u}(\chi,\tau)}$$

is enforced by the fast velocity-jump process. The effective transport of the hitchhiker population is therefore determined by the equilibrium fraction of individuals that are mobile at position χ and time τ, namely $\alpha(\chi,\tau)$.

Note that if $u(\chi,\tau)$ has reached the homogeneous density, i.e. $u(\chi,\tau)\equiv u^*\in\mathbb{R}_{>0}$, we find $\alpha(\chi,\tau)\equiv\alpha\in\mathbb{R}$, and this gives $\frac{\partial}{\partial\tau}s = \alpha\nabla_\chi^2 s$, meaning that the effective diffusion constant becomes

$$\tilde{D} = D\alpha = D\frac{\mu_0 u^*}{\mu_0 u^*+\nu}.$$

7.4 Numerical verification of the parabolic limit

To verify the analytically derived limit, we compare the PDE solution given in Equation (80) with the underlying particle-based process. The workflow of the simulation study is visualized in Figure 5, which simultaneously summarizes the scaling procedure central to the derivation of the macroscopic limit equation.

In the particle-based simulation, transporters are represented as individual particles that undergo Brownian motion with diffusion coefficient D for times in the time interval $[0, T_{\max}]$. This diffusion coefficient is determined based on chosen parameters for the Markovian Velocity-Jump Model via an algorithm that approximates the analytical expression given in Equation (47). We will introduce the algorithm in Section 7.4.1.

The carriers are initially sampled uniformly in a square domain $\Omega = [0, L]^2$. This initial distribution is used to construct a spatial density field $u(\mathbf{x}, 0)$. In the simulations, the transporters are then no longer represented as individual particles, but only through this continuum density, which evolves by diffusion in the PDE model described in Equation (80). This PDE is solved on a scaled domain in (χ, τ) coordinates as defined in Equation (73). The hitchhikers are initially initiated in the immobile state. In each time step, each hitchhiker can attach or detach, with the random variables of the run and rest durations following exponential distributions. The algorithm drawing these durations and updating the hitchhikers' states is discussed in Section 7.4.3.

The PDE given in Equation (80) will be solved numerically. The numerical scheme used will be discussed in Section 7.4.2. The domain of the stochastic simulation is scaled down to the same domain for comparison, which will be further discussed in Section 7.4.4. The metric used to determine the correspondence between the simulation and the PDE will be introduced in Section 7.4.5. The results of this comparison study are given in Section 7.4.6.

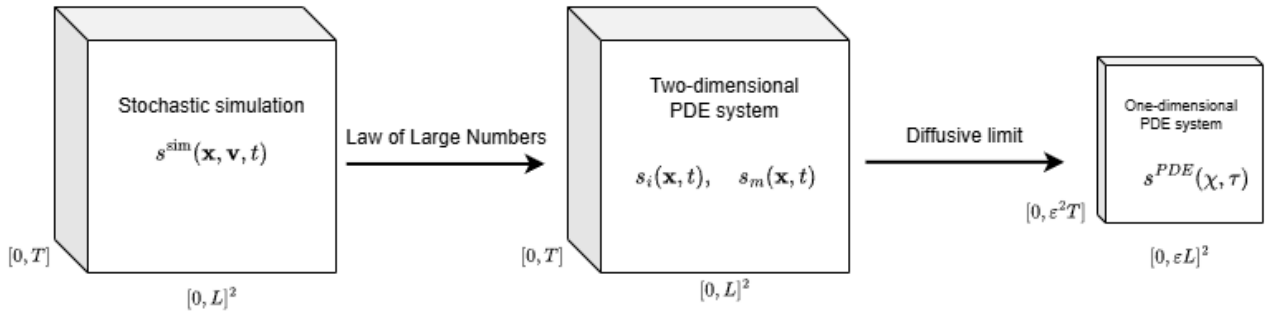


Figure 5: A schematic description of the approximation-workflow of the interactive Hitchhikers' model described in Section 7.1 in two spatial dimensions. The arrows indicate via what mechanism the stochastic process is approximated. On the domain $[0, L]^2$, the particles in the stochastic simulation are modelled by density functions s_m and s_m via the Law of Large Numbers described in Section 2.2. In the numerical simulations, the hitchhikers' velocities are no longer kept track of, and only the spatial evolution is tracked on the time domain $[0, T]$. Via the derivation described in Section 7 and Equation (80), these densities are combined to describe the complete hitchhiker population. A PDE approximates this density's evolution over time. This PDE is solved on a parabolically scaled domain, on the spatial domain $[0, \varepsilon L]^2$ and the temporal domain $[0, \varepsilon^2 T]$. After the simulation is performed on the original scales, the domains are scaled down to $[0, \varepsilon L]^2 \times [0, \varepsilon^2 T]$ in order to compare the resulting hitchhiker densities resulting from the stochastic simulation and the macroscopic PDE.

7.4.1 Numerical estimation of the diffusion tensor

Equation (80) was obtained by scaling the system given in Equation (68) by the diffusion coefficient. This constant was not prescribed in the simulations, but calculated based on the model parameter λ, which represents the mean duration of a carrier's run. This calculation is performed by numerically approximating Equation (47). This is done by discretising the velocity space and approximating the integral by a finite sum. For the pseudo-inverse $\mathcal{F}$ in Equation (47), the Moore-Penrose inverse is used [2]. The turning kernel $T(\mathbf{v}, \mathbf{v}')$ for the velocity-jump process is uniform. That is, a particle moving with velocity $\mathbf{v}'$ has an equal probability of taking on any new velocity $\mathbf{v}$ after reorientation. Because of this, the resulting diffusion tensor is isotropic, meaning all directions contribute equally to the diffusive spread. Consequently, in n dimensions, the tensor computed from the discrete approximation

$$D \approx -\frac{1}{K}\sum_{i=1}^{K}\sum_{j=1}^{K}\mathcal{F}_{ij}(\mathbf{v}_i \cdot \mathbf{v}_j)$$

must be proportional to the scaled identity matrix

$$D = D_{\text{phys}} I_n,$$

where D_{phys} is the scalar diffusion coefficient we aim to obtain. In n spatial dimensions, the trace of an isotropic tensor D then satisfies

$$\operatorname{tr}(D) = D_{\text{phys}} \operatorname{tr}(I_n) = nD_{\text{phys}},$$

so in two dimensions, we estimate the scalar diffusion coefficient by $D_{\text{phys}} = \frac{1}{2}\operatorname{tr}(D)$. We use this feature in the numerical procedure summarised in Algorithm 1. The velocity space is taken to be the annular disk $\{\mathbf{v} \in \mathbb{R}^2 : s_{\min} \leq ||\mathbf{v}||_2 \leq s_{\max}\}$. In this disk, K velocities are randomly sampled, producing a set $\{\mathbf{v}_i\}_{i=1}^{K}$ on which an estimation for the diffusion coefficient is based. The values for $K, s_{\min}$ and $s_{\max}$ defining the velocity space and the numerical procedure are given in Table 1. The used value for model parameter λ is given in Table 3 in Section 7.4.6, where the specifics of the performed simulation study are discussed.

Algorithm 1 Numerical computation of the diffusion coefficient D_{phys} in two dimensions with a uniform turning kernel, based on the analytical expression in Equation (47).

Require: $K \geq 0, \lambda, s_{\min}, s_{\max} > 0$
Ensure: D_{phys}
  Construct angles $\theta_i = \frac{2\pi i}{K}$
  Draw radii r_i uniformly in $[s_{\min}^2, s_{\max}^2]$
  Define initial velocities $\mathbf{v}_i = r_i(\cos\theta_i, \sin\theta_i)$
  Construct turning matrix $T_{ij} = \frac{1}{K}$
  Define operator $\mathcal{L} = -\lambda I_2 + \lambda T$
  Compute pseudo-inverse $\mathcal{F}$
  Initialize $D \leftarrow 0$
  for $i = 1, \ldots, K$ **do**
    for $j = 1, \ldots, K$ **do**
      $D \leftarrow D + F_{ij}(\mathbf{v}_i \cdot \mathbf{v}_j)$
    end for
  end for
  $D \leftarrow -\frac{D}{K}$
  $D_{\text{phys}} \leftarrow \frac{1}{2}\operatorname{tr}(D)$

7.4.2 Numerical discretization of the PDE system

The simulation study requires numerically solving two partial differential equations according to a numerical scheme we will outline in this section. One of the PDEs that has to be solved is the one

Table 1: The values for the parameters used in Algorithm 1 to estimate the diffusion coefficient given in Equation 47.

Parameter	Description	Value
K	Number of velocities to sample	2000
$s_{\min}$	Minimal carrier speed	0
$s_{\max}$	Maximal carrier speed	2

describing the carrier population. Every carrier is individually initiated on the grid. This initial distribution is then used to construct a density $u(\mathbf{x}, 0)$, which evolves according to the diffusion equation given in Equation (65). Since the transporters are initially uniformly distributed, u is approximately spatially homogeneous throughout the experiment, up to small numerical fluctuations. The total population of hitchhikers is modelled to evolve according to the macroscopic limit equation given in Equation (80). In this equation, we observe that the evolution of this density is dependent on the local density u of carriers via the rate parameter $\mu(u)$ that occurs in the equation for $\alpha(\chi, \tau)$. We will use $\mu(u)u = \mu_0 u$, and thereby assume that the effect of the local carrier density grows linearly with a rate constant μ_0. This equation is solved to obtain a density we will denote as s^{PDE}. This density is then compared to the density resulting from a stochastic simulation, which we will refer to as s^{sim}.

After nondimensionalising, the spatial domain becomes $[0, 1]^2$ and time is rescaled according to $\tau = \kappa t$ with $\kappa = \frac{D}{L^2}$. The hitchhiker density is then governed by the PDE

$$\partial_\tau s = -\nabla_\chi \cdot (-\alpha D \nabla_\chi s + (-D \nabla_\chi \alpha) s), \qquad \alpha(u) = \frac{\mu_0 u}{\nu + \mu_0 u}.$$

This PDE is advanced on the time interval $[0, \kappa T_{\max}]$ using an explicit finite-difference scheme with homogeneous Neumann boundary conditions. The transporter density $u(\chi, \tau)$ evolves according to Equation (65), using the scheme and boundary conditions. A schematic workflow of the approximation process including the numeric PDE solver is depicted in Figure 6.

As both PDEs are solved using the same scheme, we will give the numerical scheme for a generic scalar field $w(\mathbf{x}, t)$ defined on a two-dimensional rectangular spatial domain $\Omega = [0, L_x] \times [0, L_y]$. We let $\mathbf{x} = (x, y)^\top$. All results mentioned in this section are based on Chapter 7.2 of Vuik et al. [48]. The parameters for this discretization are given in Table 2.

We introduce a uniform Cartesian grid by

$$x_i = i\,\Delta x, \qquad i = 0, \ldots, N_x - 1, \qquad y_j = j\,\Delta y, \qquad j = 0, \ldots, N_y - 1,$$

with

$$\Delta x = \frac{L_x}{N_x - 1}, \qquad \Delta y = \frac{L_y}{N_y - 1}.$$

The temporal grid is given by

$$t^n = n\,\Delta t.$$

In the simulations, Δt was chosen conservatively as

$$\Delta t = \frac{(\min\{\Delta x, \Delta y\})^2}{8}.$$

Table 2: The values for the numerical scheme used to perform the simulation of the stochastic hitchhiker system described in Section 7.4.3 and the macroscopic limit equation given in Equation (80). The model parameters used alongside these numerical parameters are given in Table 3.

Parameter	Description	Value
L_x	Width of the rectangular domain	10
L_y	Length of the rectangular domain	10
N_x	Number of divisions on the x-axis	150
N_y	Number of divisions on the y-axis	150

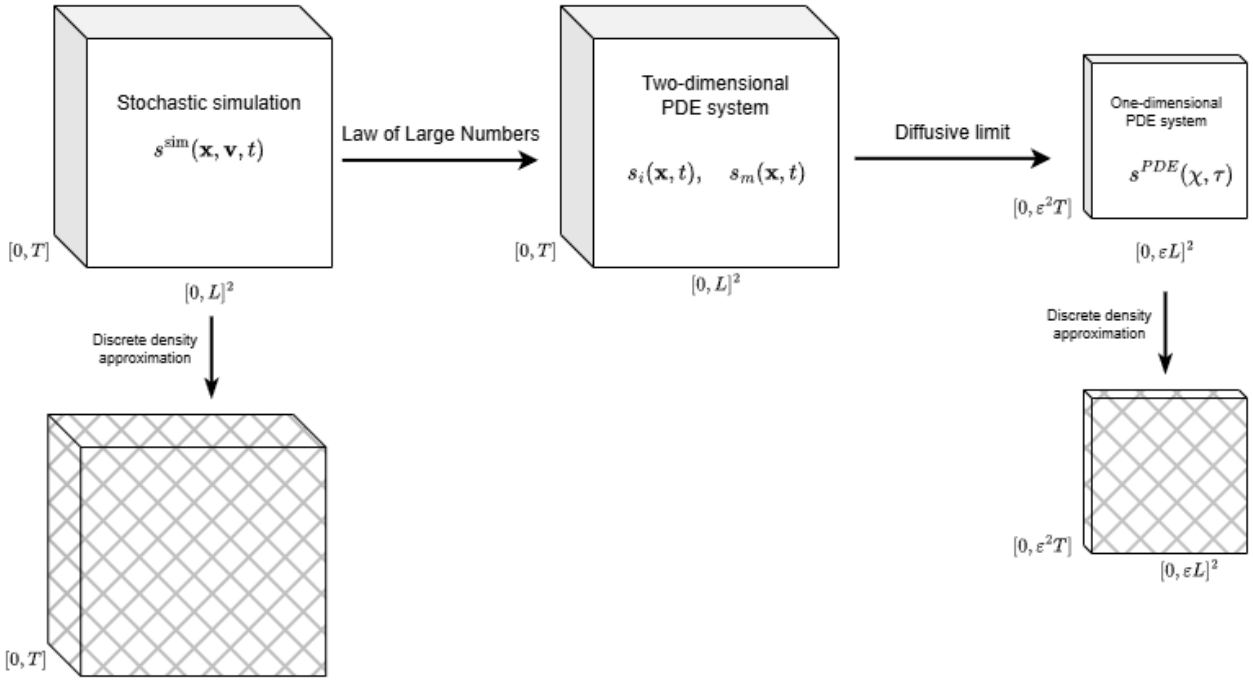


Figure 6: An extension of the workflow depicted in Figure 5, appropriated for the numerical simulations. The arrows indicate via what mechanism the source model is approximated by the target model. Via the Formal diffusion limit described in Section 7 and the Law of Large Numbers described in Section 2.2, Equation (80) approximates the particle-based Hitchhikers' Model. The total density of the hitchhikers is compared between the simulations and the solution of the one-dimensional PDE system. To this end, a discretization of both the regular and the scaled grid is applied. The discretization is imposed by dividing the two-dimensional area into smaller squares, with the coarseness defined by the parameters given in Table 2. Then, the finite difference scheme given in Equation (82) was imposed. The densities s^{sim} and s^{PDE} are calculated as the average density in each grid square.

We denote the discrete approximation at grid point (x_i, y_j) and time level t^n by

$$w_{i,j}^n \approx w(x_i, y_j, t^n).$$

For a PDE of the form

$$\frac{\partial w(\mathbf{x}, \mathbf{v}, t)}{\partial t} = f(w),$$

the numerical update is given by the forward Euler scheme

$$w_{i,j}^{n+1} = w_{i,j}^n + \Delta t \, f_h(w^n)_{i,j},$$

where f_h denotes the finite difference approximation of f. In the present setting, the main spatial operator is a diffusion-type operator.

For the standard Laplace operator, we use the second-order centered stencil

$$(\Delta_h w)_{i,j} = \frac{w_{i+1,j} - 2w_{i,j} + w_{i-1,j}}{\Delta x^2} + \frac{w_{i,j+1} - 2w_{i,j} + w_{i,j-1}}{\Delta y^2}. \tag{82}$$

For a smooth function w, Taylor expansion shows that

$$(\Delta_h w)_{i,j} = (\Delta w)(x_i, y_j) + \mathcal{O}(\Delta x^2 + \Delta y^2),$$

so the spatial discretization is second-order accurate. Combining this with the first-order time discretization yields a local truncation error of order

$$\tau_{i,j}^n = \mathcal{O}(\Delta t + \Delta x^2 + \Delta y^2).$$

Thus, the method is first-order accurate in time and second-order accurate in space.

The PDE system is solved with reflecting boundary conditions, corresponding to homogeneous Neumann conditions. These conditions express zero flux across the boundary. This choice is consistent with the reflecting boundary conditions in the particle simulation: particles that reach the boundary are reflected back into the domain, so no mass is lost. Equivalently, ghost points are introduced by copying the nearest interior value across the boundary. In the x-direction, for instance, one sets ghost values so that

$$w_{-1,j} = w_{1,j}, \qquad w_{N_x,j} = w_{N_x-2,j},$$

and analogously in the y-direction. Equivalently, at the left boundary,

$$\frac{w_{1,j} - w_{-1,j}}{2\Delta x} = 0,$$

so the discrete normal derivative vanishes. This ensures that the numerical flux through the boundary is zero and that mass is conserved, up to discretization error.

7.4.3 The particle-based stochastic simulation

While the PDE model is numerically solved, a particle-based simulation is performed in parallel. In this simulation, hitchhikers are modeled as individual particles that switch stochastically between two states: an immobile state, in which they remain fixed in space, and a mobile state, in which they undergo Brownian motion with the same diffusion coefficient D_{phys} as the transporters, which is determined according to the method described in Section 7.4.1. In the simulations, the hitchhikers' initial positions are initiated according to a two-dimensional normal distribution in the centre of the domain Ω with variance $\frac{1}{2}$. A detached hitchhiker attaches with rate $\mu_0 u(\mathbf{x}, t)$, while detachment occurs with constant rate ν. In the numerical implementation, attachment events are sampled with a Bernoulli approximation over a time step Δt, using the probability $1-\exp\big(-\mu_0 u(\mathbf{x}, t)\Delta t\big)$. Detachment times are sampled from exponential waiting-time distributions with parameter ν. At each time step, the positions of all hitchhikers are recorded, and an empirical density $s^{\text{sim}}(\mathbf{x}, t)$ is constructed by binning the particle positions onto a uniform spatial grid. This procedure is summarised in Algorithm 2.

7.4.4 Coupling of the stochastic simulation and PDE solution

The macroscopic limit given by Equation (80) is compared to the particle-based simulation by quantifying the agreement between the density of the hitchhikers in the two models over time. The particle simulation advances in physical time t, while the PDE is advanced in the nondimensional diffusive time $\tau = \kappa t$ for $\kappa = \frac{D}{L^2}$. To ensure consistency between the stochastic simulation and the PDE model, both are evolved in a coupled manner: for each simulation time step Δt, the PDE is advanced in smaller steps of size $\Delta\tau$ until the corresponding diffusive time $\tau = \frac{D}{L^2}t$ is reached. The particle positions are nondimensionalized according to $\chi = \frac{\mathbf{x}}{L}$, so that both the particle density and the PDE are represented on the domain $[0, 1]^2$. This coupling is also explained in Algorithm 2.

Algorithm 2 Hybrid simulation of the Markovian Hitchhikers' Model and PDE comparison. For the stochastic hitchhiking dynamics, transporters diffuse and hitchhikers attach and detach according to Poisson processes. Mobile hitchhikers undergo Brownian motion with diffusion coefficient D_{phys}, which is calculated with Algorithm 1. The macroscopic PDE given in Equation (80) is evolved on the scaled domain. At each physical time step, the hitchhiker positions are binned to obtain the empirical density s^{sim}, which is compared to the PDE solution s^{PDE}.

Require: $N_t, N_h, L, T_{\max}, \nu, \mu_0, D_{\text{phys}}$
Ensure: s^{sim}, s^{PDE}

Initialise transporter density $u(\mathbf{x}, 0)$ from uniformly distributed transporters
Initialise hitchhiker positions from a two-dimensional normal distribution
Draw detachment clocks $J_h^{\text{off}} \sim \text{Exp}(\nu)$
Set $t \leftarrow 0$, $\tau \leftarrow 0$, $\kappa \leftarrow D/L^2$
while $t < T_{\max}$ **do**
  Particle update
  Move all motile hitchhikers by one Brownian step with diffusion coefficient D
  Reflect particles at the boundary of $\Omega = [0, L]^2$
  Decrease detachment clocks by Δt
  Switch expired mobile particles to immobile
  for each immobile hitchhiker h **do**
    Interpolate local transporter density $u(\mathbf{x}_h, t)$
    Set $r_h^{\text{on}} \leftarrow \mu_0 u(\mathbf{x}_h, t)$
    Attach with probability $1 - \exp(-r_h^{\text{on}} \Delta t)$
    if h attaches **then**
      Sample a new detachment clock $J_h^{\text{off}} \sim \text{Exp}(\nu_{\text{off}})$
    end if
  end for
  Advance physical time: $t \leftarrow t + \Delta t$
  PDE update
  Set $\tau_{\text{target}} \leftarrow \kappa t$
  while $\tau < \tau_{\text{target}}$ **do**
    Update u by diffusion with homogeneous Neumann boundary conditions
    Compute $\alpha(\chi, \tau) \leftarrow \dfrac{\mu_0 u}{\nu_{\text{off}} + \mu_0 u}$
    Update s^{PDE} by

$$\partial_\tau s = \nabla \cdot \big(\nabla(\alpha s)\big)$$

    Advance $\tau \leftarrow \tau + \Delta\tau$
  end while
  Bin hitchhiker positions onto the grid to obtain $s^{\text{sim}}(\mathbf{x}, t)$
  Scale the hitchhiker density to $s^{\text{sim}}(\chi, \tau)$
end while

7.4.5 Density comparison

The agreement between the particle simulation and the deterministic limit is quantified by comparing the hitchhiker density over time. The difference between the particle-based density s^{sim} and the PDE solution s^{PDE} is measured in the discrete $L^2(\Omega)$ norm, as given in Equation (2). Specifically, we get

$$E_2(t) = \left(\int_\Omega \left(s^{\text{sim}}(\mathbf{x}, t) - s^{\text{PDE}}(\mathbf{x}, t) \right)^2 \, d\mathbf{x} \right)^{1/2}$$

The relative error, which is a scale-independent metric, is then calculated by normalizing the error by the L^2-norm of the PDE solution:

$$E_{2,r}(t) = \frac{E_2(t)}{||s^{PDE}(\cdot, t)||_2}. \tag{83}$$

Both densities are represented on a uniform Cartesian grid with mesh widths Δx and Δy. The integrals are therefore approximated via the corresponding Riemann sums:

$$\tilde{E}_{2,r}(t) = \frac{\left(\sum_{i=1}^{N_x} \sum_{j=1}^{N_y} (s_{ij}^{\text{sim}} - s_{ij}^{\text{PDE}})^2 \right)^{\frac{1}{2}}}{\left(\sum_{i=1}^{N_x} \sum_{j=1}^{N_y} (s_{ij}^{\text{PDE}})^2 \right)^{\frac{1}{2}}}, \tag{84}$$

where s_{ij}^{sim} and s_{ij}^{PDE} denote the average densities resulting from the simulation and the PDE solver in the grid square centred at coordinates (x_i, y_j) on the scaled domain.

7.4.6 Parameters in the simulation study

The parameters used in the simulation study are reported in Table 3. These values are chosen such that the diffusive limit holds for the system as given in Equation (72). In particular, the switching dynamics between motile and immobile states occur on a faster timescale than the macroscopic spatial redistribution of the hitchhikers. The number of hitchhikers is based on the numerical grid of the stochastic simulation, which consists of 150^2 grid points. The range of the number of transporters is chosen to be of the same order of magnitude so that the densities in the diffused limit are of $\mathcal{O}(1)$.

7.4.7 Results of the simulation study

The results of the simulations as described in Algorithm 2 for 50000 hitchhikers are shown in Figure 7. In this figure, we can compare the numerical solution of the PDE with the density obtained from the stochastic simulation at several time points. Overall, the two solutions show good qualitative agreement at all depicted time points. Compared with the simulation using 50000 hitchhikers, the results for 10000 and 25000 hitchhikers exhibit the same qualitative behaviour. Due to the smaller

Table 3: The values for the parameters used in the simulation study described in Section 7.4, where the stochastic process described in Algorithm 2 is compared to the numeric solution of the PDE given in Equation (68). The value of D is derived by approaching Equation (47) via Algorithm 1. The number of hitchhikers was varied. The grid parameters are given in Table 2.

Parameter	Description	Value
λ	Reorientation rate transporters	0.25
ν	Detachment rate hitchhikers	800
μ_0	Attachment rate coefficient hitchhikers	400
ε	Scaling parameter	0.1
D	Calculated diffusion coefficient of the transporters	3.88
$T_{\max}$	Duration of the simulation study	7.5
N_t	Number of transporters	50000
N_h	Number of hitchhikers	{10000, 25000, 50000}

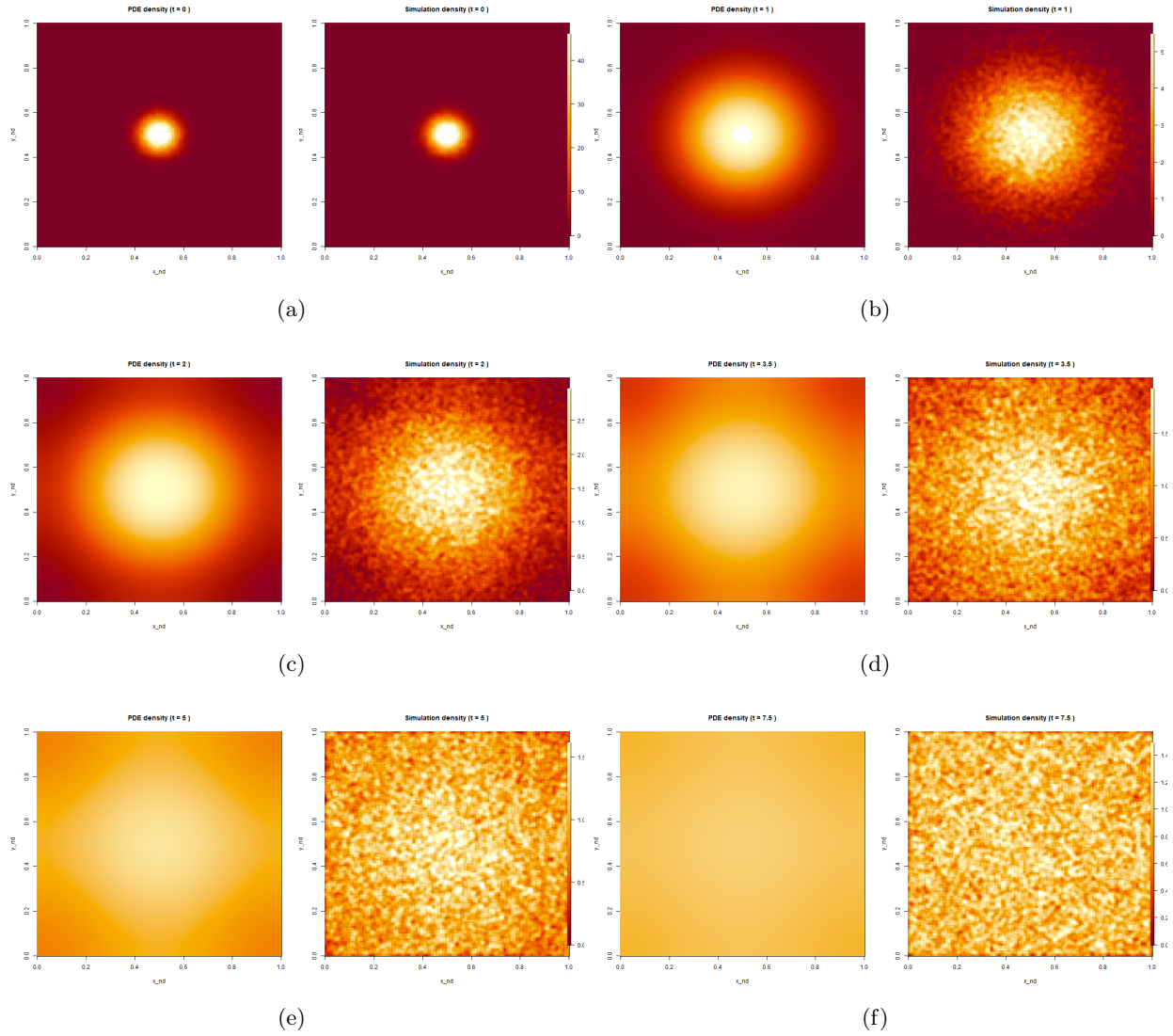


Figure 7: A comparison between the density of the hitchhikers population as a result of the macroscopic PDE limit (left) and the particle-based stochastic simulation (right) for time points (a) 0.01 (b) 1.00 (c) 2.00 (d) 3.50 (e) 5.00 (f) 7.50. The simulation was performed according to Algorithm 2 with the parameters as reported in Table 3. In the simulation for these figures, the number of hitchhikers was 50000.

number of particles, they display larger fluctuations. However, the overall density profile remains unchanged. The figures resulting from these simulations can be accessed via the link given in Appendix B.

The result of the numerical calculation of the relative L^2-error for various values of the number of hitchhikers is given in Figure 8. The error was calculated according to Equation (84). As expected from the Law of Large Numbers, the relative error decreases as the number of hitchhikers increases. This indicates that the model approximation is more accurate for higher particle densities. The error curves exhibit noisiness because each curve is obtained from a single simulation. Averaging the error over multiple independent simulations would reduce this variability and provide a smoother estimate of the expected relative error.

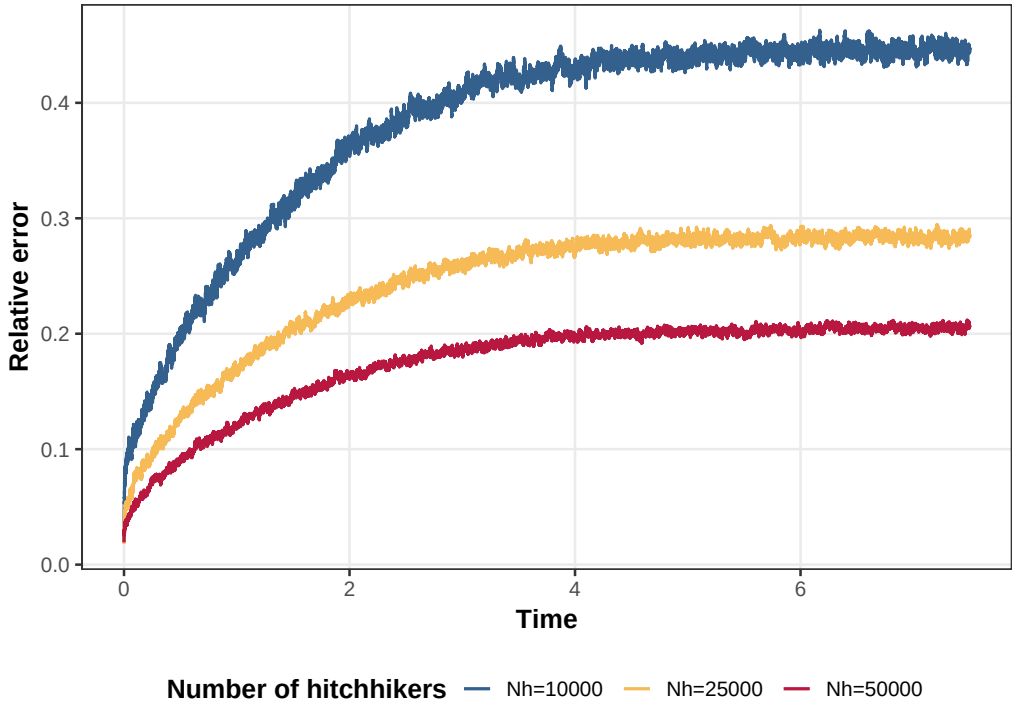


Figure 8: The relative error of the approximation of the hitchhiker density resulting from the stochastic Hitchhiking process and the deterministic PDE given in Equation (80) for 10000, 25000 and 500000 hitchhikers. A single simulation run was performed as described in Algorithm 2 with the parameters given in Table 3. The plotted error is calculated via Equation (84).

8 Mathematical background for moment equations

In Section 7, we derived a deterministic macroscopic limit equation that approximates the Markovian Hitchhikers' Model. In Section 9, we will introduce a different approximation method for this model, resulting in moment equations that capture the individual-based stochastic dynamics of the hitchhiking model with ordinary differential equations. In this section, we will first discuss moment equations in general and then consider some properties of the Mean Squared Displacement, which we will eventually use as an indicator for the accuracy of our new model approximation.

8.1 Moment equations

Moment equations summarise the main traits of a dynamic process and often have a direct physical interpretation, such as total mass, mean position, or mean velocity. Rather than tracking the full density on the underlying state space, they produce a small set of coupled differential equations that capture the essential macroscopic behaviour of the system.

In probability theory, the k-th moment of distribution of a random variable X is defined as the expected value $\mathbb{E}(X^k)$. We can apply the same idea to a density $p(\mathbf{x}, \mathbf{v}, t)$ in the Velocity-Jump model on a state space $Z = \Omega \times V$ by defining the expected value of some entity on Z, weighted by the density p. We assume that, for each t on a time interval $[0, T]$, the density is integrable over Z and that the moments considered below are finite. When $p(\cdot, \cdot, t)$ is normalized to unit mass, it may be interpreted as a probability density. In that case the moments coincide with expectations in the sense of probability theory.

For a test function $\varphi \in C^1(\overline{\Omega} \times \overline{V})$ that is measurable and integrable against the density function, we define the moment

$$M_\varphi(t) = \int_Z \varphi(z)\, p(\mathbf{x}, \mathbf{v}, t)\; dz.$$

Different choices of the test function correspond with different macroscopic quantities. For example, taking $\varphi \equiv 1$ yields the total mass, while other test functions, such as $\varphi(\mathbf{x}, \mathbf{v}) = \mathbf{x}$, $\varphi(\mathbf{x}, \mathbf{v}) = \mathbf{v}$, $\varphi(\mathbf{x}, \mathbf{v}) = \|\mathbf{x}\|^2$ or $\varphi(\mathbf{x}, \mathbf{v}) = \|\mathbf{v}\|^2$, produce first and second spatial or velocity moments.

We will work with spatial moments, velocity moments and mixed moments. This is part of a standard reduction strategy, in which a finite moment space is chosen. In general, the equation for a given moment depends on moments of higher order, leading to an infinite hierarchy of coupled equations. To obtain a finite-dimensional system, a finite collection of moments is selected and the remaining higher-order moments are approximated in terms of the selected ones. This procedure is known as moment closure and is a standard reduction technique in kinetic theory [26]. The choice of closure determines the accuracy and structural properties of the reduced moments system. A suitable closure should preserve essential features of the original system, such as mass conservation.

Moment methods are especially valuable as moments have direct physical interpretation and often evolve on the macroscopic time scale of interest. In this thesis, we employ moment equations to derive a macroscopic description of the transport process from its underlying particle-based formulation. This approach reduces the dimensionality of the problem while retaining the dominant transport behaviour. In this context, the moment method therefore act as a bridge between microscopic and macroscopic equations. Additionally, this method reduces computational costs by approximating a high-dimensional density equation with a lower-dimensional system.

8.2 Mean Squared Displacement (MSD)

A moment that is of particular interest, is the spatial second-order moment of the displacement probability distribution, which measures the spread of a particle over time by calculating the average squared distance from its initial position. Therefore, this moment is referred to as the Mean Squared Displacement (MSD). The MSD can be determined using the average over many particles or, in situations where it is difficult to obtain large-scale observations of many particles, it is convenient to

characterize the macroscopic behavior through a reduction to a single-particle observation. A time-averaged variant of the MSD allows for this. We will study both versions of the MSD and how the MSD can be used to determine the type of diffusion regime undergone by the particles in question.

8.2.1 The ensemble MSD for multiple particles

If data on the movement of many particles is available, the ensemble average displacement can be used to construct an expression for the Mean Squared Displacement. Suppose we observe $N \in \mathbb{N}$ particles at various time steps in a space in $\mathbb{R}^n$. The trajectory of each particle is modelled as a stochastic process, denoted as

$$X^{(i)} = (X_m^{(i)})_{m \in \mathbb{N}}$$

The observed position of the i-th particle after time step m is denoted as $\mathbf{x}_i(m)$. Then, fixing time zero, the displacement of the particles after the m-th step is given by

$$\Delta \mathbf{x}_i(m) = \mathbf{x}_i(m) - \mathbf{x}_i(0).$$

Using this, the empirical mean squared displacement after n steps is then given by

$$\widehat{MSD}_N(m) := \frac{1}{N} \sum_{i=1}^{N} ||\Delta \mathbf{x}_i(m)||^2,$$

where $||\cdot||$ denotes the Euclidean norm in $\mathbb{R}^n$. Assuming that the observed particles are independent and identically distributed, and $\mathbb{E}\left(||X(m) - X(0)||^2\right) < \infty$ holds, the Law of Large Numbers as discussed in Section 2.2 gives

$$\widehat{MSD}_N(m) \xrightarrow{N \to \infty} MSD(m) := \mathbb{E}\left(||X(m) - X(0)||^2\right). \tag{85}$$

Hence the empirical MSD converges to the ensemble MSD, which is the second moment of the displacement distribution.

8.2.2 The time-averaged MSD for single-particle tracking

In many experimental settings, observing a large collection of independent particles is not feasible. When it is not possible to observe multiple particles, a more useful viewpoint is to use a single long trajectory and average over time lags. In this case, the ensemble average in Equation (85) cannot be computed directly and is replaced by a time average along one trajectory. We will use the method described by Michalet [28]. In contrast to the ensemble estimator, which averages over many independent particles at a single time point, the time-averaged estimator averages over many time windows of a single trajectory.

Suppose a single particle is observed on the time interval $[0, T]$. We denote its trajectory on this interval as $X = (X_t)_{0 \le t \le T}$. For a fixed time lag $\tau > 0$, the time-averaged Mean Squared Displacement is calculated by averaging the squared displacement over all pairs of observations separated by the time lag τ, which we can write as the time-averaged MSD:

$$\widehat{MSD}_T(\tau) = \frac{1}{T - \tau} \int_0^{T-\tau} ||X(s + \tau) - X(s)||^2 \, ds.$$

When observations are available only at discrete sampling times $t_n = n\Delta t$ for $n \in \{0, 1, \ldots, M-1\}$ for some $M \in \mathbb{N}_{>1}$, the integral is approximated by the Riemann sum

$$\widehat{MSD}_M(m) = \frac{1}{M - m} \sum_{n=0}^{M-m-1} ||X_{n+m} - X_n||^2,$$

where m is the lag measured in sampling intervals. That is, we have $\tau = m\Delta t$.

Under ergodicity assumptions in the mean-squared sense, the time average converges to the expected squared displacement as the observation time increases. In particular, for every fixed time lag τ, for $T \to \infty$, we get

$$\widehat{MSD}_T(\tau) \to \mathbb{E}\left(||X(t+\tau) - X(t)||^2\right).$$

If the process has stationary increments, the distribution of the increment $X(t+\tau) - X(t)$ is independent on the starting time t. We then get

$$\mathbb{E}\left(||X(t+\tau) - X(t)||^2\right) = \mathbb{E}\left(||X(\tau) - X(0)||^2\right) = MSD(\tau). \tag{86}$$

Hence, for processes with stationary increments, the time-averaged estimator converges to the same quantity as the ensemble MSD.

In this thesis, the ensemble MSD is computed from simulations of many independent particles. To investigate whether a single tracked particle is sufficient to recover the macroscopic transport behaviour, we additionally compute the time-averaged MSD from one trajectory. Rather than fixing the total observation time, we evaluate the estimator on progressively longer trajectory segments [0,t]. This yields the family of estimators given by

$$\widehat{MSD}_t(\tau) = \frac{1}{t-\tau}\int_0^{t-\tau} ||X(s+\tau) - X(s)||^2 \, ds, \tag{87}$$

which can be compared directly with the ensemble MSD. As t increases, more displacement measurements contribute to the average, reducing the fluctuations. This allows us to study how long a single particle's trajectory must be observed before it provides a reliable approximation of the ensemble behaviour.

8.2.3 The MSD for diffusive processes

The MSD is an interesting moment to study, as for Brownian diffusion, the MSD grows linearly in time. This forms the basis for estimating diffusion coefficients from trajectory data and for distinguishing normal diffusion from anomalous diffusion. We will formally derive this property of the Mean Squared Displacement.

Theorem 8.1. *For ordinary Brownian motion in $\mathbb{R}^n$, the Mean Squared Displacement (MSD) is given by $MSD(t) = 2nDt$.*

Proof. To describe Brownian Motion, we use the definition as given in Chapter 7 in the book by Durrett [11].

Let $B = (B_t)_{t\geq 0}$ be a standard Brownian motion trajectory in $\mathbb{R}^n$. This means the following properties hold:

(P1) $B_0 = 0$ almost surely. That is, $\text{Prob}(B_0 = 0) = 1$.

(P2) The paths given by $t \mapsto B_t$ are continuous almost surely.

(P3) B has independent increments. That is, for every $s \in [0, t)$, the increment $B_t - B_s$ is independent of previous increments.

(P4) B has Gaussian increments. For all $u, t \geq 0$, we have $B_{t+u} - B_t \sim \mathcal{N}(0, u)$. That is to say, a time step u from t results in a normally distributed increment with mean 0 and variance u.

This means that for $B_t = (B_t^{(1)}, \ldots, B_t^{(n)})$, each coordinate $B_t^{(i)}$ is a one-dimensional Brownian motion process with

$$B_t^{(i)} \sim \mathcal{N}(0, t), \quad \mathbb{E}\left[B_t^{(i)}\right] = 0, \quad \mathbb{E}\left[\left(B_t^{(i)}\right)^2\right] = t. \tag{88}$$

We define the diffusion coefficient $D > 0$ such that the variance after time t in one dimension is $2Dt$. A diffusion process with the diffusion coefficient D is then modelled as

$$X_t = X_0 + \sqrt{2D}B_t.$$

From this, we calculate

$$||X_t - X_0||^2 = 2D||B_t||^2,$$

and so

$$\begin{aligned}
\mathbb{E}\left[||X_t - X_0||^2\right] &= 2D \cdot \mathbb{E}\left[||B_t||^2\right] \\
&= 2D \cdot \mathbb{E}\left[\sum_{i=1}^{n} \left(B_t^{(i)}\right)^2\right] \\
&= 2D \sum_{i=1}^{n} \mathbb{E}\left[\left(B_t^{(i)}\right)^2\right] \\
&= 2Dnt,
\end{aligned}$$

where the second equality follows from the linearity of the expectation, and the last equality follows from Equation (88).

As shown in Equation (85), we have $MSD(t) = \mathbb{E}\left[||X_t - X_0||^2\right]$, so indeed, the claim holds. We will use this result to verify the diffusive limit of the Hitchhikers model and its corresponding effective diffusion constant. □

In particular, the MSD can be fitted to a power law of the form $MSD(t) \propto t^\alpha$. The exponent α indicates the diffusion type of the system that is studied [25]. For $\alpha = 1$, there is a normal diffusion regime, as in accordance with Theorem 8.1. For $\alpha < 1$, the diffusion type is called subdiffusion, which indicates that particles may be hindered or trapped in their environment. For $\alpha > 1$, the process exhibits superdiffusion, meaning that particles spread faster than in the case of Brownian motion. This may be caused by directed transport.

9 Moment equations for the Markovian Hitchhikers' Model

We will now construct a system of moments equations for the Markovian Hitchhikers' Model given in Equation (67). Note that we now no longer assume that the hitchhikers are moving according to the diffusion equation, as we did when deriving the diffusive limit equation in Section 7. Therefore, an equation for the spread of the carrier density in space and time should also be taken into account. In n spatial dimensions, the Hitchhikers' Model then consists of three coupled delay-integro-partial differential equations with $(2n + 1)$ degrees of freedom, which makes the model complex to work with. In order to facilitate the interplay between experiments and modelling, we follow the set-up presented by Taylor-King et al. [47] to obtain a closed system of ordinary integro-differential equations. As discussed in Section 8, these moments are directly interpretable as measurable statistics that summarize the transport dynamics over time. In particular, we can use the moment equations for the Hitchhikers' Model to find an equation for the Mean Squared Displacement. To this end, we first make several assumptions on the turning kernel T in addition to Assumptions (T1) - (T6) listed in Section 4.1.

Firstly, we assume that the turning angle of the reorientating particles is on average zero. That is to say, when switching velocities, the average post-turn velocity $\overline{\mathbf{v}}(\mathbf{v}')$ has the same orientation as the pre-turn velocity $\mathbf{v}'$. We define the index of persistence ψ as the following scalar:

$$\overline{\mathbf{v}}(\mathbf{v}') = \int_V \mathbf{v} T(\mathbf{v}, \mathbf{v}') \, d\mathbf{v} = \psi \mathbf{v}'.$$

Note that for isotropic diffusion, we have $\overline{\mathbf{v}}(\mathbf{v}') = 0$ for all $\mathbf{v} \in V$. In this specific case, all $\psi \in \mathbb{R}$ suffice, but we take $\psi = 0$ when substituting for $\int_V \mathbf{v} T(\mathbf{v}, \mathbf{v}') \, d\mathbf{v}$.

The second assumption we make, is that the average mean squared speed $S_T^2(\mathbf{v}')$ is constant. That is,

$$S_T^2(\mathbf{v}') = S_T^2 = \int_V ||\mathbf{v}||^2 T(\mathbf{v}, \mathbf{v}') \, d\mathbf{v}.$$

The interpretation of this assumption is that the post-turn speed is independent of the pre-turn speed. In other words, the turning kernel T is memoryless with regard to speed.

Lastly, we assume conservation of mass in the system. That is, we have

$$N_{s_m}(t) + N_{s_i}(t) = N \in \mathbb{N},$$

where $N_{s_m}(t)$ is the number of particles in the mobile state and $N_{s_i}(t)$ is the number of particles in the immobile state at time t .

We will now derive a closed system of moments, using the general definition given in Section 8.1. Let $\varphi = \varphi(\mathbf{x}, \mathbf{v})$ be a test function and ρ a density. We calculate the expected value of φ over the space $V \times \Omega$ at time t weighted by the distribution $\rho \in \{s_i, s_m\}$ as follows:

$$Q_\rho(\varphi, t) = \int_V \int_\Omega \varphi(\mathbf{x}, \mathbf{v}) \rho(\mathbf{x}, \mathbf{v}, t) \, d\mathbf{x} \, d\mathbf{v}.$$

We then define the following moments in space and time:

- $N_\rho(t) = Q(1, t)$ is the number of particles contributing to the density ρ at time t.
- $D_\rho^2(t) = Q_\rho(||\mathbf{x}||^2, t)$ is the mean squared displacement of particles contributing to the density ρ at time t.
- $B_\rho(t) = Q_\rho(\mathbf{v} \cdot \mathbf{x}, t)$ is the mean velocity-displacement of particles contributing to the density ρ at time t.
- $V_\rho^2(t) = Q_\rho(||\mathbf{v}||^2, t)$ is the mean squared velocity of particles contributing to the density ρ at time t.

A full derivation of the moments system is given in Appendix A. This derivation results in a system of seven ordinary differential equations for these observables:

$$\begin{cases} \frac{dD^2_{s_i}(t)}{dt} & = \nu D_{s_m}(t) - \int_V \int_\Omega ||\mathbf{x}||^2 \left(\mu(u) s_i(\mathbf{x},\mathbf{v},t)\right)\ d\mathbf{x}\ d\mathbf{v}, \\ \frac{dD^2_{s_m}(t)}{dt} & = 2B_{s_m}(t) - \frac{dD^2_{s_i}(t)}{dt}, \\ \frac{dB_{s_m}(t)}{dt} & = V^2_{s_m}(t) - \frac{d}{dt}B_{s_i}(t) + \lambda(\psi-1)B_{s_m}, \\ \frac{dB_{s_i}(t)}{dt} & = \nu B_{s_m}(t) - \int_V \int_\Omega (\mathbf{v}\cdot\mathbf{x})\mu(u) s_i(\mathbf{x},\mathbf{v},t)\ d\mathbf{x}\ d\mathbf{v}, \\ \frac{dV^2_{s_m}(t)}{dt} & = -\frac{dV^2_{s_i}(t)}{dt} + \lambda S_T^2 N_{s_m}(t) - \lambda V^2_{s_m}(t), \\ \frac{dV^2_{s_i}(t)}{dt} & = \nu V^2_{s_m}(t) - \int_V \int_\Omega ||\mathbf{v}||^2 \mu(u) s_i(\mathbf{x},\mathbf{v},t)\ d\mathbf{x}\ d\mathbf{v} \\ \frac{dN_{s_m}(t)}{dt} & = -\frac{dN_{s_i}(t)}{dt} = -\nu N_{s_m}(t) + \int_V \int_\Omega \mu(u) s_i(\mathbf{x},\mathbf{v},t)\ d\mathbf{x}\ d\mathbf{v}. \end{cases} \tag{89}$$

If expressions for $u(\mathbf{x},\mathbf{v},t)$ and $s_i(\mathbf{x},t)$ are known, this system becomes closed. We will study this system analytically while assuming that $\mu(u) = \mu$ is constant and then compare the expression for the Mean Squared Displacement in this system to the MSD reported in simulations.

9.1 The case of a homogeneous carrier density

We will study the model under the assumption that u is homogeneously distributed over the spatial field. Then, we have $\mu(u) =: \mu$, and we obtain the model

$$\begin{cases} \frac{dD^2_{s_i}(t)}{dt} & = \nu D_{s_m}(t) - \mu D^2_{s_i}(t), \\ \frac{dD^2_{s_m}(t)}{dt} & = 2B_{s_m}(t) - \frac{dD^2_{s_i}(t)}{dt}, \\ \frac{dB_{s_m}(t)}{dt} & = V^2_{s_m}(t) - \frac{d}{dt}B_{s_i}(t) + \lambda(\psi-1)B_{s_m}, \\ \frac{dB_{s_i}(t)}{dt} & = \nu B_{s_m}(t) - \mu B_{s_i}(t), \\ \frac{dV^2_{s_m}(t)}{dt} & = -\frac{dV^2_{s_i}(t)}{dt} + \lambda S_T^2 N_{s_m}(t) - \lambda V^2_{s_m}(t), \\ \frac{dV^2_{s_i}(t)}{dt} & = \nu V^2_{s_m}(t) - \mu V^2_{s_i}(t), \\ \frac{dN_{s_m}(t)}{dt} & = -\frac{dN_{s_i}(t)}{dt} = -\nu N_{s_m}(t) + \mu N_{s_i}(t). \end{cases} \tag{90}$$

With this, the average Mean Squared Displacement (MSD) of the particles in the system can be described over time, via the definition $D^2 = D^2_{s_m} + D^2_{s_i}$. This approximation of the MSD provides information on the limit behavior of the system, as we have discussed in Section 8.2.3 . We will study the system given in Equation (90) analytically and then compare the results with the MSD determined from the simulations as described in Section 7.4.

9.1.1 The long-term MSD

Because the system in Equation (90) consists of first-order linear ordinary differential equations, its long-time behaviour can be characterised analytically. In particular, the equilibrium values of all moments can be obtained in closed form. Writing these equilibrium values explicitly in terms of the model parameters enables us to infer information on the microscopic transport process from macroscopic observations.

With $N_{s_i}(t) = N_0 - N_{s_m}(t)$, we see that the equation for N_{s_m} reduces to an uncoupled scalar linear ODE:

$$\frac{d}{dt}N_{s_m}(t) = \mu N_0 - (\mu+\nu)N_{s_m}(t),$$

for which we find the solution

$$N_{s_m}(t) = \frac{\mu}{\mu+\nu}N_0 + \left(N_{s_m}(0) - \frac{\mu}{\mu+\nu}N_0\right)\exp(-(\mu+\nu)t).$$

This explicit solution shows that there is a globally asymptotically stable equilibrium, as $\mu, \nu > 0$ holds. We define the equilibrium value as follows:

$$N_{s_m,\text{eq}} := \lim_{t\to\infty} N_{s_m(t)} = \frac{\mu}{\mu+\nu}N_0.$$

The subsystem, consisting of the ODEs for the second velocity moments $V_m^2(t)$ and $V_i^2(t)$, is linear and driven by N_{s_m}. Since $N_{s_m}(t) \to N_{s_m,\text{eq}}$ exponentially, the velocity moments also converge exponentially to a constant value. At equilibrium, we find

$$\nu V_{s_m,\text{eq}}^2 - \mu V_{s_i,\text{eq}}^2 = 0, \qquad \lambda S_T^2 N_{s_m,\text{eq}} - \lambda V_{s_m,\text{eq}}^2 = 0,$$

which gives

$$\begin{aligned} V_{s_m,\text{eq}}^2 &= S_T^2 N_{s_m,\text{eq}} \\ &= S_T^2 \frac{\mu}{\mu+\nu}N_0. \end{aligned}$$

Similarly, for the mixed-moment equations, we find the unique and stable equilibrium

$$\begin{aligned} B_{s_m,\text{eq}} &= \frac{V_{s_m,\text{eq}}^2}{-\lambda(\psi-1)} \\ &= \frac{-S_T^2\mu}{\lambda(\psi-1)(\mu+\nu)}N_0. \end{aligned}$$

This convergence also takes place with exponential speed, as forced by the convergence of $V_{s_m}^2$ to $V_{s_m,\text{eq}}^2$. We then define the total second moment

$$D^2(t) := D_{s_i}^2(t) + D_{s_m}^2(t).$$

Adding the two second-moment equations gives

$$\frac{d}{dt}D^2(t) = 2B_{s_m}(t).$$

Since $B_{s_m}(t) \to B_{s_m,\text{eq}}$ exponentially, we obtain

$$\frac{d}{dt}D^2(t) = 2B_{s_m,\text{eq}} + O(e^{-ct}) \qquad \text{for some } c > 0,$$

and therefore, integrating gives

$$D^2(t) = D^2(0) + (2B_{s_m,\text{eq}})t + O(1) \qquad \text{as } t \to \infty.$$

Dividing by the conserved total number of particles N_0 and grouping the initial value $D^2(0)$ and the $\mathcal{O}(1)$-addition in the derived expression for $D^2(t)$, the Mean Squared Displacement for the system in equilibrium is given by

$$\begin{aligned} MSD(t) &= \frac{2B_{s_m,\text{eq}}}{N_0}t + \mathcal{O}(1) \\ &= \frac{-2S_T^2\mu}{\lambda(\psi-1)(\mu+\nu)}t + \mathcal{O}(1). \end{aligned} \tag{91}$$

9.1.2 The MSD for two-dimensional Brownian motion

The expression given in Equation (91) holds for general cases of the Markovian Hitchhikers' Model. We will use this to compare analytical results with results from the simulations performed in Section 7.4. Here, the particles moved in a two-dimensional space and followed Brownian motion. Since, according to Theorem 8.1, in two spatial dimensions Brownian motion obeys $\mathrm{MSD}(t) \sim 4D_{\text{eff}}t$, and the index of persistence is $\psi = 0$, we get

$$MSD(t) = \frac{S_T^2 \mu}{\lambda(\mu+\nu)} t + \mathcal{O}(1).$$

We see the effective diffusion constant is given by $D_{\text{eff}} = \frac{S_T^2 \mu}{2\lambda(\mu+\nu)}$. We recognize the fraction $\frac{\mu}{\mu+\nu}$ as the parameter α from Section 7.3 in the case of a homogeneous carrier population. In this case, we have derived the estimator

$$\hat{\alpha} = D_{\text{eff}} \frac{2\lambda}{S_T^2}. \tag{92}$$

This relation provides a connection between microscopic dynamics and macroscopic observable transport. In particular, the fraction of time hitchhikers spend in the mobile state can be experimentally identified in several ways. If the attachment and detachment rates can be measured directly, one can directly estimate α via $\alpha = \frac{\mu}{\mu+\nu}$. In certain contexts, the attachment and detachment parameters μ and ν might be difficult to directly quantify, as these processes take place on a small scale. However, if only particle trajectories are available, the effective diffusion coefficient D_{eff} can be estimated from the long-time slope of the Mean Squared Displacement. Then, provided the reorientation rate λ and the mean squared speed are known, α can be estimated with Equation (92). Finally, when the diffusion coefficient of the carriers is known independently, α may also be obtained from the ratio between the effective diffusion coefficient of the hitchhiking population and that of the carriers. These equivalent estimators make the model applicable across a range of experimental settings, where different quantities may be measurable depending on the available measurement techniques.

In the next subsection, we will simulate the model given in Equation (90) and compare the estimator for α in Equation (92) to the true value as defined by the simulation parameters.

9.2 Numerical verification of the MSD from the moment system for the Markovian case

To validate the moment system derived in Section 9.1, we compare its prediction for the Mean Squared Displacement (MSD) with the MSD obtained from stochastic particle simulations.

We consider the case of a spatially uniform transporter density. Consequently, the attachment rate $\mu(u) = \mu$ becomes constant, and therefore the mobile fraction $\alpha(\mathbf{x}, t) = \frac{\mu}{\mu+\nu}$ is also constant. The parameter values used in the simulations are listed in Table 3 in Section 7.4.

The constant transporter density u_0 is obtained numerically by uniformly distributing the transporters over the domain, constructing the corresponding density field on the grid, smoothing this field, and computing its spatial average. Using the outcome $u_0 \approx 0.99$ and the parameters in Table 3 yields $\alpha = \frac{\mu_0 u_0}{\nu+\mu_0 u_0} \approx 0.330$.

The system of moment equations was solved numerically using the LSODA solver in R, with all moments initially set to zero. From the slope of the MSD curve, the effective diffusion coefficient D_{eff} was determined. Applying the estimator derived in Equation (92) gives $\hat{\alpha} \approx 0.341$, which differs by approximately 3% from the value implied by the model parameters.

The resulting prediction for the MSD was compared with the ensemble-averaged MSD obtained from the stochastic simulations. The result of this comparison is shown in Figure 9. During the initial transient, the two curves differ noticeably, indicating an influence of the initial conditions before the process reaches its diffusive regime. However, at later times, all curves seem to correspond well,

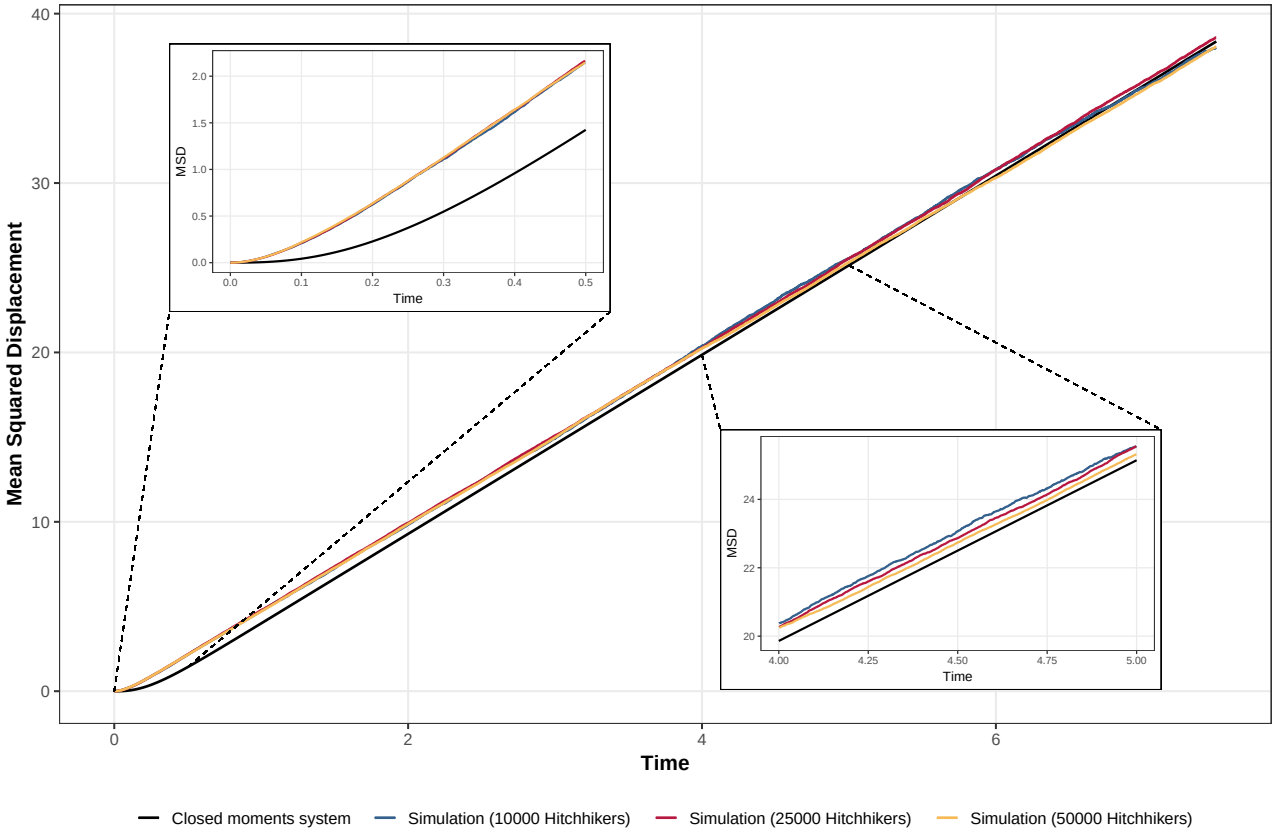


Figure 9: A comparison of the MSD over time based on the results of the simulation described in Algorithm 2 for 10000, 25000 and 50000 hitchhikers and for the closed moment system given in Equation (90). The population of carriers is assumed to be homogeneously distributed. A zoomed-in panel of the curves is given in the two panels, highlighting the behavior of the curves in the time intervals $[0, 0.5]$ and $[4, 5]$. The parameters used in the simulation are given in Table 3.

indicating that the moments system eventually approximates the MSD of the stochastic system rather well. We note that the MSD curves for the stochastic simulations with 10000, 25000 and 50000 hitchhikers are very similar on the full time interval considered, indicating that all values are sufficiently high to estimate the MSD. The agreement between the stochastic simulations and the moment system confirms that the asymptotic expression derived in Section 9.1 accurately describes the long-time ensemble MSD of the hitchhiking process.

9.3 Numerical verification of the time-averaged single-particle MSD

The previous subsection considered the ensemble MSD obtained by averaging over many particles. However, in many settings, it is impractical to obtain trajectories for a large number of particles. We therefore investigate whether the effective diffusive coefficient can be recovered from a single trajectory using the time-averaged MSD by implementing a single-particle tracking experiment. A single particle was simulated according to the agent-based hitchhiking model, with the process as described in Algorithm 2. Rather than computing the displacement from the initial position, the trajectory was analysed using a fixed time lag $\tau = 0.05$. For each observation time T, the mean squared displacement was estimated by averaging the squared displacements over all pairs of positions separated by the lag τ that were available up to time T. Consequently, the estimate is based on progressively more displacement samples as the observation time increases.

Figure 10 shows the resulting estimate in black, together with the ensemble mean squared displacement at the same lag τ as the red curve. The latter curve is obtained from the experiment for 50000 hitchhikers described in Section 7.4. The black curve evolves as the observation time increases and

more displacement measurements become available. In contrast, the ensemble MSD at a fixed lag is independent of the observation time and is therefore visualised as the constant value of the function $MSD(\tau)$ defined in Equation (86). Initially, the single-particle estimate fluctuates substantially because only a limited number of displacement measurements are available. As the observation time increases, the number of displacement samples grows, reducing the sampling variability. The estimated MSD therefore eventually seems to gradually stabilize around a value that agrees with the ensemble MSD at the same lag time.

This agreement demonstrates that, for the hitchhiking process considered here, sufficiently long single-particle trajectories can provide an accurate estimate of the ensemble MSD at fixed lag time. Consequently, the effective diffusion coefficient may be inferred from long single-particle observations without requiring simultaneous measurements of many particles.

When combining results of the numerical experiments visualized in Figures 9 and 10, we conclude that the analytical moment system, the ensemble-averaged MSD obtained from many stochastic trajectories, and the time-averaged MSD obtained from a single sufficiently long trajectory all recover a similar effective diffusion coefficient. Consequently, the effective diffusion coefficient may be estimated either analytically, from an ensemble of trajectories, or from a single long trajectory. Since we had derived that $D_{\text{eff}} = \alpha D$ with D denoting the carriers' diffusion coefficient and α as in Equation (78), these approaches provide information regarding the underlying attachment and detachment dynamics.

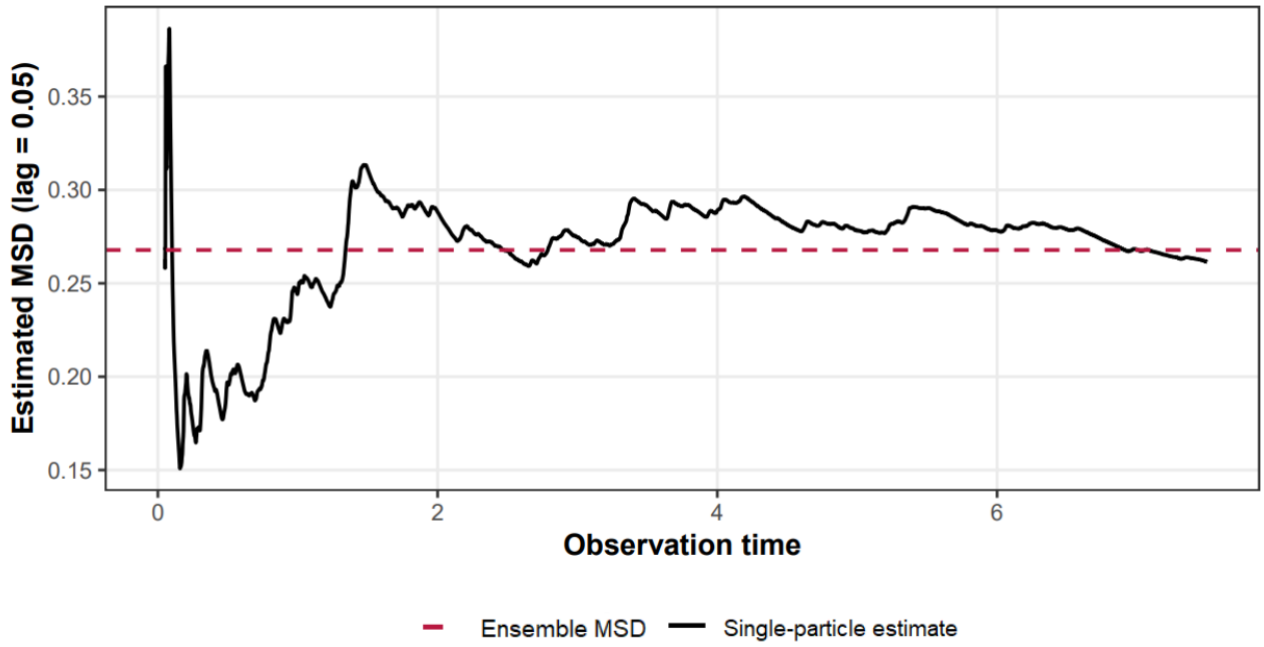


Figure 10: The time-adjusted single-particle mean-squared displacement (MSD) estimated at a lag of $\tau = 0.05$ for the Markovian Hitchhiking Model with homogeneously distributed carriers. The black curve shows the time-averaged MSD computed from an individual trajectory, obtained from an agent-based simulation as described in Algorithm 2. The rate parameters for the hitchhiking process are given in Table 3. At each time point, the MSD is calculated according to Equation (87). The red dashed line indicates the ensemble-averaged MSD across 50000 particles, calculated according to Equation (86). The evolution of the hitchhiker density during this simulation is visualized in Figure 7.

10 Discussion and conclusions

The goal of this thesis was to formulate a mathematical model for applications in carrier-mediated transport dynamics, a phenomenon observed in many fields within the natural sciences. This was achieved through an extension of the classical Velocity-Jump Model, which describes the evolution of a density function over space and time based on individual-level properties of these particles. Building upon this Velocity-Jump Model, we derived a two-population transport model that captures this hitchhiking mechanism and bridges individual movement rules with collective macroscopic behaviour.

From the analysis performed on the Hitchhikers' Model under the assumption of Markovian dynamics, three main conclusions can be drawn. First, using a parabolic scaling, the Markovian Hitchhikers' Model can be approximated by a diffusion-advection equation in which the hitchhikers' effective diffusion coefficient equals the carrier diffusion coefficient, scaled by the fraction of mobile hitchhikers. The diffusion coefficient of the carriers can be analytically derived based on the average time before a reorientation takes place. Second, a moment closure procedure provides an alternative approximation of the stochastic process. The resulting expression for the Mean Squared Displacement can also be used to derive the effective diffusion coefficient of the hitchhiker population. Third, numerical experiments demonstrate that the effective diffusion coefficient in the Hitchhikers' Model can be recovered consistently using three different approaches: the diffusion limit, the ensemble Mean Squared Displacement, and the time-averaged MSD of a single sufficiently long trajectory. The agreement between simulations of the approaches confirm that the continuum limit equation and the moment system approximate the long-term behaviour of the underlying stochastic process well.

This work showcases that the same transport process can be described at multiple scales. We formulated a stochastic agent-based model that captures the individual attachment and detachment events. Then, we scaled this model using a parabolic limit, which resulted in a continuum description of the particle density. Lastly, we built a moment system that yields analytical expressions for experimentally observable quantities. The numerical experiments demonstrate that these descriptions are consistent within the considered parameter regime. Consequently, the attachment and detachment dynamics may be inferred from observations in several independent ways. When both carrier and hitchhiker diffusion coefficients are measurable, the mobile fraction follows directly from their ratio. Alternatively, the effective diffusion coefficient may be recovered from the long-time growth of the ensemble MSD or from the time-averaged MSD of a sufficiently long single trajectory.

10.1 Practical application of the Hitchhikers' Model

The results obtained in this thesis may be useful in applications where attachment and detachment events cannot be observed directly, while particle trajectories can be reported. In such situations, the analytical relationship between the effective diffusion coefficient and the attachment dynamics provides a means of estimating microscopic rate parameters based on macroscopic observations. This insight can help drive experimental design or the design of observational studies.

All practical results presented in this work are obtained by comparing simulations to analytical results. However, in practice, a new complication can be found in fitting the model to data. An organism's trajectory is often not completely available. Instead, reports only denote when and how often an organism is observed at certain spots, and information on the actual trajectory is inferred based on that data [36]. A main concern in inference techniques is the assumption that reorientations take place at all observed time points, which means that they are imposed rather than inferred. From the inferred trajectory, no useful information on the organism's behavior or the actual movement paths can be obtained. In recent years, efforts have been made to construct Bayesian inference techniques for the trajectories and parameters in a Velocity-Jump model [5, 8, 9].

10.2 Limitations of the study and recommendations for future work

The work represented relies on several simplifying assumptions. In the analytical work, explicit expressions for the diffusion coefficients were only obtained for spatially homogeneous transporter densities.

However, in practice, populations often contain non-negligible variances. Extending the results to spatially heterogeneous environments poses a challenge. This limitation also holds for the classical Velocity-Jump Model, as is discussed in more detail in Section 5. Another assumption made in the derivation of the Hitchhikers' Model was that the carriers move independently of the hitchhikers. However, it is plausible that carrying cargo affects speed, persistence or reorientation behaviour. Incorporating such feedback would lead to bidirectional coupling between the two populations. Experimental and observational studies could provide insight into whether it is reasonable to neglect this effect in certain contexts. Furthermore, the analytical expression for the MSD relies on the assumption of constant attachment rates. Finding similar closed systems for spatially heterogeneous or non-Markovian settings requires more complicated mathematical techniques.

The simplifications made in the analysis also limited the scope of the simulation study. Although a general non-Markovian Hitchhikers' Model was formulated, the analytical and numerical investigations presented in this thesis focused primarily on its Markovian reduction. However, the simulation set-up presented in this study is general enough that independent numerical experiments could still yield interesting results and potentially offer additional insight that could guide analytical work. We propose the study of the diffusion limits and the moments reduction for non-Markovian attachment dynamics as an interesting direction for future work.

Also due to the simplifications, the frameworks developed in this thesis are particularly applicable to systems in which neither hitchhikers nor carriers directly interact with other particles of their own type, nor respond to any external cues. Should an application require it, we do believe that the latter can be implemented in this model, using the technique outlined by Erban and Othmer [14]. However, the lack of particle-particle interactions limits the models' application. This feature is difficult to include in the model. The rigorous derivation of transport equations for interacting populations is an active area of research [7, 15, 27, 45].

Lastly, no systematic sensitivity analysis is performed on the model parameters due to the high computational power necessary for the agent-based simulations. All the rate parameters listed in Table 3 were fixed throughout the experiments. Therefore, it remains unclear how well the derived approximations work for other parameter regimes.

10.3 Novelty

The central novelty of this thesis lies in the formulation and analysis of a mathematical transport framework in which immotile particles move by hitchhiking on mobile carriers. To our knowledge, such a model has not been reported yet and we believe that this model is a useful addition to the study of carrier-mediated transport, which occurs in many contexts within the natural sciences. The exact mechanisms involved in this process are often difficult to directly observe and measure. By deriving and studying a moment-closed system that captures these dynamics, our work provides a mechanistic baseline against which more complex interaction structures can be compared. The general methodology of deriving equations through renewal processes with auxiliary variables is based on the work of by Taylor-King et al., who applied this for an extension of the Velocity-Jump Model with a resting state [47]. However, the full derivation of their model is not explicitly given. By elucidating all steps, the work in this thesis establishes a methodological template for extending such models to more complex biological scenarios. The model presented by Taylor-King et al. differs from ours in two main aspects. First, the former model describes only one population which can enter and leave a mobile state, whereas we incorporate a one-way interaction between two populations. Second, the reorientation process of the moving particles differs. In particular, the hitchhikers in our model can have several reorientations during their mobile period, whereas the particles modelled by Taylor-King et al. reorientate every time they leave the immobile period.

A second novelty concerns the Markovian Hitchhikers' Model. One striking limitation of this model is caused by the assumption of exponentially distributed running times. We have shown that this assumption can be relaxed by providing a more general formulation of the Velocity-Jump Model. According to our literature search, this general framework has not been reported yet.

More broadly, this work illustrates how mathematics can help in understanding attachment and detachment dynamics. By combining stochastic simulations, analysis, continuum modelling, and moment methods, it establishes a direct connection between these microscopic dynamics and experimentally measurable quantities such as the diffusion rate and particle trajectories. It showcases that with the right model, large-scale observations can be insightful in elucidating small-scale mechanisms. In particular, the effective diffusion coefficient provides a direct link between experimentally determinable particle trajectories and the underlying stochastic processes. This provides a theoretical basis for studying carrier-mediated transport and for future models incorporating memory effects, particle interactions, and more traits of realistic biological behaviour.

Appendices

A Deriving moment equations for the Markovian Hitchhikers' Model

We will derive the moment system given in Equation (89), which is an approximation for a Markovian Hitchhikers process with pure diffusive carriers.

Recall that we have

$$\begin{cases} \frac{\partial}{\partial t} s_i(\mathbf{x}, \mathbf{v}, t) = \nu s_m(\mathbf{x}, \mathbf{v}, t) - \mu(u) s_i(\mathbf{x}, \mathbf{v}, t), \\ \frac{\partial}{\partial t} s_m(\mathbf{x}, \mathbf{v}, t) = -\nu s_m(\mathbf{x}, \mathbf{v}, t) + \mu(u) \int_V T(\mathbf{v}, \mathbf{v}') s_i(\mathbf{x}, \mathbf{v}', t)\ d\mathbf{v}' \\ \qquad\qquad - \mathbf{v} \cdot \nabla_{\mathbf{x}} s_m(\mathbf{x}, \mathbf{v}, t) \\ \qquad\qquad - \lambda s_m(\mathbf{x}, \mathbf{v}, t) + \lambda \int_V T(\mathbf{v}, \mathbf{v}') s_m(\mathbf{x}, \mathbf{v}', t)\ d\mathbf{v}'. \end{cases} \tag{93}$$

The assumption of conserved mass immediately gives the differential equations for $N_{s_m}(t)$ and $N_{s_i}(t)$:

$$\frac{dN_{s_m}(t)}{dt} = -\frac{dN_{s_i}(t)}{dt} = -\nu N_{s_m}(t) + \int_V \int_\Omega \mu(u) s_i(\mathbf{x}, \mathbf{v}, t)\ d\mathbf{x}\ d\mathbf{v}. \tag{94}$$

A.1 Deriving ODEs for the second spatial moments

We derive the ordinary differential equation for the mean squared displacement of the motile hitchhikers. We first observe that we can substitute the partial differential equation for $s_m(\mathbf{x}, \mathbf{v}, t)$ in the definition for $\frac{dD^2_{s_m}}{dt}$ as follows:

$$\begin{aligned} \frac{dD^2_{s_m}(t)}{dt} &= \frac{d}{dt} \left(\int_V \int_\Omega ||\mathbf{x}||^2 s_m(\mathbf{x}, \mathbf{v}, t)\ d\mathbf{x}\ d\mathbf{v} \right) \\ &= \int_V \int_\Omega ||\mathbf{x}||^2 \frac{\partial}{\partial t} s_m(\mathbf{x}, \mathbf{v}, t)\ d\mathbf{x}\ d\mathbf{v} \\ &= \int_V \int_\Omega ||\mathbf{x}||^2 \Big(-\nu s_m(\mathbf{x}, \mathbf{v}, t) - \mathbf{v} \cdot \nabla_{\mathbf{x}} s_m(\mathbf{x}, \mathbf{v}, t) + \mu(u) s_i \\ &\qquad\qquad - \lambda s_m(\mathbf{x}, \mathbf{v}, t) + \lambda \int_V T(\mathbf{v}, \mathbf{v}') s_m(\mathbf{x}, \mathbf{v}', t)\ d\mathbf{v}' \Big)\ d\mathbf{x}\ d\mathbf{v}. \end{aligned}$$

We use the linearity of the integral operators and write

$$\frac{dD^2_{s_m}(t)}{dt} = I_1 + I_2 + I_3 + I_4 + I_5,$$

with

$$\begin{aligned} I_1 &:= -\nu \int_V \int_\Omega ||\mathbf{x}||^2 s_m(\mathbf{x}, \mathbf{v}, t)\ d\mathbf{x}\ d\mathbf{v}, \\ I_2 &:= -\int_V \int_\Omega ||\mathbf{x}||^2 \mathbf{v} \cdot \nabla_{\mathbf{x}} s_m(\mathbf{x}, \mathbf{v}, t)\ d\mathbf{x}\ d\mathbf{v}, \\ I_3 &:= \int_V \int_\Omega ||\mathbf{x}||^2 \left(\mu(u) s_i\right)\ d\mathbf{x}\ d\mathbf{v}. \\ I_4 &:= -\lambda \int_V \int_\Omega ||\mathbf{x}||^2 s_m(\mathbf{x}, \mathbf{v}, t)\ d\mathbf{x}\ d\mathbf{v}, \\ I_5 &:= \lambda \int_V \int_\Omega ||\mathbf{x}||^2 \int_V T(\mathbf{v}, \mathbf{v}') s_m(\mathbf{x}, \mathbf{v}', t)\ d\mathbf{v}'\ d\mathbf{x}\ d\mathbf{v}, \end{aligned}$$

We see that, per definition,

$$I_1 = -\nu D^2_{s_m}(t).$$

We use the Divergence Theorem [1] to rewrite the integral I_2. To this end, we define the vector field given by

$$F(\mathbf{x}) := ||\mathbf{x}||^2 \mathbf{v} s_m(\mathbf{x}, \mathbf{v}, t),$$

which has divergence

$$\begin{aligned}\nabla_{\mathbf{x}} \cdot F &= \nabla_{\mathbf{x}} \cdot \left(||\mathbf{x}||^2 \mathbf{v} s_m\right) \\ &= ||\mathbf{x}||^2 \mathbf{v} \cdot \nabla_{\mathbf{x}} s_m + s_m \mathbf{v} \nabla_{\mathbf{x}} ||\mathbf{x}||^2.\end{aligned}$$

As $\nabla_{\mathbf{x}} \cdot \mathbf{v} = 0$, we get

$$\int_\Omega ||\mathbf{x}||^2 \mathbf{v} \cdot \nabla_{\mathbf{x}} s_m \, d\mathbf{x} = \int_\Omega \nabla_{\mathbf{x}} \cdot F \, d\mathbf{x} - \int_\Omega s_m \mathbf{v} \cdot \nabla_{\mathbf{x}} ||\mathbf{x}||^2 \, d\mathbf{x}.$$

We now apply the Divergence Theorem to find

$$\int_\Omega \nabla_{\mathbf{x}} \cdot F \, d\mathbf{x} = \int_{\partial\Omega} F \cdot n \, dS = 0,$$

where the equality to zero follows from the assumption that $\Omega = \mathbb{R}^d$. Thus, with $\nabla_{\mathbf{x}} ||\mathbf{x}||^2 = 2\mathbf{x}$, we find

$$\int_\Omega ||\mathbf{x}||^2 \mathbf{v} \cdot \nabla_{\mathbf{x}} s_m \, d\mathbf{x} = -2 \int_\Omega s_m(\mathbf{x}, \mathbf{v}, t) \cdot (\mathbf{v} \cdot \mathbf{x}).$$

Thus, substituting this in Equation (A.1), gives

$$\begin{aligned}I_2 &= 2 \int_V \int_\Omega (\mathbf{v} \cdot \mathbf{x}) s_m(\mathbf{x}, \mathbf{v}, t) \, d\mathbf{x} \, d\mathbf{v} \\ &= 2 B_{s_m}(t).\end{aligned}$$

To express I_3 in terms of the observables, we derive $\frac{d}{dt} D^2_{s_i}(t)$:

$$\begin{aligned}\frac{d}{dt} D^2_{s_i}(t) &= \frac{d}{dt} \left(\int_V \int_\Omega ||\mathbf{x}||^2 s_i(\mathbf{x}, \mathbf{v}, t) \, d\mathbf{x} \, d\mathbf{v} \right) \\ &= \int_V \int_\Omega ||\mathbf{x}||^2 \left(\nu s_m(\mathbf{x}, \mathbf{v}, t) - \mu(u) s_i(\mathbf{x}, \mathbf{v}, t) \right) \, d\mathbf{x} \, d\mathbf{v} \\ &= \nu D^2_{s_m}(t) - I_3.\end{aligned}$$

Therefore, we find

$$I_3 = -\frac{d}{dt} D^2_{s_i}(t) + \nu D^2_{s_m}(t).$$

We immediately see that per definition,

$$I_4 = -\lambda D^2_{s_m}(t).$$

For I_5, we find

$$\begin{aligned}I_5 &= \lambda \int_V \int_\Omega ||\mathbf{x}||^2 \int_V T(\mathbf{v}, \mathbf{v}') s_m(\mathbf{x}, \mathbf{v}', t) \, d\mathbf{v}' \, d\mathbf{x} \, d\mathbf{v} \\ &= \lambda \int_V \int_\Omega ||\mathbf{x}||^2 \int_V \left(T(\mathbf{v}, \mathbf{v}') \, d\mathbf{v} \right) s_m(\mathbf{x}, \mathbf{v}', t) \, d\mathbf{x} \, d\mathbf{v} \\ &= \lambda \int_V \int_\Omega ||\mathbf{x}||^2 s_m(\mathbf{x}, \mathbf{v}', t) \, d\mathbf{x} \, d\mathbf{v} \\ &= \lambda D^2_{s_m}(t),\end{aligned}$$

where we apply Fubini's theorem [42] and Assumption (T2).

Thus, we derived

$$\frac{dD^2_{s_m}(t)}{dt} = 2B_{s_m}(t) - \frac{d}{dt}D^2_{s_i}(t).$$

A.2 Deriving ODEs for mixed moments

We find

$$\begin{aligned}\frac{dB_{s_m}(t)}{dt} = \int_V \int_\Omega (\mathbf{v}\cdot\mathbf{x})\Big(&- \nu s m(\mathbf{x},\mathbf{v},t) - \mathbf{v}\cdot\nabla_{\mathbf{x}} s_m(\mathbf{x},\mathbf{v},t) \\ &- \lambda s_m(\mathbf{x},\mathbf{v},t) + \lambda\int_V T(\mathbf{v},\mathbf{v}') s_m(\mathbf{x},\mathbf{v},t)\ d\mathbf{v}' + \mu(u)s_i\Big)\ d\mathbf{x}\ d\mathbf{v} \\ = I_1 + I_2 + I_3 + I_4 + I_5\end{aligned}$$

for

$$\begin{aligned} I_1 &:= -\nu\int_V\int_\Omega (\mathbf{v}\cdot\mathbf{x}) s_m(\mathbf{x},\mathbf{v},t)\ d\mathbf{x}\ d\mathbf{v}, \\ I_2 &:= -\int_V\int_\Omega (\mathbf{v}\cdot\mathbf{x})\,(\mathbf{v}\cdot\nabla_{\mathbf{x}} s_m(\mathbf{x},\mathbf{v},t))\ d\mathbf{x}\ d\mathbf{v}, \\ I_3 &:= -\lambda\int_V\int_\Omega (\mathbf{v}\cdot\mathbf{x}) s_m(\mathbf{x},\mathbf{v},t)\ d\mathbf{x}\ d\mathbf{v}, \\ I_4 &:= \lambda\int_V\int_\Omega (\mathbf{v}\cdot\mathbf{x})\int_V T(\mathbf{v},\mathbf{v}') s_m(\mathbf{x},\mathbf{v}',t)\ d\mathbf{v}'\ d\mathbf{x}\ d\mathbf{v}, \\ I_5 &:= \int_V\int_\Omega (\mathbf{v}\cdot\mathbf{x})\mu(u) s_i(\mathbf{x},\mathbf{v},t)\ d\mathbf{x}\ d\mathbf{v}. \end{aligned}$$

We immediately find

$$I_1 = -\nu B_{s_m}(t),$$

As in Section A.1, we apply the Divergence Theorem to find

$$\begin{aligned} I_2 &= -\int_V\int_\Omega (\mathbf{v}\cdot\mathbf{x})\,(\mathbf{v}\cdot\nabla_{\mathbf{x}} s_m(\mathbf{x},\mathbf{v},t))\ d\mathbf{x}\ d\mathbf{v} \\ &= \int_V\int_\Omega s_m(\mathbf{x},\mathbf{v},t)\,(\mathbf{v}\cdot\nabla_{\mathbf{x}}(\mathbf{v}\cdot\mathbf{x}))\ d\mathbf{x}\ d\mathbf{v} \\ &= \int_V\int_\Omega s_m(\mathbf{x},\mathbf{v},t)\,(\mathbf{v}\cdot\mathbf{v})\ d\mathbf{x}\ d\mathbf{v} \\ &= \int_V\int_\Omega s_m(\mathbf{x},\mathbf{v},t)||\mathbf{v}||^2\ d\mathbf{x}\ d\mathbf{v} \\ &= V^2_{s_m}(t). \end{aligned}$$

We see that

$$I_3 = -\lambda B_{s_m}(t).$$

For I_4, we apply Fubini's Theorem to find

$$\begin{aligned} I_4 &= \lambda\int_V\int_\Omega (\mathbf{v}\cdot\mathbf{x})\int_V T(\mathbf{v},\mathbf{v}') s_m(\mathbf{x},\mathbf{v}',t)\ d\mathbf{v}'\ d\mathbf{x}\ d\mathbf{v} \\ &= \lambda\int_V\int_\Omega \mathbf{x}\cdot\int_V \mathbf{v}\cdot T(\mathbf{v},\mathbf{v}')\ d\mathbf{v} s_m(\mathbf{x},\mathbf{v}',t)\ d\mathbf{x}\ d\mathbf{v}' \\ &= \lambda\int_V\int_\Omega \mathbf{x}\cdot(\psi\mathbf{v}') s_m(\mathbf{x},\mathbf{v}',t)\ d\mathbf{x}\ d\mathbf{v}' \\ &= \lambda\psi\int_V\int_\Omega (\mathbf{x}\cdot\mathbf{v}') s_m(\mathbf{x},\mathbf{v}',t)\ d\mathbf{x}\ d\mathbf{v}' \\ &= \lambda\psi B_{s_m}(t). \end{aligned}$$

We find I_5 by calculating $\frac{dB_{s_i}(t)}{dt}$:

$$\begin{aligned}\frac{dB_{s_i}(t)}{dt} &= \frac{d}{dt}\left(\int_V \int_\Omega (\mathbf{v}\cdot\mathbf{x}) s_i(\mathbf{x},\mathbf{v},t)\ d\mathbf{x}\ d\mathbf{v}\right)\\ &= \int_V \int_\Omega (\mathbf{v}\cdot\mathbf{x})\left(\nu s_m(\mathbf{x},\mathbf{v},t) - \mu(u) s_i(\mathbf{x},\mathbf{v},t)\right)\ d\mathbf{x}\ d\mathbf{v}\\ &= \int_V \int_\Omega (\mathbf{v}\cdot\mathbf{x})\left(\nu s_m(\mathbf{x},\mathbf{v},t)\right)\ d\mathbf{x}\ d\mathbf{v} - \int_V \int_\Omega (\mathbf{v}\cdot\mathbf{x})\left(\mu(u) s_i(\mathbf{x},\mathbf{v},t)\right)\ d\mathbf{x}\ d\mathbf{v}\\ &= \nu B_{s_m}(t) - I_5.\end{aligned}$$

Thus, we find

$$I_5 = -\frac{dB_{s_i}(t)}{dt} + \nu B_{s_m}(t).$$

So, we find

$$\begin{aligned}\frac{dB_{s_m}(t)}{dt} &= I_1 + I_2 + I_3 + I_4 + I_5\\ &= V^2_{s_m}(t) + \lambda(\psi - 1) B_{s_m}(t) - \frac{dB_{s_i}(t)}{dt}\end{aligned}$$

A.3 Deriving ODEs for the second velocity moments

$$\begin{aligned}\frac{dV^2_{s_m}(t)}{dt} &= \frac{d}{dt}\left(\int_V \int_\Omega ||\mathbf{v}||^2 s_m(\mathbf{x},\mathbf{v},t)\ d\mathbf{x}\ d\mathbf{v}\right)\\ &= \int_V \int_\Omega ||\mathbf{v}||^2 \Big(- \nu s_m(\mathbf{x},\mathbf{v},t) - \mathbf{v}\cdot\nabla_{\mathbf{x}} s_m(\mathbf{x},\mathbf{v},t) - \lambda s_m(\mathbf{x},\mathbf{v},t)\\ &\quad + \lambda \int_V T(\mathbf{v},\mathbf{v}') s_m(\mathbf{x},\mathbf{v},t)\ d\mathbf{v}' + \mu(u) s_i(\mathbf{x},\mathbf{v},t)\Big)\ d\mathbf{x}\ d\mathbf{v}\\ &= I_1 + I_2 + I_3 + I_4 + I_5,\end{aligned}$$

with

$$\begin{aligned}I_1 &:= -\nu \int_V \int_\Omega ||\mathbf{v}||^2 s_m(\mathbf{x},\mathbf{v},t)\ d\mathbf{x}\ d\mathbf{v},\\ I_2 &:= -\int_V \int_\Omega ||\mathbf{v}||^2 \left(\mathbf{v}\cdot\nabla_{\mathbf{x}} s_m(\mathbf{x},\mathbf{v},t)\right)\ d\mathbf{x}\ d\mathbf{v},\\ I_3 &:= -\lambda \int_V \int_\Omega ||\mathbf{v}||^2 \left(s_m(\mathbf{x},\mathbf{v},t)\right)\ d\mathbf{x}\ d\mathbf{v},\\ I_4 &:= \lambda \int_V \int_\Omega ||\mathbf{v}||^2 \left(\int_V T(\mathbf{v},\mathbf{v}') s_m(\mathbf{x},\mathbf{v},t)\ d\mathbf{v}'\right)\ d\mathbf{x}\ d\mathbf{v},\\ I_5 &:= \int_V \int_\Omega ||\mathbf{v}||^2 \left(\mu(u) s_i(\mathbf{x},\mathbf{v},t)\right)\ d\mathbf{x}\ d\mathbf{v},\end{aligned}$$

We see that

$$I_1 = -\nu V^2_{s_m}(t).$$

Using the Divergence Theorem, we find

$$\begin{aligned}I_2 &= -\int_V \int_\Omega ||\mathbf{v}||^2 \left(\mathbf{v}\cdot\nabla_{\mathbf{x}} s_m(\mathbf{x},\mathbf{v},t)\right)\ d\mathbf{x}\ d\mathbf{v}\\ &= \int_V \int_\Omega ||\mathbf{v}||^2 \left(s_m(\mathbf{x},\mathbf{v},t)\cdot\nabla_{\mathbf{x}}\mathbf{v}\right)\ d\mathbf{x}\ d\mathbf{v}\\ &= 0,\end{aligned}$$

as $\nabla_{\mathbf{x}}\mathbf{v} = 0$.

We see

$$I_3 = -\lambda V_{s_m}^2(t).$$

We find

$$\begin{aligned} I_4 &= \lambda \int_V \int_\Omega ||\mathbf{v}||^2 \left(\int_V T(\mathbf{v}, \mathbf{v}') s_m(\mathbf{x}, \mathbf{v}, t) \, d\mathbf{v}' \right) d\mathbf{x} \, d\mathbf{v} \\ &= \lambda \int_V \int_\Omega \left(\int_V ||\mathbf{v}||^2 T(\mathbf{v}, \mathbf{v}') \, d\mathbf{v} s_m(\mathbf{x}, \mathbf{v}, t) \right) d\mathbf{x} d\mathbf{v}' \\ &= \lambda S_T^2 N_{s_m}(t), \end{aligned}$$

where we apply Fubini's Theorem and the assumption that $\int_V ||\mathbf{v}||^2 T(\mathbf{v}, \mathbf{v}') \, d\mathbf{v} \equiv S_T^2$.

We also find

$$\begin{aligned} \frac{dV_{s_i}^2(t)}{dt} &= \int_V \int_\Omega ||\mathbf{v}||^2 \left(\nu s_m(\mathbf{x}, \mathbf{v}, t) - \mu(u) s_i(\mathbf{x}, \mathbf{v}, t) \right) \\ &= \nu V_{s_m}^2(t) - I_5. \end{aligned}$$

Thus, we have

$$I_5 = -\frac{dV_{s_i}^2(t)}{dt} + \nu V_{s_m}^2(t).$$

So, we get

$$\frac{dV_{s_m}^2(t)}{dt} = -\lambda V_{s_m}^2(t) + \lambda S_T^2 N_{s_m}(t) - \frac{dV_{s_i}^2(t)}{dt}.$$

B Code availability and computational details

The code used in this thesis is accessible via the online repository https://github.com/MaiNguyen0310/Thesis_Model_Approximation. This repository additionally contains the extra figures from simulations that were not included in this thesis and a .drawio file used to create the schematic figures presented without citation.

All code was run on a laptop equipped with an AMD Ryzen 7 6800U CPU (8 cores, 16 threads, 2.7–4.7 GHz) and 16 GB RAM, using R Version 4.4.2.

References

[1] Robert Adams and Christopher Essex. *Calculus*. en. 10th ed. Boston, MA: Addison Wesley, May 2021.

[2] João Carlos Alves Barata and Mahir Saleh Hussein. "The Moore–Penrose Pseudoinverse: A Tutorial Review of the Theory". In: *Brazilian Journal of Physics* 42.1-2 (Dec. 2011), pp. 146–165. ISSN: 1678-4448. DOI: 10.1007/s13538-011-0052-z. URL: http://dx.doi.org/10.1007/s13538-011-0052-z.

[3] Andrew W. Bartlow and Salvatore J. Agosta. "Phoresy in animals: review and synthesis of a common but understudied mode of dispersal". In: *Biological Reviews* 96.1 (2020), pp. 223–246. ISSN: 1469-185X. DOI: 10.1111/brv.12654. URL: http://dx.doi.org/10.1111/brv.12654.

[4] Howard C. Berg. *Random Walks in Biology: New and Expanded Edition*. Princeton University Press, Nov. 2018. ISBN: 9780691000640. DOI: 10.2307/j.ctv7r40w6. URL: http://dx.doi.org/10.2307/j.ctv7r40w6.

[5] Paul G. Blackwell. *Bayesian inference for velocity-jump models for movement*. 2025. DOI: 10.48550/ARXIV.2509.21226. URL: https://arxiv.org/abs/2509.21226.

[6] Jake Bowley, Craig Baker-Austin, Adam Porter, Rachel Hartnell, and Ceri Lewis. "Oceanic Hitchhikers – Assessing Pathogen Risks from Marine Microplastic". In: *Trends in Microbiology* 29.2 (Feb. 2021), pp. 107–116. ISSN: 0966-842X. DOI: 10.1016/j.tim.2020.06.011. URL: http://dx.doi.org/10.1016/j.tim.2020.06.011.

[7] José Antonio Carrillo and Shuchen Guo. "Interacting particle approximation of cross-diffusion systems". In: *Nonlinearity* 39.2 (Feb. 2026), p. 025009. ISSN: 1361-6544. DOI: 10.1088/1361-6544/ae3655. URL: http://dx.doi.org/10.1088/1361-6544/ae3655.

[8] Arianna Ceccarelli, Alexander P. Browning, and Ruth E. Baker. "Approximate Solutions of a General Stochastic Velocity-Jump Model Subject to Discrete-Time Noisy Observations". In: *Bulletin of Mathematical Biology* 87.5 (Mar. 2025). ISSN: 1522-9602. DOI: 10.1007/s11538-025-01437-x. URL: http://dx.doi.org/10.1007/s11538-025-01437-x.

[9] Arianna Ceccarelli, Alexander P. Browning, Tai Chaiamarit, Ilan Davis, and Ruth E. Baker. "A likelihood-based Bayesian inference framework for the calibration of and selection between stochastic velocity-jump models". In: *Journal of the Royal Society Interface* 23.236 (Mar. 2026). ISSN: 1742-5662. DOI: 10.1098/rsif.2025.0866. URL: http://dx.doi.org/10.1098/rsif.2025.0866.

[10] Jenna R. Christensen and Samara L. Reck-Peterson. "Hitchhiking Across Kingdoms: Cotransport of Cargos in Fungal, Animal, and Plant Cells". In: *Annual Review of Cell and Developmental Biology* 38.1 (Oct. 2022), pp. 155–178. ISSN: 1530-8995. DOI: 10.1146/annurev-cellbio-120420-104341. URL: http://dx.doi.org/10.1146/annurev-cellbio-120420-104341.

[11] Rick Durrett. *Probability: Theory and Examples*. Cambridge University Press, Apr. 2019. ISBN: 9781108473682. DOI: 10.1017/9781108591034. URL: http://dx.doi.org/10.1017/9781108591034.

[12] Radek Erban and Jan and Haskovec. "From individual to collective behaviour of coupled velocity jump processes: A locust example". In: *Kinetic amp; Related Models* 5.4 (2012), pp. 817–842. ISSN: 1937-5077. DOI: 10.3934/krm.2012.5.817. URL: http://dx.doi.org/10.3934/krm.2012.5.817.

[13] Radek Erban and Hans G. Othmer. "From Individual to Collective Behavior in Bacterial Chemotaxis". In: *SIAM Journal on Applied Mathematics* 65.2 (Jan. 2004), pp. 361–391. ISSN: 1095-712X. DOI: 10.1137/s0036139903433232. URL: http://dx.doi.org/10.1137/S0036139903433232.

[14] Radek Erban and Hans G. Othmer. "Taxis equations for amoeboid cells". In: *Journal of Mathematical Biology* 54.6 (Feb. 2007), pp. 847–885. ISSN: 1432-1416. DOI: 10.1007/s00285-007-0070-1. URL: http://dx.doi.org/10.1007/s00285-007-0070-1.

[15] Gonzalo Galiano and Virginia Selgas. "On a cross-diffusion segregation problem arising from a model of interacting particles". In: *Nonlinear Analysis: Real World Applications* 18 (Aug. 2014), pp. 34–49. ISSN: 1468-1218. DOI: 10.1016/j.nonrwa.2014.02.001. URL: http://dx.doi.org/10.1016/j.nonrwa.2014.02.001.

[16] Carlos García-Robledo, Diego Dierick, and Konstantine Manser. "Electric transportation and electroreception in hummingbird flower mites". In: *Proceedings of the National Academy of*

Sciences 122.5 (Jan. 2025). ISSN: 1091-6490. DOI: 10.1073/pnas.2419214122. URL: http://dx.doi.org/10.1073/pnas.2419214122.

[17] Enrico Gavagnin and Christian A. Yates. "Modeling persistence of motion in a crowded environment: The diffusive limit of excluding velocity-jump processes". In: *Physical Review E* 97.3 (Mar. 2018). ISSN: 2470-0053. DOI: 10.1103/physreve.97.032416. URL: http://dx.doi.org/10.1103/PhysRevE.97.032416.

[18] Dzung Ha. *Functional Analysis Volume 1: A gentle introduction*. Matrix Editions, Jan. 2006. ISBN: 978-0-9715766-1-2.

[19] Efrat Hagai, Reut Dvora, Tal Havkin-Blank, Einat Zelinger, Ziv Porat, Stefan Schulz, and Yael Helman. "Surface-motility induction, attraction and hitchhiking between bacterial species promote dispersal on solid surfaces". In: *The ISME Journal* 8.5 (Dec. 2013), pp. 1147–1151. ISSN: 1751-7370. DOI: 10.1038/ismej.2013.218. URL: http://dx.doi.org/10.1038/ismej.2013.218.

[20] Paul Richard Halmos and Viakalathur Shankar Sunder. *Bounded Integral Operators on L2 Spaces*. Springer Berlin Heidelberg, 1978. ISBN: 9783642670169. DOI: 10.1007/978-3-642-67016-9. URL: http://dx.doi.org/10.1007/978-3-642-67016-9.

[21] Noel Hawkins. *Crab hitches a ride on the back of a jellyfish*. https://www.bbc.com/news/videos/c0krq7dj840o. Accessed: 2026-03-15. 2024.

[22] Thomas Hillen and Hans G. Othmer. "The Diffusion Limit of Transport Equations Derived from Velocity-Jump Processes". In: *SIAM Journal on Applied Mathematics* 61.3 (2000), pp. 751–775. ISSN: 00361399. URL: http://www.jstor.org/stable/3061749.

[23] Thomas Hillen and Amanda Swan. "The Diffusion Limit of Transport Equations in Biology". In: *Mathematical Models and Methods for Living Systems*. Springer International Publishing, 2016, pp. 73–129. ISBN: 9783319426792. DOI: 10.1007/978-3-319-42679-2_2. URL: http://dx.doi.org/10.1007/978-3-319-42679-2_2.

[24] Colin J. Ingham, Oren Kalisman, Alin Finkelshtein, and Eshel Ben-Jacob. "Mutually facilitated dispersal between the nonmotile fungus Aspergillus fumigatus and the swarming bacterium Paenibacillus vortex". In: *Proceedings of the National Academy of Sciences* 108.49 (Nov. 2011), pp. 19731–19736. ISSN: 1091-6490. DOI: 10.1073/pnas.1102097108. URL: http://dx.doi.org/10.1073/pnas.1102097108.

[25] Constanza Kettmayer, Enrico Gratton, and Laura C Estrada. "Comparison of MSD analysis from single particle tracking with MSD from images. Getting the best of both worlds". In: *Methods and Applications in Fluorescence* 12.1 (Oct. 2023), p. 015001. ISSN: 2050-6120. DOI: 10.1088/2050-6120/acfd7e. URL: http://dx.doi.org/10.1088/2050-6120/acfd7e.

[26] Christian Kuehn. "Moment Closure - A Brief Review". In: (2015). DOI: 10.48550/ARXIV.1505.02190. URL: https://arxiv.org/abs/1505.02190.

[27] Yue Li, Li Chen, and Zhipeng Zhang. "Convergence Towards the Population Cross-Diffusion System from Stochastic Many-Particle System". In: *Communications in Mathematical Research* 40.1 (Dec. 2023), pp. 43–63. ISSN: 1674-5647. DOI: 10.4208/cmr.2023-0002. URL: http://dx.doi.org/10.4208/cmr.2023-0002.

[28] Xavier Michalet. "Mean square displacement analysis of single-particle trajectories with localization error: Brownian motion in an isotropic medium". In: *Physical Review E* 82.4 (Oct. 2010). ISSN: 1550-2376. DOI: 10.1103/physreve.82.041914. URL: http://dx.doi.org/10.1103/PhysRevE.82.041914.

[29] Saurabh S. Mogre, Jenna R. Christensen, Cassandra S. Niman, Samara L. Reck-Peterson, and Elena F. Koslover. "Hitching a Ride: Mechanics of Transport Initiation through Linker-Mediated Hitchhiking". In: *Biophysical Journal* 118.6 (Mar. 2020), pp. 1357–1369. ISSN: 0006-3495. DOI: 10.1016/j.bpj.2020.01.024. URL: http://dx.doi.org/10.1016/j.bpj.2020.01.024.

[30] Andy Morffew. *Hummingbird Flower Mites*. https://www.flickr.com/photos/andymorffew/48875780486. Accessed: 2026-03-15. 2019.

[31] Alise R Muok, Dennis Claessen, and Ariane Briegel. "Microbial hitchhiking: how Streptomyces spores are transported by motile soil bacteria". In: *The ISME Journal* 15.9 (Mar. 2021), pp. 2591–

2600. ISSN: 1751-7370. DOI: 10.1038/s41396-021-00952-8. URL: http://dx.doi.org/10.1038/s41396-021-00952-8.

[32] Alise R Muok, Dennis Claessen, and Ariane Briegel. "Microbial hitchhiking: how Streptomyces spores are transported by motile soil bacteria". In: *The ISME Journal* 15.9 (Mar. 2021), pp. 2591–2600. ISSN: 1751-7370. DOI: 10.1038/s41396-021-00952-8. URL: http://dx.doi.org/10.1038/s41396-021-00952-8.

[33] H. G. Othmer, S. R. Dunbar, and W. Alt. "Models of dispersal in biological systems". In: *Journal of Mathematical Biology* 26.3 (1988), pp. 263–298. ISSN: 1432-1416. DOI: 10.1007/bf00277392. URL: http://dx.doi.org/10.1007/BF00277392.

[34] Hans G. Othmer and Thomas Hillen. "The Diffusion Limit of Transport Equations II: Chemotaxis Equations". In: *SIAM Journal on Applied Mathematics* 62.4 (Jan. 2002), pp. 1222–1250. ISSN: 1095-712X. DOI: 10.1137/s0036139900382772. URL: http://dx.doi.org/10.1137/S0036139900382772.

[35] Kevin J. Painter and Jonathan A. Sherratt. "Modelling the movement of interacting cell populations". In: *Journal of Theoretical Biology* 225.3 (Dec. 2003), pp. 327–339. ISSN: 0022-5193. DOI: 10.1016/s0022-5193(03)00258-3. URL: http://dx.doi.org/10.1016/S0022-5193(03)00258-3.

[36] Toby A. Patterson, Alison Parton, Roland Langrock, Paul G. Blackwell, Len Thomas, and Ruth King. "Statistical modelling of individual animal movement: an overview of key methods and a discussion of practical challenges". In: *AStA Advances in Statistical Analysis* 101.4 (2017), pp. 399–438. ISSN: 1863-818X. DOI: 10.1007/s10182-017-0302-7. URL: http://dx.doi.org/10.1007/s10182-017-0302-7.

[37] Derek Ping, Tong Wang, David T Fraebel, Sergei Maslov, Kim Sneppen, and Seppe Kuehn. "Hitchhiking, collapse, and contingency in phage infections of migrating bacterial populations". In: *The ISME Journal* 14.8 (May 2020), pp. 2007–2018. ISSN: 1751-7370. DOI: 10.1038/s41396-020-0664-9. URL: http://dx.doi.org/10.1038/s41396-020-0664-9.

[38] Murray H. Protter and Charles B. Morrey. "Differentiation under the Integral Sign. Improper Integrals. The Gamma Function". In: *Intermediate Calculus*. Springer New York, 1985, pp. 421–453. ISBN: 9781461210863. DOI: 10.1007/978-1-4612-1086-3_8. URL: http://dx.doi.org/10.1007/978-1-4612-1086-3_8.

[39] A. G. Ramm. "A Simple Proof of the Fredholm Alternative and a Characterization of the Fredholm Operators". In: *The American Mathematical Monthly* 108.9 (Nov. 2001), pp. 855–860. ISSN: 1930-0972. DOI: 10.1080/00029890.2001.11919820. URL: http://dx.doi.org/10.1080/00029890.2001.11919820.

[40] John Salogiannis and Samara L. Reck-Peterson. "Hitchhiking: A Non-Canonical Mode of Microtubule-Based Transport". In: *Trends in Cell Biology* 27.2 (Feb. 2017), pp. 141–150. ISSN: 0962-8924. DOI: 10.1016/j.tcb.2016.09.005. URL: http://dx.doi.org/10.1016/j.tcb.2016.09.005.

[41] Tahoura Samad, Nicole Billings, Alona Birjiniuk, Thomas Crouzier, Patrick S Doyle, and Katharina Ribbeck. "Swimming bacteria promote dispersal of non-motile staphylococcal species". In: *The ISME Journal* 11.8 (Apr. 2017), pp. 1933–1937. ISSN: 1751-7370. DOI: 10.1038/ismej.2017.23. URL: http://dx.doi.org/10.1038/ismej.2017.23.

[42] R.L. Schilling. *Measures, Integrals and Martingales*. Measures, Integrals and Martingales. Cambridge University Press, 2017. ISBN: 978-1-316-62024-3. URL: https://books.google.nl/books?id=sdAoDwAAQBAJ.

[43] Christian Schmeiser, Gaël Raoul, and Vincent Calvez. "Confinement by biased velocity jumps: Aggregation of escherichia coli". In: *Kinetic and Related Models* 8.4 (2015), pp. 651–666. ISSN: 1937-5093. DOI: 10.3934/krm.2015.8.651. URL: http://dx.doi.org/10.3934/krm.2015.8.651.

[44] Abhishek Shrivastava, Visha K. Patel, Yisha Tang, Susan Connolly Yost, Floyd E. Dewhirst, and Howard C. Berg. "Cargo transport shapes the spatial organization of a microbial community". In: *Proceedings of the National Academy of Sciences* 115.34 (Aug. 2018), pp. 8633–8638. ISSN: 1091-6490. DOI: 10.1073/pnas.1808966115. URL: http://dx.doi.org/10.1073/pnas.1808966115.

[45] Angela Stevens. "The Derivation of Chemotaxis Equations as Limit Dynamics of Moderately Interacting Stochastic Many-Particle Systems". In: *SIAM Journal on Applied Mathematics* 61.1 (Jan. 2000), pp. 183–212. ISSN: 1095-712X. DOI: 10.1137/s0036139998342065. URL: http://dx.doi.org/10.1137/S0036139998342065.

[46] Michael Stoll and Ronald van Luijk. *Linear Algebra II*. Lecture notes, unpublished manuscript. 2024.

[47] Jake P. Taylor-King, E. Emiel van Loon, Gabriel Rosser, and S. Jon Chapman. "From Birds to Bacteria: Generalised Velocity Jump Processes with Resting States". In: *Bulletin of Mathematical Biology* 77.7 (2015), pp. 1213–1236. ISSN: 1522-9602. DOI: 10.1007/s11538-015-0083-7. URL: http://dx.doi.org/10.1007/s11538-015-0083-7.

[48] Kees Vuik, Fred Vermolen, Martin van Gijzen, and Thea Vuik. *Numerical Methods for Ordinary Differential Equations*. TU Delft OPEN Publishing, Feb. 2023. ISBN: 9789463666657. DOI: 10.5074/t.2023.001. URL: http://dx.doi.org/10.5074/t.2023.001.

[49] P. Signe White, Levi Morran, and Jacobus de Roode. "Phoresy". In: *Current Biology* 27.12 (2017), R578–R580. ISSN: 0960-9822. DOI: 10.1016/j.cub.2017.03.073. URL: http://dx.doi.org/10.1016/j.cub.2017.03.073.

[50] Thomas Witelski and Mark Bowen. "Fast/slow Dynamical Systems". In: *Methods of Mathematical Modelling*. Springer International Publishing, 2015, pp. 201–213. ISBN: 9783319230429. DOI: 10.1007/978-3-319-23042-9_10. URL: http://dx.doi.org/10.1007/978-3-319-23042-9_10.